M.A.M. SCHOOL OF ENGINEERING

(Autonomous)

(Accredited by NAAC || Approved by AICTE || Affiliated to Anna University) Trichy – Chennai Trunk Road, Siruganur, Tiruchirappalli – 621 105

UG CURRICULUM

(I to VIII SEMESTERS)

B.TECH. INFORMATION TECHNOLOGY

Choice Based Credit System (CBCS)

(For the students admitted during the Academic year 2024- 25 and onwards)

REGULATIONS 2024

M.A.M SCHOOL OF ENGINEERING

(AUTONOMOUS)

REGULATIONS 2024

CHOICE BASED CREDIT SYSTEM

B.TECH. INFORMATION TECHNOLOGY

I. PROGRAM EDUCATIONAL OBJECTIVES (PEOs)

1. Demonstrate technical competence with analytical and critical thinking to understand and meet the diversified requirements of industry, academia and research.

2. Exhibit technical leadership, team skills and entrepreneurship skills to provide business solutions to real world problems

3. Work in multi-disciplinary industries with social and environmental responsibility, work ethics and adaptability to address complex engineering and social problems

4. Pursue lifelong learning, use cutting edge technologies and involve in applied research to design optimal solutions

II. PROGRAM OUTCOMES (POs)

PO1	Engineering Knowledge: Apply the knowledge of mathematics ,science, engineering fundamentals, and an engineering specialization to the solution of complex engineering problems.
PO2	Problem Analysis: Identify, formulate, review research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences ,and engineering sciences.
PO3	Design/Development of Solutions: Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for the public healthand safety, and the cultural, societal, and environmental considerations.
PO4	Conduct Investigations of Complex Problems: Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions.
PO5	Modern Tool Usage:Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modeling to complex engineering activities with an understanding of the limitations.

PO6	The Engineer and Society:Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to the professional engineering practice.
PO7	Environment and Sustainability:Understand the impact of the professional engineering solutions in societal and environmental contexts and demonstrate the knowledge of, and need for sustainable development.
PO8	Ethics: Apply ethical principles and commit to professional ethics and responsibilities and norms of the engineering practice.
PO9	Individual and Teamwork: Function effectively as an individual, and as member or leader in diverse teams, and in multi-disciplinary settings.
PO10	Communication: Communicate effectively on complex engineering activities with the engineering community and with society at large, such as, being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.
PO11	Project Management and Finance: Demonstrate knowledge and understanding of the engineering and management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.
PO12	Life-long Learning :Recognize the need for, and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change.

III. PROGRAM SPECIFIC OUTCOMES (PSOs)

PSO1	Have proficiency in programming skills to design, develop and apply appropriate techniques, to solve complex engineering problems..
PSO2	Have knowledge to build, automate and manage business solutions using cutting edge technologies.
PSO3	Have excitement towards research in applied computer technologies

CURRICULUM

M.A.M SCHOOL OF ENGINEERING

DEPARTMENT OF INFORMATION TECHNOLOGY

REGULATIONS 2024

CHOICE BASED CREDIT SYSTEM

(Students admitted from the Academic Year 2024 – 25 onwards) I TO VIII SEMESTERS

CURRICULUM

Induction Program (Mandatory)	3 weeks duration
Induction program for students to be offered right at the start of the first year	• Physical activity • Creative Arts • Universal Human Values • Literary • Proficiency Modules • Lectures by Eminent People • Visits to local Areas • Familiarization to Dept./Branch& Innovations

B.TECH. INFORMATION TECHNOLOGY

SEMESTER I

S. No	Course Code	Course	L	T	P	C	Maximum Marks			Category
							CA	ES	Total	
THEORY COURSES										
1.	24HS101	Communicative English	3	0	0	3	40	60	100	HS
2.	24BS101	Matrices & Calculus	3	1	0	4	40	60	100	BS
3.	24ES104	Programming in C	3	0	0	3	40	60	100	ES
4.	24HS102	Heritage of Tamil	1	0	0	1	40	60	100	HS
THEORY COURSES WITH LABORATORY COMPONENT										
5.	24BS102	Engineering Chemistry	3	0	2	4	50	50	100	BS
LABORATORY COURSES										
6.	24HS103	Communicative English Laboratory	0	0	2	1	60	40	100	HS
7.	24ES105	Programming in C Lab	0	0	4	2	60	40	100	ES
8.	24ES106	Engineering Practices lab	0	0	2	1	60	40	100	ES
9.	24ES107	Workshop Practices Lab	0	0	2	1	60	40	100	ES
		TOTAL	**13**	**1**	**12**	**20**				

SEMESTER II										
THEORY COURSES										
S.No	Course Code	Course	L	T	P	C	Maximum Marks			Category
							CA	ES	Total	
THEORY COURSES										
1.	24BS202	Discrete Mathematics	3	1	0	4	40	60	100	BS
2.		Language Elective	2	0	0	2	40	60	100	HS
3.	24HS201	Tamils and Technology	1	0	0	1	40	60	100	HS
4.	24ES201	Design Thinking	2	0	0	2	40	60	100	ES
5.	24ES208	Python Programming	3	0	0	3	40	60	100	ES
THEORY COURSES WITH LABORATORY COMPONENT										
6.	24BS203	Physics for Engineers	3	0	2	4	50	50	100	BS
7.	24ES210	Data Structures & Algorithms	3	0	2	4	50	50	100	ES
LABORATORY COURSES										
8.	24ES209	Python Programming Laboratory	0	0	4	2	60	40	100	ES
9.	24ES205	Engineering Drawing	0	0	4	2	60	40	100	ES
10.	24TP201	Aptitude Skills and Communication skills I	0	0	2	1	100		100	EEC
		TOTAL	17	1	14	25				

SEMESTER III										
THEORY COURSES										
S.No	Course Code	Course	L	T	P	C	Maximum Marks			Category
							CA	ES	Total	
THEORY COURSES										
1.	24BS301	Statistics &Numerical Methods	3	1	0	4	40	60	100	BS
2.	24AD301	Fundamentals of AI	3	0	0	3	40	60	100	PC
3.	24CS302	Object Oriented Programming	3	0	0	3	40	60	100	PC
4.	24AD303	Design and Analysis of Algorithms	3	0	0	3	40	60	100	PC
THEORY COURSES WITH LABORATORY COMPONENT										
6.	24CS305	Digital Principles and System Design	3	0	2	4	50	50	100	PC
7.	24IT301	Data science using R	3	0	2	4	50	50	100	PC
LABORATORY COURSES										
8.	24 AD306	AI Lab	0	0	4	2	60	40	100	PC
9.	24 CS307	Object Oriented Programming Lab	0	0	4	2	60	40	100	PC
10.	24TP301	Aptitude Skills and Communication skills II	0	0	2	1	100	0	100	EEC
		TOTAL	18	1	14	26				

SEMESTER IV										
THEORY COURSES										
S.No	Course Code	Course	L	T	P	C	Maximum Marks			Category
							CA	ES	Total	
THEORY COURSES										
1.	24ECE01	Microprocessors and Microcontrollers	3	0	0	3	40	60	100	PC
2.	24CS402	Database Management Systems	3	0	0	3	40	60	100	PC
3.	24IT401	Computer Architecture	3	0	0	3	40	60	100	PC
4.	24MC401	Environmental Science	3	0	0	0	40	60	100	MC
5.	24MC402	Disaster Management and Preparedness	3	0	0	2	40	60	100	MC
THEORY COURSES WITH LABORATORY COMPONENT										
6.	24AD403	Operating Systems	3	0	2	4	50	50	100	PC
7.	24CS405	JAVA Programming	3	0	2	4	50	50	100	PC
LABORATORY COURSES										
8.	24ECE02	Microprocessors and Microcontrollers Lab	0	0	4	2	60	40	100	PC
9.	24CS407	Database Management Systems Lab	0	0	4	2	60	40	100	PC
10.	24TP401	Aptitude Skills III & Technical Skills I	0	0	2	1	100	0	100	EEC
		TOTAL	18	1	14	23				

SEMESTER V										
THEORY COURSES										
S.No	Course Code	Course	L	T	P	C	Maximum Marks			Category
							CA	ES	Total	
THEORY COURSES										
1.	24AD501	Internet of Things	3	0	0	3	40	60	100	PC
2.	24AD501	Computer Networks	3	0	0	3	40	60	100	PC
3.	-	Professional Elective-I	3	0	0	3	40	60	100	PE
4.	-	Professional Elective-II	3	0	0	3	40	60	100	PE
5.	-	Open Elective-I	3	0	0	3	40	60	100	OE
THEORY COURSES WITH LABORATORY COMPONENT										
6.	24AD502	Big Data Analytics	3	0	2	4	50	50	100	PC
LABORATORY COURSES										
7.	24AD503	Computer Networks Lab	0	0	4	2	60	40	100	PC
8.	24AD504	IOT Laboratory	0	0	0	2	60	40	100	PC
9.	24TP501	Aptitude Skills IV & Technical Skills II	0	0	2	1	100	-	100	EEC
		TOTAL	18	0	8	24				

SEMESTER VI

THEORY COURSES

S.No	Course Code	Course	L	T	P	C	Maximum Marks CA	ES	Total	Category
THEORY COURSES										
1.	24HS601	Principles of Management	3	0	0	3	40	60	100	HS
2.	24CS602	Mobile Application and Development	3	0	0	3	40	60	100	PC
3.	-	Professional Elective–III	3	0	0	3	40	60	100	PE
4.	-	Professional Elective–IV	3	0	0	3	40	60	100	PE
5.	-	Open Elective–II	3	0	0	3	40	60	100	OE
THEORY COURSES WITH LABORATORY COMPONENT										
6.	24AD603	Deep Learning	3	0	2	4	50	50	100	PC
LABORATORY COURSES										
7.	24CS604	Mobile App Development Lab	0	0	4	2	60	40	100	PC
8.	24TP601	Aptitude Skills V & Technical Skills III	0	0	2	1	100	-	100	EEC
		TOTAL	18	0	8	22				

SEMESTER VII

THEORY COURSES

S.No	Course Code	Course	L	T	P	C	Maximum Marks CA	ES	Total	Category
THEORY COURSES										
1.	24HS701	AI in Ethics	3	0	0	3	40	60	100	HS
2.	24IT701	Distributed Systems	3	0	0	3	40	60	100	PC
3.	-	Professional Elective–V	3	0	0	3	40	60	100	PE
4.	-	Open Elective–III	3	0	0	3	40	60	100	OE
LABORATORY COURSES										
5.	24AD605	Internship / Mini-Project	0	0	4	2	60	40	100	EEC
		TOTAL	12	0	4	14				

SEMESTER VIII

THEORY COURSES

S.No	Course Code	Course	L	T	P	C	Maximum Marks CA	ES	Total	Category
LABORATORY COURSES										
1.	24IT801	Project Work	0	0	20	10	60	40	100	EEC
		TOTAL	0	0	20	10				

PROFESSIONAL ELECTIVE COURSES						
S.No	Course Code	Course	L	T	P	C
VERTICAL I						
1.	24CSX16	Augmented Reality and Virtual Reality	3	0	0	3
2.	24CSX11	Game Development	3	0	0	3
3.	24ADX01	UI/UX Design	3	0	0	3
4.	24ADX02	Computational Thinking	3	0	0	3
5.	24ADX03	Data Virtualization Tools	3	0	0	3
6.	24ADX04	Natural Processing Language	3	0	0	3

PROFESSIONAL ELECTIVE COURSES						
S.No	Course Code	Course	L	T	P	C
VERTICAL II						
1.	24ADX05	Web Technologies	3	0	0	3
2.	24ADX06	Full Stack Development	3	0	0	3
3.	24ADX07	Data Exploration Techniques	3	0	0	3
4.	24ADX08	AI in Commerce	3	0	0	3
5.	24ADX09	AI Agents	3	0	0	3
6.	24ADX10	Generative AI	3	0	0	3

PROFESSIONAL ELECTIVE COURSES						
S.No	Course Code	Course	L	T	P	C
VERTICAL III						
1.	24CSX17	Security Assessment and Risk Analysis	3	0	0	3
2.	24CSX18	Ethical Hacking	3	0	0	3
3.	24CSX19	Digital Mobile Forensics	3	0	0	3
4.	24CSX20	Cryptocurrency and Block Chain Technologies	3	0	0	3
5.	24CSX21	Security and Privacy in Cloud	3	0	0	3
6.	24CSX22	Web Application Security	3	0	0	3

PROFESSIONAL ELECTIVE COURSES						
S.No	Course Code	Course	L	T	P	C
VERTICAL IV						
1.	24ITX01	Cloud Service Management	3	0	0	3
2.	24ITX02	Robotics Process Automation	3	0	0	3
3.	24ITX03	AI based Conversational System	3	0	0	3
4.	24ITX04	Big Data Integration and Processing	3	0	0	3
5.	24ITX05	Predictive Analytics	3	0	0	3
6.	24ITX06	3D Modelling & Animation	3	0	0	3

PROFESSIONAL ELECTIVE COURSES						
S.No	Course Code	Course	L	T	P	C
VERTICAL V						
1.	24CSX33	AI in Drone Development	3	0	0	3
2.	24CSX34	Computer Vision	3	0	0	3
3.	24CSX35	Edge Computing	3	0	0	3
4.	24CSX36	Explainable AI	3	0	0	3
5.	24CSX37	Human Computer Interaction	3	0	0	3
6.	24CSX38	Image Processing and Analytics	3	0	0	3

S.No.	Category	Credits Per Semester								Total Credit	Credits in%
		I	II	III	IV	V	VI	VII	VIII		
1	HS	5	3				3	3		14	8.6
2	BS	8	8	4	4					24	14.7
3	ES	7	13							20	12.3
5	PC			20	18	14	9	3		64	39.3
6	PE					6	6	3		15	9.2
7	OE					3	3	3		9	5.5
8	EEC		1	1	1	1	1	2	10	17	10.4
9	MC				0					0	0.0
Total		20	25	25	23	24	22	14	10	163	100

HS–Humanities and Social Science

BS–Basic Science

ES–Engineering Science

PC–Professional Core

PE –Professional Elective

OE–Open Elective

EEC–Employability Enhancement Course

MC–Mandatory course

CA–Continuous Assessment

ES–End Semester Examination

<table>
<tr><td>R 2024</td><td colspan="5">SCIENCE & HUMANITIES</td><td>SEMESTER: I</td></tr>
<tr><td rowspan="2">24HS101</td><td rowspan="2">COMMUNICATIVE ENGLISH - I</td><td>L</td><td>T</td><td>P</td><td>C</td><td rowspan="2">HS</td></tr>
<tr><td>3</td><td>0</td><td>0</td><td>3</td></tr>
<tr><td colspan="7">COMMON TO: ALL PROGRAMS</td></tr>
</table>

COURSE OBJECTIVES:

The objectives of learning this course are to:

- ✓ Enable learners to use words appropriately in their communication.
- ✓ Enhance learners' grammatical accuracy in communication.
- ✓ Develop learners ability to read and listen to texts in English.
- ✓ Strengthen the communication skills of the learners.
- ✓ Help learners write appropriately in professional contexts

COURSE OUTCOMES:

At the end of this course, students are able to

CO1: Understand the basic grammatical structures and apply them in right context
CO2: Identify and report cause and effects in events, industrial processes through technical texts.
CO3: Apply appropriate words in a professional context.
CO4: Interpret information presented in tables, charts and other graphic forms.
CO5: Draft effective resumes in the context of job search.

<table>
<tr><td>UNIT: I</td><td>BASICS OF LANGUAGE</td><td>9</td></tr>
<tr><td colspan="3">Reading - Reading brochures (technical context), telephone messages advertisements, user manuals. Writing - Sequential Writing – connecting ideas using transitional words (Jumbled Sentences), Grammar – basics; parts of speech, Simple Tenses – Form, Function and Meaning;
Vocabulary - Synonyms; One word substitution</td></tr>
<tr><td>Pedagogical Tools</td><td colspan="2">Black board, chalk, group discussion, role play, youtube videos, NPTEL videos</td></tr>
<tr><td>UNIT: II</td><td>INTRODUCTION TO FUNDAMENTALS OF COMMUNICATION</td><td>9</td></tr>
<tr><td colspan="3">Reading - Reading biographies, travelogues, newspaper reports, Writing -Cause and Effect Essays, Grammar: Continuous Tenses, Subject-Verb Agreement, Idioms; Vocabulary: Antonyms,
Language puzzles.</td></tr>
<tr><td>Pedagogical Tools</td><td colspan="2">Black board, chalk, group discussion, role play, youtube videos, NPTEL videos</td></tr>
<tr><td>UNIT: III</td><td>NARRATION AND SUMMATION</td><td>9</td></tr>
<tr><td colspan="3">Reading – Reading advertisements, Case Studies, Writing- Check-list, Instructions. Grammar:
Perfect Tenses, Imperatives; Adjectives, Vocabulary: Language Games/ Group Discussion.</td></tr>
<tr><td>Pedagogical Tools:</td><td colspan="2">Black board, chalk, group discussion, role play, youtube videos, NPTEL videos</td></tr>
<tr><td>UNIT: IV</td><td>REPORTING OF EVENTS AND RESEARCH</td><td>9</td></tr>
<tr><td colspan="3">Reading –Newspaper articles; Writing – Recommendations, Transcoding Grammar – Reported
Speech, Pronouns - Possessive & Relative pronouns, Vocabulary: Oral Presentation.</td></tr>
<tr><td>Pedagogical Tools</td><td colspan="2">Black board, chalk, group discussion, role play, youtube videos, NPTEL videos</td></tr>
<tr><td>UNIT: V</td><td>THE ABILITY TO PUT IDEAS OR INFORMATION COGENTLY</td><td>9</td></tr>
<tr><td colspan="3">Reading – Company profiles, Statement of Purpose, (SOP), an excerpt of interview with professionals; Writing – Job / Internship application – Cover letter & Resume; Grammar – Numerical adjectives, Relative Clauses. Degrees of comparison, Phrasal Verbs; Vocabulary: Informal Vocabulary and Formal Substitutes.</td></tr>
<tr><td>Pedagogical Tools</td><td colspan="2">Black board, chalk, group discussion, role play, youtube videos, NPTEL videos</td></tr>
<tr><td colspan="3" align="right">Total Periods :45</td></tr>
</table>

TEXT BOOKS:				
Sl.No	**Authors**	**Title of the Book**	**Publisher**	**Year of publication**
1	Raymond, Murphy	English Grammar in Use (5^{th} Edition)	Cambridge Press: New York	2019
2	Dr. KN. Shoba, and Dr. Lourdes Joevani	English for Science & Technology	Cambridge University Press	2021

REFERENCE BOOKS:				
Sl.No	**Authors**	**Title of the Book**	**Publisher**	**Year of publication**
1	Meenakshi Raman & Sangeeta Sharma	Technical Communication Principles And Practices	Oxford Univ. Press	2016
2	Lakshmi Narayanan	A Course Book on Technical English	Scitech Publications (India) Pvt. Ltd.	2017
3	Kulbhusan Kumar	Effective Communication Skill	R S Salaria, Khanna Publishing House.	2018

WEB LEARNING RESOURCES:
1 https://store.acolad.com/products/english-for-engineering
2 https://www.cambridge.es/en/catalogue/business-english/other-titles/cambridge-english-for/engineering
3 https://shipcon.eu.com/english-for-engineers/
4 https://www.udemy.com/course/english-for-engineers/
5 https://store.acolad.com/products/english-for-engineering

CO – PO – PSO MAPPING

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO 10	PO 11	PO 12	PS O1	PS O2	PS O3
CO1	-	-	-	-	-	1	1	-	-	-	-	3	-	-	-
CO2	-	3	-	-	-	-	3	3	-	3	-	3	-	-	-
CO3	-	-	-	-	2	-	2	-	-	3	-	3	-	-	-
CO4	-	-	-	-	-	3	-	1	2	3	-	3	-	-	-
CO5	-	-	-	-	-	-	-	-	-	3	3	3	-	-	-
AVG	-	3	-	-	2	2	2	2	2	3	3	3	-	-	-

<table>
<tr><td>R 2024</td><td colspan="5">SCIENCE & HUMANITIES</td><td>SEMESTER: I</td></tr>
<tr><td rowspan="2">24BS101</td><td rowspan="2">MATRICES AND CALCULUS</td><td>L</td><td>T</td><td>P</td><td>C</td><td rowspan="2">BS</td></tr>
<tr><td>3</td><td>1</td><td>0</td><td>4</td></tr>
<tr><td colspan="7">COMMON TO: ALL DEPARTMENTS</td></tr>
</table>

COURSE OBJECTIVES:

The objectives of learning this course are to:

- Develop the use of matrix algebra techniques that is needed by engineers for practical applications.
- Familiarize the student with functions of several variables. this is needed in many branches of engineering.
- Make the students understand various techniques of integration.
- Acquaint the student with mathematical tools needed in evaluating multiple integrals and their applications.
- Make the student acquire sound knowledge of techniques in solving ordinary differential equations that model engineering problems.

COURSE OUTCOMES:

At the end of this course, students are able to

CO1: Apply the knowledge of matrices with the concepts of eigenvalues to study their problems in core areas

CO2: Apply the basic techniques and theorems function of several variables in other areas of mathematics

CO3: Apply different methods of integration in solving practical problems.

CO4: Apply multiple integral ideas in solving areas, volumes and other practical problems.

CO5: Solve basic application problems described by second and higher order linear differential equations with constant coefficients.

UNIT: I	**MATRICES**	**9+3**
Eigen values and Eigenvectors of a real matrix - Properties of Eigen values and Eigenvectors (without proof) - Statement and applications of Cayley- Hamilton theorem (without proof) - Diagonalization of matrices- Reduction of a quadratic form to canonical form by orthogonal transformation-Nature of quadratic forms.		
Pedagogical Tools	Chalk & Board, PPT, NPTEL video, you tube video, Group Discussion	
UNIT: II	**FUNCTIONS OF SEVERAL VARIABLES**	**9+3**
Partial derivatives - Total derivative - Jacobian and properties - Taylor's series expansion for function of two variables - Extreme values of functions of two variables - Lagrange multipliers method.		
Pedagogical Tools	Chalk & Board, PPT, NPTEL video, you tube video, Group Discussion	
UNIT: III	**INTEGRAL CALCULUS**	**9+3**
Definite and indefinite integrals - Substitution rule - Techniques of Integration: Integration by parts, Trigonometric integrals, Trigonometric substitutions, Integration of rational functions by Partial fraction, Integration of irrational functions		
Pedagogical Tools	Chalk & Board, PPT, NPTEL video, you tube video, Group Discussion	
UNIT: IV	**MULTIPLE INTEGRALS**	**9+3**
Double integrals - Change of order of integration - Double integrals in polar coordinates - Triple integrals - Applications in area and volume (except spherical , cylindrical coordinates)		
Pedagogical Tools	Chalk & Board, PPT, NPTEL video, you tube video, Group Discussion	
UNIT: V	**ORDINARY DIFFERENTIAL EQUATIONS**	**9+3**
Second and higher order linear differential equations with constant coefficients - Variable coefficients - Euler Cauchy equation - method of variation parameters.		
Pedagogical Tools	Chalk & Board, PPT, NPTEL video, you tube video, Group Discussion	
		Total Periods :60

TEXT BOOKS:

Sl. No	Authors	Title of the Book	Publisher	Year of publication
1	Kreyszig.E	Advanced Engineering Mathematics	John Wiley and sons, New Delhi	2016
2	Grewal B.S	Higher Engineering Mathematics	Khanna Publishers, New Delhi	2018
3	James Stewart	Calculus : Early Transcendentals	Cengage Learning, New Delhi	2015

REFERENCE BOOKS:

Sl. No	Authors	Title of the Book	Publisher	Year of Publication
1	Bali.N, M.Goyal And Watkins.C	Advanced Engineering Mathematics	Lakshmi Publications, New Delhi	2015
2	Ramana B.V	Higher Engineering Mathematics	McGraw Hill Education, New Delhi	2016
3	Narayanan.S, Manicavasagam Pillai.T.K	Calculus	S.Vishwanathan Publishers, Chennai	2009

WEB LEARNING RESOURCES:

1 https://nptel.ac.in/courses/111108157
2 https://nptel.ac.in/courses/111104125
3 https://nptel.ac.in/courses/111105121
4 https://nptel.ac.in/courses/111104085
5 https://nptel.ac.in/courses/111104521
6 https://www.brainkart.com/subject/Matrices-and-Calculus_454/
7 https://youtu.be/i8FukKfMKCI
8 https://youtu.be/wRR715lkK-E
9 https://youtu.be/iGJxxlyqrRM
10 https://youtu.be/yyc4yhlFATk
11 https://youtu.be/Ziu0y2kWTCM

CO – PO – PSO MAPPING

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO 10	PO 11	PO 12	PS O1	PS O2	PS O3
CO1	3	3	1	1	-	-	-	-	-	-	-	3	-	-	-
CO2	3	3	1	1	-	-	-	-	-	-	-	3	-	-	-
CO3	3	3	1	1	-	-	-	-	-	-	-	3	-	-	-
CO4	3	3	1	1	-	-	-	-	-	-	-	3	-	-	-
CO5	3	3	3	3	-	-	-	-	-	-	-	2	-	-	-
AVG	3	3	1	1	-	-	-	-	-	-	-	3	-	-	-

<table>
<tr><td>R 2024</td><td colspan="5">COMPUTER SCIENCE AND ENGINEERING</td><td>SEMESTER:I</td></tr>
<tr><td rowspan="2">24ES104</td><td rowspan="2">PROGRAMMING IN C</td><td>L</td><td>T</td><td>P</td><td>C</td><td rowspan="2">ES</td></tr>
<tr><td>3</td><td>0</td><td>0</td><td>3</td></tr>
<tr><td colspan="7">Common to CSE, IT AND AI&DS Departments</td></tr>
</table>

COURSE OBJECTIVES:

The objectives of learning this course are:

- To understand the constructs of C Language.
- To develop C Programs using basic programming constructs
- To develop C programs using arrays and strings
- To develop modular applications in C using functions
- To develop applications in C using pointers and structures
- To do input/output and file handling in C

COURSE OUTCOMES:

At the end of this course, students are able to:
CO1: Develop simple applications in C using basic constructs
CO2: Design and implement applications using arrays and strings
CO3: Develop and implement modular applications in C using functions.
CO4: Develop applications in C using structures and pointers.
CO5: Design applications using sequential and random access file processing

UNIT: I	BASICS OF C PROGRAMMING	9

Algorithm, and Flowchart for problem solving with Sequential Logic Structure – Applications of C Language - Structure of C program - C programming: Data Types - Constants – Operators - Input/Output statements, Assignment statements – Control flow statements – Preprocessor directives - Compilation process, Library Functions.

Pedagogical Tools	Black board, chalk, Group Discussion, Role Play, Youtube Videos,Nptel videos.

UNIT: II	ARRAYS AND STRINGS	9

Introduction to Arrays: Declaration, Initialization - Passing Arrays to Functions – Multidimensional Arrays - String operations – NULL Character - Reading and Writing a String – Processing the Strings – Character arithmetic – Searching and Sorting of Strings - Selection sort, linear and binary search.

Pedagogical Tools	Black board, chalk, Group Discussion, Role Play, Youtube Videos,Nptel videos.

UNIT: III	FUNCTIONS AND POINTERS	9

Modular programming - Function prototype, function call, Built-in functions (string functions, math functions) - I/O functions - (Formatted scanf() & printf(), getchar (), putchar (), getche(), gets(), puts()) – Recursion – Pointers – Pointer operators – Pointer arithmetic – Arrays and pointers – Array of pointers – Pointers as Function Arguments, Functions Returning Pointers - Parameter passing.

Pedagogical Tools	Black board, chalk, Group Discussion, Role Play, Youtube Videos,Nptel videos.

UNIT: IV	FILE PROCESSING	9

Files – Input/ Output Operations on Files - Error Handling During I/O Operations - Types of file processing: Sequential access, Random access – Sequential access file - Random access file - Command line arguments

Pedagogical Tools	Black board, chalk, Group Discussion, Role Play, Youtube Videos,Nptel videos.

UNIT: V	STRUCTURES AND UNION	9
Structure - Nested structures – Pointer and Structures – Array of structures – Self referential structures – Dynamic memory allocation – typedef – Union - STORAGE CLASSES: Storage classes-Automatic variables - External variables - Static variables.		
Pedagogical Tools	Black board, chalk, Group Discussion, Role Play, Youtube Videos,Nptel videos.	
		Total Periods : 45

TEXT BOOKS:				
Sl. No	**Authors**	**Title of the Book**	**Publisher**	**Year of publication**
1	E Balagurusamy	Programming in ANSI C	Tata McGraw Hill	2010
2	Yashwant Kanetkar	Let us C	Notion Press	2020
3	ReemaThareja	Programming in C	Oxford University Press	2016
REFERENCE BOOKS:				
Sl. No	**Authors**	**Title of the Book**	**Publisher**	**Year of publication**
1	Paul Deitel and Harvey Deitel	C How to Program with an Introduction to C++	BPB Publications	2018
2	Kernighan, B.W and Ritchie,D.M	The C Programming language	Pearson Education	2015
3	Byron S. Gottfried	Schaum's Outline of Theory and Problems of Programming with C	McGraw-Hill Education	21996

WEB LEARNING RESOURCES:
1. https://en.cppreference.com/w/c/language
2. https://www.programiz.com/c-programming
3. https://www.w3schools.com/c/
4. https://www.geeksforgeeks.org/c-programming-language/
5. https://www.javatpoint.com/c-programming-language-tutorial

CO PO PSO MAPPING

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO 10	PO11	PO12	PSO1	PSO2	PSO3
CO1	2	2	2	1	2	1	1	1	2	-	3	2	2	2	-
CO2	2	3	2	1	2	1	1	1	2	-	3	2	2	2	-
CO3	3	2	2	1	3	1	1	1	2	-	3	3	2	2	-
CO4	2	3	3	1	2	1	2	1	2	-	3	2	2	3	-
CO5	2	3	3	1	2	1	-	-	2	1	2	2	2	2	-
AVG	2	3	3	1	2	1	1.5	1	2	1	2.5	2.5	2	2.5	-

R 2024	SCIENCE & HUMANITIES					SEMESTER: I
24HS102	தமிழy மர(/ Heritage of Tamil	L	T	P	C	HS
		1	0	0	1	

COMMON TO: ALL PROGRAMS

COURSE OBJECTIVES:

The objectives of learning this course are to

- ✓ Learn the Extensive literature of classical tamil
- ✓ Review the fine arts heritage of tamil culture
- ✓ Realize the contribution of tamil in Indian freedom struggle

COURSE OUTCOMES:

At the end of this course, students are able to

CO1: Understand the weaving and ceramic technology of ancient tamil people nature.

CO2: Understand the construction technology, building materials in sangam period and case studies.

CO3: Infer the metal process, coin and beads manufacturing with relevant archaeological evidence.

CO4: Realize the agriculture methods, irrigation technology and pearl diving.

CO5: Apply the knowledge of scientific tamil and tamil computing.

UNIT: I	LANGUAGE AND LITERATURE	3

Dravidian Languages – Tamil as a Classical Language - Classical Literature in Tamil – Distributive Justice in Sangam Literature - Management Principles in Thirukural - Tamil Epics and Impact of Buddhism & Jainism in Tamil Land - Bakthi Literature Azhwars and Nayanmars - Forms of minor Poetry - Development of Modern literature in Tamil - Contribution of Bharathiyar and Bharathidhasan

Pedagogical Tools	Board & Chalk, PPT, NPTEL video, you tube video, Group Discussion

UNIT: II	HERITAGE - ROCK ART PAINTINGS TO MODERN ART – SCULPTURE	3

Hero stone to modern sculpture - Bronze icons - Tribes and their handicrafts - Art of temple car making - - Massive Terracotta sculptures, Village deities, Thiruvalluvar Statue at Kanyakumari, Making of musical instruments - Mridangam, Parai, Veenai, Yazh and Nadhaswaram - Role of Temples in Social and Economic Life of Tamils.

Pedagogical Tools	Chalk & Board, PPT, NPTEL video, you tube video, Group Discussion

UNIT: III	FOLK AND MARTIAL ARTS	3

Therukoothu, Karakattam, VilluPattu, KaniyanKoothu, Oyillattam, Leather Puppetry, Silambattam, Valari, Tiger dance - Sports and Games of Tamils.

Pedagogical Tools	Chalk & Board, PPT, NPTEL video, you tube video, Role Play

UNIT: IV	THINAI CONCEPT OF TAMILS	3

Flora and Fauna of Tamils & Agam and Puram Concept from Tholkappiyam and Sangam Literature - Aram Concept of Tamils - Education and Literacy during Sangam Age - Ancient Cities and Ports of Sangam Age -Export and Import during Sangam Age - Overseas Conquest of Cholas.

Pedagogical Tools	Chalk & Board, PPT, NPTEL video, you tube video, Group Discussion

UNIT: V	CONTRIBUTION OF TAMILS TO INDIAN NATIONAL MOVEMENT AND INDIAN CULTURE	3

Contribution of Tamils to Indian Freedom Struggle - The Cultural Influence of Tamils over the other parts of India – Self-Respect Movement - Role of Siddha Medicine in Indigenous Systems of Medicine – Inscriptions & Manuscripts – Print History of Tamil Books.

Pedagogical Tools	Chalk & Board, PPT, NPTEL video, you tube video, Group Discussion

Total Periods :15

TEXT CUM REFERENCE BOOKS:				
Sl. No	Authors	Title of the Book	Publisher	Year of publication
1	Dr.K.K.Pillay	tamilnadu history people and culture	Tamilnadu Textbook and Education works Corporation	2019
2	EL Sundaram	Computer Tamil	Vikatanprasuram	2016
3	Dr.S.Singaravelu	Social Life of the Tamils - The Classical Period	International Institute of Tamil Studies.	2001
4	Dr.S.V.Subatamanian, Dr.K.D. Thirunavukkarasu	Historical Heritage of the Tamils	International Institute of Tamil Studies	2010
5	Dr.M.Valarmathi	The Contributions of the Tamils to Indian Culture	International Institute of Tamil Studies	2001
6		Keeladi - 'Sangam City Civilization on the banks of river Vaigai'	Department of Archaeology& Tamil Nadu Text Book and Educational Services Corporation, Tamil Nadu	2019
7	Dr. K. K. Pillay	Studies in the History of India with Special Reference to Tamil Nadu	The Author	1979
8		Porunai Civilization	Department of Archaeology & Tamil Nadu Text Book and Educational Services Corporation, Tamil Nadu	2019
9	R.Balakrishnan	Journey of Civilization Indus to Vaigai	RMRL	2019
10	Dr.K.K.Pillay	Social Life of Tamils	A joint publication of TNTB & ESC and RMRL	1975

WEB LEARNING RESOURCES:
https://youtu.be/8J3UJXu4JZ0?si=ekqrc_x3J79C_MwI
https://www.youtube.com/live/WbnNQM2LNQA?si=S5YS3vXjIotIuDxp
https://www.youtube.com/live/10Z7NdBPAYU?si=Xbvjmr9wzfQBCHH6
https://www.youtube.com/live/xkrRTmvPsbY?si=Xdj6zDOA-Wl7Vu9j
https://youtu.be/ByHvsH0I080?si=O2HnEcVubA8tb5h8

CO – PO – PSO MAPPING

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO 10	PO 11	PO 12	PS O1	PS O2	PS O3
CO1	-	-	-	-	-	-	3	3	-	2	-	3	-	-	-
CO2	-	-	-	-	-	-	3	3	-	2	-	3	-	-	-
CO3	-	-	-	-	-	-	3	3	-	2	-	3	-	-	-
CO4	-	-	-	-	-	-	3	3	-	2	-	3	-	-	-
CO5	-	-	-	-	-	-	3	3	-	2	-	3	-	-	-
AVG	-	-	-	-	-	-	3	3	-	2	-	3	-	-	-

R 2024	SCIENCE & HUMANITIES					SEMESTER: I
24BS102	ENGINEERING CHEMISTRY	L	T	P	C	BS
		3	0	2	4	

COMMON TO: AI & DS, CSE, ECE and IT

COURSE OBJECTIVES:

The objectives of learning this course are to:

- ✓ Inculcate sound understanding of water quality parameters and water treatment techniques.
- ✓ Introduce the basic concepts and applications of phase rule and alloys.
- ✓ Facilitate the understanding of different types of fuels, their preparation, properties and combustion characteristics.
- ✓ Familiarize the students with the different energy sources, operating principles, working processes and applications of energy conversion and storage devices.
- ✓ Impart knowledge on the basic principles and preparatory methods of nonmaterial.

COURSE OUTCOMES:

At the end of this course, students are able to

CO1: Understand the quality of water from quality parameter data, analyze and propose the suitable treatment methodologies to treat water.

CO2: Recognize different forms of energy resources and apply them for suitable applications in energy sectors.

CO3: Apply the knowledge of phase rule and alloys for material selection requirements.

CO4: Analyze and recommend suitable fuels for engineering processes and applications.

CO5: Apply basic concepts of nano science and nanotechnology in designing the synthesis of Nano materials.

UNIT: I	WATER TECHNOLOGY	9
Water: Sources, impurities and water quality parameters, Hardness of water – types – expression of hardness – units, Boiler troubles: Scale and sludge, Priming &foaming. Need for water treatment, Treatment of boiler feed water: Internal treatment (phosphate, colloidal, sodium aluminate and calgon conditioning) and External treatment (Ion exchange or demineralization and zeolite process), Municipal water treatment: primary treatment and disinfection (UV, Ozonation, break-point chlorination). Desalination of brackish water: Reverse Osmosis.		
Pedagogical Tools	Chalk & Board, PPT, NPTEL Videos, youtube videos, Group Discussion	

UNIT: II	ENERGY SOURCES AND STORAGE DEVICES	9
Nuclear energy: light water nuclear power plant, breeder reactor. Solar energy conversion: Principle, working and applications of solarcells; Recent developments in solar cell materials. Wind energy; Basic Electrochemical Terminologies, Batteries: Types of batteries, Primary battery (dry cell), Secondary battery (lead acid battery and lithium-ion-battery);Electric vehicles– working principles; Fuel cells: H_2-O_2 fuel cell, Bio Fuel,microbial fuel cell; Super capacitors: Storage principle, types and examples.		
Pedagogical Tools	Chalk & Board, PPT, NPTEL Videos, youtube videos, Group Discussion	

UNIT: III	PHASE RULE AND ALLOYS	9
Phase rule: Introduction, definition of terms with examples. One component system - water system, sulphur system; Reduced phase rule; Construction of a simple eutectic phase diagram – Two component system: lead-silver system-Pattinson's process, $FeCl_3$-H_2O system. Alloys: Introduction- Definition- properties of alloys- significance of alloying, Alloys-Nichrome and stainless steel (18/8) – heat treatment of steel. Introduction to composites – definition-types-uses.		
Pedagogical Tools	Chalk & Board, PPT, NPTEL Videos, youtube videos, Group Discussion	

UNIT: IV	FUELS AND COMBUSTION	9
Fuels: Introduction: Classification of fuels; Coal and coke: Analysis of coal (proximate and ultimate), Carbonization, Manufacture of metallurgical coke (Otto Hoffmann method). Petroleum and Diesel: Manufacture of synthetic petrol (Bergius process), Property - Knocking, Power alcohol and biodiesel (trans - esterification). Combustion of fuels: Introduction: Calorific value - higher and lower calorific values, Flue gas analysis-ORSAT Method.CO_2 emission and carbon footprint.		
Pedagogical Tools	Chalk & Board, PPT, NPTEL Videos, youtube videos, Group Discussion, Role Play	

UNIT: V	NANO TECHNOLOGY	9
Basics: Distinction between molecules, nanomaterials and bulk materials; Size-dependent properties (optical, electrical, mechanical and magnetic); Types of nanomaterials: Definition, properties and uses of - nanoparticle, nanocluster, nanorod, nanowire and nanotube. Preparation of nanomaterials: sol-gel, laser ablation, chemical vapour deposition, Analytical techniques- SEM, TEM, Applications of nanomaterials		
Pedagogical Tools	Chalk & Board, PPT, NPTEL Videos, youtube videos, Group Discussion	

	Total Periods :45
Practical Exercises: (Any six experiments to be conducted)	**Total Periods:30**
1. Preparation of Na_2CO_3 as a primary standard and determination of types and amount of alkalinity in water sample	
2. Determination of total, temporary & permanent hardness of water by EDTA method.	
3. Determination of chloride content of water sample by Argentometric method.	
4. Estimation of sodium /potassium present in water using a flame photometer.	
5. Estimation of copper content of the given solution by Iodometry	
6. Determination of strength of given hydrochloric acid using pH meter.	
7. Determination of strength of acids in a mixture of acids using conductivity meter.	
8. Estimation of iron content of the given solution using potentiometer	
9. Estimation of Nickel in steel	
	Total Periods :75

TEXT BOOKS:

Sl. No	Authors	Title of the Book	Publisher	Year of publication
1	P.C.Jain and Monica Jain	Engineering Chemistry	16th Edition,Dhanpat Rai Publishing Company (P) Ltd, New Delhi	2018
2	S.S. Dara	A Text book of Engineering Chemistry	S.Chand Publishing,12thEdition	2018
3	Vairam S, Kalyani P and Suba Ramesh	Engineering Chemistry	2nd Edition, Wiley India Pvt. Ltd., New Delhi	2014
4	J Mendham RC Denn MJK Thomas David J Barnes	Vogel's Textbook of Quantitative Chemical Analysis	Pearson Education	2018

REFERENCE BOOKS:

Sl. No	Authors	Title of the Book	Publisher	Year of publication
1	B.S. Murty, P. Shankar, Baldev Raj, B. B. Rath and James Murday	Text book of nano science and nanotechnology	Universities Press-IIM Series in Metallurgy and Materials Science	2018
2	Shikha Agarwal	Engineering Chemistry-Fundamentals and Applications	Cambridge University Press, Delhi, Second Edition	2019
3	O.G.Palanna	Engineering Chemistry	McGraw Hill Education (India) Private Limited, 2ndEdition	2017
4	Prasanta Rath	Engineering Chemistry	Cengage Learning India, Pvt., Ltd., Delhi. 1st Edition	2015

WEB LEARNING RESOURCES:

https://nptel.ac.in/courses/105106119 (Unit 1)
https://nptel.ac.in/courses/103103206 (Unit 2)
https://www.brainkart.com>article phase rule (Unit 3)
https://nptel.ac.in/courses/113/104/113104008/ (Unit 4)
https://nptel.ac.in/courses/104103019 (Unit 5)
https://www.brainkart.com/subject/engineering-chemistry_264/ (All Units)
https://www.youtube.com/watch?v=4RDA_B_dRQ0(Reverse Osmosis)
https://www.youtube.com/watch?v=XUzpG1-rJLA Bergius Process)
https://www.youtube.com/watch?v=2bDf7JSRvf8

https://www.youtube.com/watch?v=Pme64aNaE5A (Otto-Hoffmman Method)
https://www.youtube.com/watch?v=VxMM4g2Sk8U (Lithium ion Batteries)

CO – PO – PSO MAPPING

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO 10	PO 11	PO 12	PS O1	PS O2	PS O3
CO1	3	2	2	1	-	1	1	-	-	-	-	1	-	-	-
CO2	3	1	2	1	-			-	-	-	-	2	-	-	-
CO3	3	1	-	-	-	-	-	-	-	-	-	-	-	-	-
CO4	3	1	1	-	-	1	2	-	-	-	-	-	-	-	-
CO5	2	1	1		-	-	-	-	-	-	-	-	-	-	-
AVG	3	1	2	1	-	1	2	-	-	-	-	2	-	-	-

R 2024	SCIENCE & HUMANITIES					SEMESTER: I
24HS103	COMMUNICATIVE ENGLISH LABORATORY	L	T	P	C	BS
		0	0	2	2	
COMMON TO: ALL PROGRAMS						

COURSE OBJECTIVES:

The objectives of learning this course are to:

- ✓ Improve the communicative competence of learners
- ✓ Help learners use language effectively in academic /work contexts
- ✓ Develop various listening strategies to comprehend various types of audio materials like
- ✓ Build on students' English language skills by engaging them in listening, speaking
- ✓ Use language efficiently in expressing their opinions via various media.

COURSE OUTCOMES:

At the end of this course, students are able to

CO1: Identify varied group discussion skills and apply them to take part in effective

CO2: Listen to and understand different points of view in a discussion

CO3: Speak fluently and accurately in formal and informal communicative contexts

CO4: Describe products and processes and explain their uses and purposes clearly and accurately

CO5: Express their opinions effectively in both formal and informal discussions

LIST OF EXPERIMENTS

1. Write about a self introduction for your future job opportunities
2. Write a telephonic conversation between a father and a son on “career”
3. Write a product description for a fire extinguisher
4. Give any one product user manual
5. Prepare a TED talk about artificial intelligence
6. Describe a famous person’s inspirational you heard before in your life
7. Write about panel discussion
8. Write your view and opinion the solve the water scarcity

Total Periods :30

CO – PO – PSO MAPPING

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	-	-	-	-	-	1	1	-	-	-	-	3	-	-	-
CO2	-	3	-	-	-	-	3	3	-	3	-	3	-	-	-
CO3	-	-	-	-	2	-	2	-	-	3	-	3	-	-	-
CO4	-	-	-	-	-	3	-	1	2	3	-	3	-	-	-
CO5	-	-	-	-	-	-	-	-	-	3	3	3	-	-	-
AVG	-	3	-	-	-	1	-	1	1	-	3	3	-	-	-

R 2024	COMPUTER SCIENCE AND ENGINEERING					SEMESTER:I
24ES105	PROGRAMMING IN C LABORATORY	L	T	P	C	ES
		0	0	4	2	

Common to CSE,IT and AI&DS Departments

COURSE OBJECTIVES:

The objectives of learning this course are:

- To familiarize with C programming constructs.
- To develop programs in C using basic constructs.
- To develop programs in C using arrays.
- To develop applications in C using strings, pointers, functions.
- To develop applications in C using structures.
- To develop applications in C using file processing.

COURSE OUTCOMES:

At the end of this course, students can able to

CO1: Demonstrate knowledge on C programming constructs.
CO2: Develop programs in C using basic constructs.
CO3: Develop programs in C using arrays and strings.
CO4: Develop applications in C using pointers, functions.
CO5: Develop applications in C using structures.
CO6: Develop applications in C using file processing.

LIST OF EXPERIMENTS:

Group A:

1. Write a C Program to find the sum of digits.
2. Write a C Program to check whether a given number is Armstrong or not.
3. Write a C Program to check whether a given number is Prime or not.
4. Write a C Program to generate the Fibonacci series.
5. Write a C Program to display the given number is Adam number or not.
6. Write a C Program to print reverse of the given number and string.
7. Write a C Program to find minimum and maximum of ‘n’ numbers using array.
8. Write a C Program to arrange the given number in ascending order.
9. Write a C Program to add and multiply two matrices.
10. Write a C Program to calculate NCR and NPR.
11. Write a C program to count the total number of vowels or consonants in a string.
12. Write a C program in C to read a sentence and replace lowercase characters with uppercase and vice versa.

Group B:

1. Write a C Program to find the grade of a student using else if ladder.
2. Write a C Program to implement the various string handling function.
3. Write a C Program to create an integer file and displaying the even numbers only.
4. Write a C Program to calculate quadratic equation using switch-case.
5. Write a C Program to count number of characters, words and lines in a text file.
6. Write a C Program to generate student mark list using array of structures.
7. Write a C Program to create and process the student mark list using file
8. Write a C Program to create and process pay bill using file
9. Write a C Program to create and process inventory control using file
10. Write a C Program to create and process electricity bill using file
11. Write a C Program to illustrate how a file stored on the disk is read.
12. Write a C program to read the file and store the lines in an array.

Note: One Question from Group A and another one Question from Group B is compulsory for End Semester Examination.

Total Periods : 60

CO PO PSO MAPPING

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PS01	PS02	PS03
CO1	2	2	2	1	2	1	1	1	2	-	3	2	2	2	-
CO2	2	3	2	1	2	1	1	1	2	-	3	2	2	2	-
CO3	3	2	2	1	3	1	1	1	2	-	3	3	2	2	-
CO4	2	3	3	1	2	1	2	1	2	-	3	2	2	3	-
CO5	2	3	3	1	2	1	-	-	2	1	2	2	2	2	-
CO6	2	2	2	1	2	1	1	1	2	-	3	2	2	2	-
AVG	2	3	2	1	2	1	1	1	2	1	3	2	2	2	-

R 2024	GENERAL ENGINEERING					SEMESTER: I
		L	T	P	C	
24 ES 106	ENGINEERING PRACTICES LABORATORY	0	0	2	1	ES
COMMON TO ALL BRANCHES AIDS, CSE, BME, ECE, IT and EEE						

COURSE OBJECTIVES:

The main objectives of this course are to:

- Study the various basic domestic wiring circuits and measure the electrical parameters.
- Impart the Knowledge about the stair case wiring, wiring layout and its connections
- Impart the knowledge of various basic electronic components .
- Know about Solder and test simple electronic circuits; Assemble and test simple electronic components on PCB.
- Study about the operation of various Boolean operations in electronics.

COURSE OUTCOMES:

At the end of this course, students are able to:

CO1:Wire various electrical joints in common household electrical wire work.
CO2:Understand the stair case wiring, wiring layout and its connections
CO3:Measure the electrical quantities using ammeter, voltmeter,wattmeter and energy meter
CO4:Study the construction, working principle and wiring of single phase energy meter.
CO5:Solder and test simple electronic circuits; Assemble and test simple electronic components on PCB.

LIST OF EXPERIMENTS:

I ELECTRICAL ENGINEERING PRACTICE

1. Residential house wiring using switches, fuse, indicator, lamp and energy meter.
2. Fitting and Installation of household appliances- LED TV,Fan
3. Stair case wiring.
4. Measurement of electrical quantities – voltage, current, power & power factor in RLC circuit.
5. Measurement of energy using single phase energy meter.

II ELECTRONIC ENGINEERING PRACTICE

1. Study of Electronic components and equipments – Resistor, colour coding, Measurement of AC signal parameter (peak-peak, rms period, frequency) using CRO.
2. Verification of logic gates AND, OR, EX-OR and NOT.
3. Generation of Clock Signal.
4. Soldering simple electronic circuits and checking continuity.
5. Assembling and testing electronic components on a small PCB.

Total Periods :30

CO PO PSO MAPPING:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PS01	PS02	PS03
CO1	3	2	-		1	1	1	-	-	-	-	2	2	1	1
CO2	3	2	-		1	1	1	-	-	-	-	2	2	1	1
CO3	3	2	-		1	1	1	-	-	-	-	2	2	1	1
CO4	3	2	-		1	1	1	-	-	-	-	2	2	1	1
CO5	3	2	-		1	1	1	-	-	-	-	2	2	1	1
AVG	3	2	-		1	1	1	-	-	-	-	2	2	1	1

1-Low, 2-Medium, 3-High

R 2024	MECHANICAL ENGINEERING					SEMESTER: I
24ES107	WORKSHOP PRACTICE LABORATORY	L	T	P	C	PC
		0	0	2	1	

COMMON TO: AI&DS, BME, CSE, ECE, EEE and IT

COURSE OBJECTIVES:

The main objectives of this course are to:

- Practice few basic engineering operations in welding, and sheet metal works.
- Make the specified skills in fitting operations.
- Perform few basic operations to produce wooden joints
- Make pipe connections for household applications.

COURSE OUTCOMES:

Upon completion of this course, the students will be able to:

CO1-Draw pipe line plan; lay and connect various pipe fittings used in common household plumbing work

CO2Saw; plan; make joints in wood materials used in common household wood work.

CO3-Weld various joints in steel plates using arc welding work;

CO4-Make a tray out of metal sheet using sheet metal work.

CO5-Prepare metal joints using fitting tools

PRACTICAL EXERCISES:

1. Plumbing Works: Hands-on-exercise: Basic pipe connections – Mixed pipe material connection – Pipe connections with different joining components for pumping water from sump to overhead tank and pipe connections from overhead tank to bath shower and wash basin.
2. Carpentry using modern tools only: Hands-on-exercise: Wood work, joints such as T, Mortise and Tenon and Dove Tail.
3. Welding: Preparation of butt joints, lap joints and T- joints by Arc welding and Gas welding
4. Sheet Metal Work: Model making – Trays and funnels.
5. Fitting: Preparation of Square fitting and V – fitting models.

Total Periods: 30

CO PO PSO MAPPING:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PS01	PS02	PS03
CO1	3	2	-	-	1	1	1	-	-	-	-	2	2	1	1
CO2	3	2	-	-	1	1	1	-	-	-	-	2	2	1	1
CO3	3	2	-	-	1	1	1	-	-	-	-	2	2	1	1
CO4	3	2	-	-	1	1	1	-	-	-	-	2	2	1	1
CO5	3	2	-	-	1	1	1	-	-	-	-	2	2	1	1
AVG	3	2	-	-	1	1	1	-	-	-	-	2	2	1	1

R 2024	DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING					SEMESTER: II
24ES201	**DESIGN THINKING**	**L**	**T**	**P**	**C**	**ES**
		2	**0**	**0**	**2**	

COMMON TO ALL BRANCHES

COURSE OBJECTIVES:

The objectives of learning this course are to:

- ✓ Provide the new ways of creative thinking.
- ✓ Learn the innovation cycle of Design Thinking process for developing innovative products which useful for a student in preparing for an engineering career and to apply them for the prospective entrepreneurial activities.

COURSE OUTCOMES:

Upon completion of this course, the students will be able to:

- ✓ CO1 Understand the Concept of Design Thinking through its principles.
- ✓ CO2 Learn the tools and techniques of Design Thinking and to apply them in real life cases.
- ✓ CO3 Understand the different stages of Structured Models used in Design thinking.
- ✓ CO4 Apply the perspectives of design thinking in the entrepreneurial activities.
- ✓ CO5 Learn from the real world case studies about how to apply the concept of design thinking in product development.

UNIT: I	OVERVIEW OF DESIGN THINKING	6

Introduction to Design Thinking – Conceptual Understanding, Evolution of Design Thinking, Attributes, Principles (Human Rule, Ambiguity Rule, Re-Design Rule and Tangibility Rule) – Cycle of Design Thinking – Resources (3Ps) – Applications.

Pedagogical Tools	Chalk & Board, PPT, Brainstorming, Flipped Class Room

UNIT: II	TOOLS AND TECHNIQUES FOR DESIGN THINKING	6

Personas, Visualization, Stakeholder Mapping, Journey Mapping, Mind Mapping, Star Bursting, Divergent Thinking, Convergent Thinking, Ethnography, Brainstorming, Story Telling, Role Playing, User Interviews. (All these techniques shall be taught only to level of understanding the core concept)

Pedagogical Tools	Chalk & Board, PPT, Brainstorming, Group Discussion, Case Study Method.

UNIT: III	DESIGN THINKING MODELS	6

Double Diamond Model – Phases of Discover, Define, Develop and Deliver – Feedback Mechanism.
Stanford 5 Phase Model – Empathize, Define, Ideate, Prototype and Test.

Pedagogical Tools	Chalk & Board, PPT, Empathy Interviews & User Research

UNIT: IV	DESIGN THINKING FOR ENTREPRENEURS	6

Idea of Growth Design, Comparison of Growth Design and Product Design, Growth Process Model : What is? - What if? - What Wows? - What Works, Principle of Optimism.
Ethics in Design Thinking : 5 Approaches – Utilitarian, Rights, Fairness, Common Good and Virtue - Ethical Issues – Ethical Design Test.

Pedagogical Tools	Chalk & Board, PPT, NPTEL video, you tube video

UNIT: V	CASE STUDIES	6

1. Why Patients were not visiting a healthcare center for a free health checkup - Karnataka Health Promotion Trust
2. Why Sales Officers were not accessing help even though it was available and were still abandoning sales when a difficult objection was raised in a sales call -Shriram Life Insurance Corporation.
3. My City Savior APP - Government of Odisha.
4. Designing of a Banking APP – Kotak Mahindra Bank

Pedagogical Tools	Chalk & Board, PPT, Brainstorming, TEDx like public Speech

	TOTAL PERIODS	**30**

TEXT BOOKS:				
Sl. No	**Authors**	**Title of the Book**	**Publisher**	**Year of publication**
1	E Bala Guruswamy, Bindu Vijayakumar	Design Thinking – A Business Perspective	McGraw Hill Education (India) Private Limited.	2024
REFERENCE BOOK:				
1	David Lee	Design Thinking In the Class Room	Ulysses Press	2018
WEB LEARNING RESOURCES:				
1 NPTEL				
1.https://youtu.be/6-5J6YTrYf4?si=WE9MLo-2tbccTWNG				
2.https://youtu.be/4nTh3AP6knM?si=rdEHE4yGxSJ4zDji				
3.https://youtu.be/j6Ro7TPzRoo?si=wa75cakOWyR0dSZC				
4.https://youtu.be/DmLVfQfxtPU?si=q6NyR6yCmir3Y2ia				
5.https://youtu.be/OE2ooXUEAwc?si=A3yYLYTOKvuYx_Cn				

CO – PO – PSO MAPPING

CO	PO												PSO		
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	3	2			2							2			
CO2	3	2			2							2			
CO3	3	2			2							2			
CO4	3	2			2							2			
CO5	3	2			2							2			
Avg	3	2			2							2			

R 2024	SCIENCE & HUMANITIES					SEMESTER: II
24BS202	DISCRETE MATHEMATICS	L	T	P	C	BS
		3	1	0	4	

COMMON TO: AI&DS, CSE, IT

COURSE OBJECTIVES:

The objectives of learning this course are:

- ✔ Extend student's logical and mathematical maturity and ability to deal with abstraction.
- ✔ Understand the basic concepts of combinatorics.
- ✔ Understand the basic concepts of graph theory.
- ✔ Familiarize the applications of algebraic structures.
- ✔ Understand the concepts and significance of lattices and Boolean algebra which are widely used in computer science and engineering.

COURSE OUTCOMES:

At the end of this course, students are able to

CO1: Have knowledge of the concepts needed to test the logic of a program.
CO2: Be aware of the counting principles.
CO3: Have knowledge of graph theory applications in computer science engineering.
CO4: Have an understanding in identifying structures on many levels.
CO5: Be exposed to concepts and properties of algebraic structures such as groups, rings and fields.

UNIT: I	**LOGIC AND PROOFS**	9+3
Propositional logic – Propositional equivalences - Predicates and quantifiers – Nested quantifiers – Rules of inference - Introduction to proofs – Proof methods and strategy.		
Pedagogical Tools	Chalk & Board, PPT, NPTEL video, you tube video	
UNIT: II	**COMBINATORICS**	9+3
Mathematical induction– The basics of counting – The pigeonhole principle – Permutations and combinations – Recurrence relations – Solving linear recurrence relations – Generating functions – Inclusion and exclusion principle and its applications.		
Pedagogical Tools	Chalk & Board, PPT, NPTEL video, you tube video	
UNIT: III	**GRAPHS**	9+3
Graphs and graph models – Graph terminology and special types of graphs – Matrix representation of graphs and graph isomorphism – Connectivity – Euler and Hamilton paths		
Pedagogical Tools	Chalk & Board, PPT, NPTEL video, you tube video	
UNIT: IV	**ALGEBRAIC STRUCTURES**	9+3
Algebraic systems – Semi groups and monoids - Groups – Subgroups – Homomorphism's – Normal subgroup and cosets – Lagrange's theorem – Definitions and examples of Rings and Fields.		
Pedagogical Tools	Chalk & Board, PPT, NPTEL video, you tube video	
UNIT: V	**LATTICES AND BOOLEAN ALGEBRA**	9+3
Partial ordering – Posets – Lattices as posets – Properties of lattices - Lattices as algebraic systems – Sub lattices– Some special lattices – Boolean algebra – Sub Boolean Algebra.		
Pedagogical Tools	Chalk & Board, PPT, NPTEL video, you tube video	
	Total Periods:60	

TEXT BOOKS:

Sl.No	Authors	Title of the Book	Publisher	Year of publication
1	Rosen. K.H.	Discrete Mathematics and its Applications	7th Edition, Tata McGraw Hill Pub. Co. Ltd., New Delhi, Special Indian Edition	2017

2	Tremblay. J.P. and Manohar. R	Discrete Mathematical Structures with Applications to Computer Science	Tata McGraw Hill Pub. Co. Ltd, New Delhi, 30th Reprint	2011
REFERENCE BOOKS:				
1	Grimaldi. R.P.	Discrete and Combinatorial Mathematics: An Applied Introduction	5th Edition, Pearson Education Asia, Delhi.	2013
2	Koshy. T.	Discrete Mathematics with Applications	Elsevier Publications.	2006
3	Lipschutz. S. and Mark Lipson	Discrete Mathematics	Schaum's Outlines, Tata McGraw Hill Pub. Co. Ltd., New Delhi, 3rd Edition	2010

WEB LEARNING RESOURCES:
1 https://www.brainkart.com/subject/Discrete-Mathematics_94/
2 https://nptel.ac.in/courses/111104026
3 https://nptel.ac.in/courses/111106050
4 https://nptel.ac.in/courses/111106052
5 https://nptel.ac.in/courses/111106086
6 https://nptel.ac.in/courses/111106102
7 https://nptel.ac.in/courses/111106137
8 https://youtu.be/HipVU5vz3Q8?si=eJd9WokGLaWgA30R
9 https://youtu.be/wsvPWTDZXT0?si=5v1SJPl3O4yAe5_z

CO – PO – PSO MAPPING

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	3	3	2	-	-	-	-	-	-	-	-	2	-	-	-
CO2	3	3	-	-	-	-	-	-	-	-	-	-	-	-	-
CO3	-	3	2	-	-	2	-	-	-	3	-	-	-	-	-
CO4	-	2	2	2	-	-	-	-	-	-	-	-	-	-	-
CO5	-	2	2	2	-	-	-	-	-	2	-	-	-	-	-
Avg	3	3	2	2	-	2	-	-	-	3	-	2	-	-	-

<table>
<tr><td>R 2024</td><td colspan="5">SCIENCE & HUMANITIES</td><td>SEMESTER: II</td></tr>
<tr><td rowspan="2">24HS204</td><td rowspan="2">German</td><td>L</td><td>T</td><td>P</td><td>C</td><td rowspan="2">HS</td></tr>
<tr><td>2</td><td>0</td><td>0</td><td>2</td></tr>
<tr><td colspan="7">COMMON TO: ALL PROGRAMS</td></tr>
</table>

COURSE OBJECTIVES:

The objectives of learning this course are:

- ✔ To enable learners use words appropriately in their communication.
- ✔ To develop learners ability to read and listen to texts in German.
- ✔ To strengthen the communication skills of the learners.

COURSE OUTCOMES:

At the end of this course, students can able to

CO1: understand the German Language and read
CO2: understand the German Language and listen
CO3: understand the German Language and speak
CO4 :understand the German Language and write

UNIT: I	Reading	8

i. The pupils recognize the following types of text: dialogue; interview; advertisement; programme of a performance (cinema, theatre, concert, sport); a television and radio programme; notice; folder page of books, of audio cassettes, of videocassettes and of CDs; articles in dictionaries and lexica; a form to be filled in; menu; poem, short story, diary, comics, picture novel, greeting card, personal letter, e- mail letter, announcement, invitation.

ii. The pupils can understand the following types of text globally and/or selectively: leaflet, catalogue, label, transport timetable, city map, a programme of a performance (cinema, theatre, concert, sport), T.V. & radio programme, advertisement, notice, article in a dictionary and lexicon, menu, personal letter, e-mail letter, columns in a newspaper and magazine, comics, cuttings of reports, poem, short story, short texts of information.

iii. The pupils understand in detail the type of problem and the instructions in the text book as well as short announcements, signs denoting advice and forbiddings, simple forms, invitations and greeting cards.

iv. The pupils make use of the following strategies while reading:
- they recognise the correlation between text and picture.
- they recognise personal names, numbers and dates.
- they recognise the meaning of punctuation marks and text typography.
- they establish the correlation between the title of a text and main points of information.
- they recognise the parts of speech and clauses, word roots, prefixes, suffixes and endings of words of those learnt as well as internationalisms.
- they look for and mark main points of information in a text.
- they recognise the communicative function of the types of text listed under point (i).
- they work with word card indexing.
- they perceive the foreign culture in that they take a critical look at their own culture in the process.
- they make use of the knowledge, skills and strategies which they have acquired in the lessons of their mother language or their first foreign language, when deducing pieces of information from text or making connections between them.

v. The pupils can handle reference works (e.g., dictionaries, grammars).

Pedagogical Tools	Board & Chalk, PPT, youtube videos

UNIT: II	Listening	8

The pupils are in a position to understand different German language texts globally or in detail through a direct contact or over the media. The texts should follow the standards of level A1 of the *Framework* and observe the phonetical and intonation variants of the German language. Of special significance in the training for the skill of *listening* is the inclusion of sight perception.

i. The pupils understand questions and instructions of the teacher during the lesson.

ii. The pupils can create correlations between hearing texts and pictures.

iii. The pupils can understand short dialogues between two or several partners who refer to themes and situations already dealt with.

iv. The pupils can understand short everyday and especially tourist- related information (e.g., at the post office, in a travel agency, at the railway station / airport).

v. The pupils infer main announcements from conversations on themes and situations already dealt with.

vi. The pupils can infer selective information from news, advertisements and programme information on Radio or in T.V. as well as from easy descriptive texts.

vii. The pupils can understand short literary forms like poems and songs on the basis of directed explanation.

viii. The pupils make use of the following strategies while listening:

- they put forward hypotheses and examine them in the light of the intention of the statement of various types of text.
- they recognise intonation models, linguistic and metalinguistic means of expressing affirmation and negation.
- they make use of already known models of word building.
- theyrecognise the communicative function of varied types of text.
- they work with a dialogue – diagram.
- they draw up the construction plan of a text they have heard.

Pedagogical Tools	Board & Chalk, PPT, youtube videos

UNIT: III	**Speaking**	**7**

The pupils realize in their statements ways of speaking which are mentioned in the subsequent part entitled *Contents*.

i. The pupils reproduce the phonetic and intonation pattern correctly.

ii. The pupils ask and answer questions in connection with the themes and situations already dealt with.

iii. The pupils participate in conversation with their teacher and / or with their classmates in the course of the lesson.

iv. The pupils hold short conversations with one or several partners (known or unknown) in the sphere of the themes and situations already dealt with.

v. The pupils make short telephone calls.

vi. The pupils make short announcements in connection with themes already handled.

vii. The pupils make use of appropriate patterns of behaviour (mimics, gesticulations, body distance or nearness, etc) during conversation.

viii. The pupils can make use of the following strategies while speaking:
- they ask for and themselves provide additional / explanatory information.
- they signal lack of understanding and demand from their partner an appropriate reaction.
- they direct the conversation according to their own interests and / or change the subject.
- they make use of clichés in order, e.g., to cope more easily with situations in which they are under pressure of time.
- they make use of paralinguistic means.

Pedagogical Tools	Board & Chalk, PPT, youtube videos

UNIT: IV	Writing		7
i. The pupils fill in tables with key words according to a text they have read or heard. ii. The pupils fill in easy forms, write greeting cards, invitations and short personal announcements. iii. The pupils lay down vocabulary cards according to a preset pattern. iv. The pupils write short texts to photos and pictures. v. The pupils make use of the following strategies while writing: - they employ preset patterns and examples with sense. - they use reference works for self correction of mistakes.			
Pedagogical Tools	Board & Chalk, PPT, youtube videos		
	TOTAL PERIODS:30		
TEXT CUM REFERENCE BOOKS:			
The aims, methods and contents, as they are formulated in the syllabus for German as a second foreign language for level 1 (**A1**), are to be adopted in the textbook for this level. While the autonomy of the school in the choice of the textbook and related material is respected, choice is to be made of a work which contains the following basic text material.			
3.1. Pupils' book which contains the learning material obligatory for level 1, as well as the grammar overview and an alphabetical word list; 3.2. Work book with exercises, which supplement the learning material of the pupil's book and makes possible a differentiation within the class of pupils and various social forms (single, partner, group work) during the lesson. The book contains tests which help the periodical control of the learning process and success; 3.3. Teacher's book in which the concept of the pupil's book is presented, methodological tips given and alternative lesson schemes suggested, additional cultural (*Landeskunde*) and linguistic information included, as well as indications of possible forms of control and assessment of performance. It includes also l listening comprehension texts, exercises on cassette, keys to the tests and vocabulary to each unit; 3.4. Cassettes with listening comprehension texts from the pupil's book and where possible phonetic and grammar tests as well as further authentic texts which contribute towards the development of listening comprehension. 3.5. I.T. Material which instills in the pupil an awareness of the German-speaking world and encourages him/her to make use of interactive exercises with partners abroad and in one's own country (e-mail) and to satisfy the desire to research and increase one's knowledge of certain aspects of topics treated in class (internet). This medium should make up for the lack of actual resources at school and complete the overall picture of the German-speaking media.			

CO – PO – PSO MAPPING

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO 10	PO 11	PO 12	PS O1	PS O2	PS O3
CO1	3	1	1	2	-	-	-	-	-	-	-	-	-	-	-
CO2	1	3	2	1	-	-	-	-	-	-	-	-	-	-	-
CO3	1	2	3	1	-	-	-	-	-	-	-	-	-	-	-
CO4	2	1	1	3	-	-	-	-	-	-	-	-	-	-	-
AVG	1.75	1.75	1.75	1.75	-	-	-	-	-	-	-	-	-	-	-

R 2024	Languages					SEMESTER: II
24HS205	Italian	L	T	P	C	HS
		2	0	0	2	
COMMON TO: ALL BRANCHES						

COURSE OBJECTIVES:

The objectives of learning this course are:

Ø To enable learners use words appropriately in their communication.

Ø To develop learners ability to read and listen to texts in Italian Language.

Ø To strengthen the communication skills of the learners.

COURSE OUTCOMES:

At the end of this course, students can able to

CO1: understand the Italian Language- basics of day-to-day conversation such as talking about your likes, and dislikes, knowing the numbers, alphabet, habitual actions, and more. Learn grammar and its usage

CO2:communicate in simple terms aspects of his/her background, immediate environment & matters in areas of immediate basic need.

UNIT: I	Beginner Level A1	15

Learn the basics of day-to-day conversation such as talking about your likes, and dislikes, knowing the numbers, alphabet, habitual actions, and more. Learn grammar and its usage.

Topics

- Introducing yourself
- Saying hello and goodbye
- Nationality
- Asking and Saying how one is
- Apologizing
- Spelling one's name
- Ordering Food
- Reading simple menu
- Asking and telling time

Grammar

- Personal Subject Pronouns
- Definite and indefinite articles
- Nouns
- Adjectives
- Present Tense of a regular verb
- 🎬 Interrogatives

Tools Required:	Board & Chalk, PPT, youtube videos

UNIT: II	Beginner Level A2	15

Learn to communicate in simple terms aspects of his/her background, immediate environment & matters in areas of immediate basic need

Topics

- Booking a table at a restaurant
- Understanding a menu
- Understanding simple city directions
- Expressing agreements/disagreements
- Adjectives
- Some Italian recipes
- Some expressions of place
- Talking about past events

- Writing a greeting card

Grammar
- The verb sapere and potere
- More about the verb piacere
- Prepositions in and a
- Regular and Irregular participles
- The present perfect
- The Adverb fa
- More interrogatives

Tools Required:	Board & Chalk, PPT, youtube videos
	TOTAL PERIODS :30

TEXT CUM REFERENCE BOOKS:

Italian Language Textbooks

1. **Nuovo Espresso 1 (A1-A2)** *(Alma Edizioni)*
   - Covers greetings, introductions, ordering food, and city directions.
   - Grammar focus on articles, present tense, passato prossimo, and prepositions.
   - Includes listening exercises, cultural notes, and interactive practice.
2. **Italian Grammar in Practice (A1-A2)** *(Susanna Nocchi)*
   - Practical grammar explanations with exercises.
   - Good for mastering verbs like *sapere*, *potere*, and *piacere*.
3. **Practice Makes Perfect: Basic Italian** *(Alessandra Visconti)*
   - Focus on conversational phrases, simple dialogues, and essential grammar.
   - Great for pronunciation and everyday vocabulary like time, directions, and ordering food.
4. **Progetto Italiano Junior 1** *(Edilingua)* – if teaching younger learners.

Supplementary Online Resources:
- **BBC Languages – Italian:** Interactive lessons for beginners.
- **Duolingo/Busuu:** For extra vocabulary practice.
- **ItalianPod101:** Great for listening and pronunciation practice.

CO – PO – PSO MAPPING

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	3	2	2	3	-	-	-	-	-	-	-	-	-	-	-
CO2	2	3	3	2	-	-	-	-	-	-	-	-	-	-	-
Avg	2.5	2.5	2.5	2.5	-	-	-	-	-	-	-	-	-	-	-

<table>
<tr><td>R 2024</td><td colspan="5">Languages</td><td>SEMESTER: II</td></tr>
<tr><td rowspan="2">24HS203</td><td rowspan="2">Japanese</td><td>L</td><td>T</td><td>P</td><td>C</td><td rowspan="2">HS</td></tr>
<tr><td>2</td><td>0</td><td>0</td><td>2</td></tr>
<tr><td colspan="7">COMMON TO: ALL BRANCHES</td></tr>
</table>

COURSE OBJECTIVES:

The objectives of learning this course are:
Ø To enable learners use words appropriately in their communication.
Ø To develop learners ability to read and listen to texts in Japanese.
Ø To strengthen the communication skills of the learners.

COURSE OUTCOMES:

COURSE OUTCOMES:
At the end of this course, students can able to
CO1 understand the Japanese Language - Topics & Vocabulary
CO2 understand the Japanese Language -Grammar
CO3 understand the Japanese Language - Cultural Content
CO4 understand the Japanese Language - Skills Work

Module 1:	**Topics & Vocabulary**	**8**

- Introduce yourself with greetings in Japanese
- Exchanging business card in Japanese
- Asking about of services in stores
- Shopping
- Describing about the whereabouts of things and people
- Transportation
- Time and numbers – telling and asking the time, counting cardinal numbers
- Everyday objects
- Places – shops, important buildings
- Daily life – routines, free time
- Job
- Home
- Culture
- Existence of People and Things
- 🎬 Ordinal numbers

Tools Required:	Board & Chalk, PPT, youtube videos

Module 2:	**Grammar**	**8**

- Basic Japanese grammar rules – particles : か (ka), は (wa), の (no), と (to), を (o),に (ni),も(mo), が (ga), や (ya) .
- Present, Past, Future
- Pronouns – subject, object, possessive
- Singular vs. Plural
- Word order – sentence, question, negative
- Question formation
- 🎬 Modal verbs

Tools Required:	Board & Chalk, PPT, youtube videos

Module 3:	**Cultural Content**	**7**

- Three writing systems in Japanese (Hiragana, Katakana, Kanji)
- How to bow
- Japanese currency

(Recommended by Ist BOS held on 05.09.24 & Approved by Ist Academic Council held on 25.11.24)

• Shops in Japan • Transportation • Excursions to Japanese spas (温泉onsen)		
Tools Required:	Board & Chalk, PPT, youtube videos	
Module 4:	**Skills Work**	**7**
• Lots of speaking-inc. situational exercises & interaction • Basic pronunciation rules • Listening activities • Numbers and Counters rules • Writing practice (Hiragana)		
Tools Required:	Board & Chalk, PPT, youtube videos	
		TOTAL PERIODS : 30

TEXT CUM REFERENCE BOOKS:

1. **Genki I: An Integrated Course in Elementary Japanese** (Eri Banno et al.)
   - o Covers self-introductions, shopping, daily routines, and transportation.
   - o Introduces particles, sentence structure, and essential grammar.
   - o Includes cultural notes, listening exercises, and hiragana practice.
2. **Minna no Nihongo Shokyu I**
   - o Great for practical conversations like shopping and asking for services.
   - o Strong grammar foundation with exercises on particles and verb conjugations.
   - o Requires a translation guide unless you're familiar with Japanese.
3. **Japanese for Busy People I** (AJALT)
   - o Focused on conversational skills and real-life scenarios like business card exchange.
   - o Simple grammar explanations and cultural context.

Supplementary Resources:

- *NHK World: Easy Japanese* (free online lessons with dialogues and videos)
- *Tae Kim's Guide to Japanese Grammar* (online resource for grammar concepts)

CO – PO – PSO MAPPING

	PO 1	PO2	PO 3	PO4	PO5	PO 6	PO 7	PO 8	PO9	PO 10	PO 11	PO 12	PSO1	PSO2	PSO 3
CO1	3	1	1	2	-	-	-	-	-	-	-	-	-	-	-
CO2	1	3	1	2	-	-	-	-	-	-	-	-	-	-	-
CO3	1	1	3	2	-	-	-	-	-	-	-	-	-	-	-
CO4	1	1	1	3	-	-	-	-	-	-	-	-	-	-	-
Avg	1.5	1.5	1.5	2.25	-	-	-	-	-	-	-	-	-	-	-

<table>
<tr><td rowspan="1">R 2024</td><td colspan="5">SCIENCE & HUMANITIES</td><td>SEMESTER: II</td></tr>
<tr><td rowspan="2">24HS201</td><td rowspan="2">Tamils and Technology</td><td>L</td><td>T</td><td>P</td><td>C</td><td rowspan="2">HS</td></tr>
<tr><td>1</td><td>0</td><td>0</td><td>1</td></tr>
<tr><td colspan="7">COMMON TO: ALL PROGRAMS</td></tr>
</table>

COURSE OBJECTIVES:

The objectives of learning this course are to:

- ✔ Learn weaving, ceramic and construction technology of Tamil.
- ✔ Understand the agriculture, irrigation and manufacturing technology of tamil.
- ✔ Realize the development of scientific Tamil and computing.

COURSE OUTCOMES:

At the end of this course, students can able to :

CO1: Understand the weaving and ceramic technology of ancient Tamil people nature.

CO2: Understand the construction technology, building materials in sangam period and case studies.

CO3: Infer the metal process, coin and beads manufacturing with relevant archaeological evidence.

CO4: Realize the agriculture methods, irrigation technology and pearl diving.

CO5: Apply the knowledge of scientific Tamil and Tamil computing.

UNIT: I	**WEAVING AND CERAMIC TECHNOLOGY**	**3**
Weaving Industry during Sangam Age – Ceramic technology – Black and Red Ware Potteries (BRW) – Graffiti on Potteries.		
Pedagogical Tools	Black board, chalk, Group Discussion, Role Play, Youtube Videos,Nptel videos .	
UNIT: II	**DESIGN AND CONSTRUCTION TECHNOLOGY**	**3**
Designing and Structural construction House & Designs in household materials during Sangam Age - Building materials and Hero stones of Sangam age – Details of Stage Constructions in Silapathikaram - Sculptures and Temples of Mamallapuram - Great Temples of Cholas and other worship places - Temples of Nayaka Period - Type study Madurai Meenakshi Temple)- ThirumalaiNayakarMahal - Chettinad Houses, Indo - Saracenic architecture at Madras during British Period		
Pedagogical Tools	Black board, chalk, Group Discussion, Role Play, Youtube Videos,Nptel videos .	
UNIT: III	**MANUFACTURING TECHNOLOGY**	**3**
Art of Ship Building - Metallurgical studies - Iron industry - Iron smelting,steel -Copper and gold- Coins as source of history - Minting of Coins – Beads making-industries Stone beads - Glass beads - Terracotta beads -Shell beads/ bone beats - Archeological evidences - Gemstone types described in SilapathikaramTherukoothu, Karakattam, VilluPattu, KaniyanKoothu, Oyilattam, Leather Puppetry, Silambattam, Valari, Tiger dance - Sports and Games of Tamils.		
Pedagogical Tools	Black board, chalk, Group Discussion, Role Play, Youtube Videos,Nptel videos .	
UNIT: IV	**AGRICULTURE AND IRRIGATION TECHNOLOGY**	**3**
Flora and Fauna of Tamils &Agam and Puram Concept from Tholkappiyam and Sangam Literature - Aram Concept of Tamils - Education and Literacy during Sangam Age - Ancient Cities and Ports of Sangam Age - Export and Import during Sangam Age - Overseas Conquest of Cholas		
Pedagogical Tools	Black board, chalk, Group Discussion, Role Play, Youtube Videos,Nptel videos .	
UNIT: V	**SCIENTIFIC TAMIL & TAMIL COMPUTING**	**3**
Development of Scientific Tamil - Tamil computing – Digitalization of Tamil Books – Development of Tamil Software – Tamil Virtual Academy – Tamil Digital Library – Online Tamil Dictionaries – Sekai Project.		

Pedagogical Tools	Black board, chalk, Group Discussion, Role Play, Youtube Videos,Nptel videos.

Total Periods :15

TEXT CUM REFERENCE BOOKS:

Sl.No	Authors	Title of the Book	Publisher	Year of publication
1	Dr.K.K.Pillay	Tamilnadu history people and culture	Tamilnadu Textbook and Education works Corporation	2019
2	EL Sundaram	Computer Tamil	Vikatanprasuram	2016
3	Dr.S.Singaravelu	Social Life of the Tamils - The Classical Period	International Institute of Tamil Studies.	2001
4	Dr.S.V.Subatamanian, Dr.K.D. Thirunavukkarasu	Historical Heritage of the Tamils	International Institute of Tamil Studies	2010
5	Dr.M.Valarmathi	The Contributions of the Tamils to Indian Culture	International Institute of Tamil Studies.	2001
6	Dr. R. Sivanantham	Keeladi - 'Sangam City Civilization on the banks of river Vaigai'	Department of Archaeology & Tamil Nadu Text Book and Educational Services Corporation, Tamil Nadu	2019
7	Dr.K.K.Pillay	Studies in the History of India with Special Reference to Tamil Nadu	This Author	1979
8		Porunai Civilization	Department of Archaeology & Tamil Nadu Text Book and Educational Services Corporation, Tamil Nadu	2019
9	R.Balakrishnan	Journey of Civilization Indus to Vaigai	RMRL	2019
10	Dr.K.K.Pillay	Social Life of Tamils	A joint publication of TNTB & ESC and RMRL	1975

WEB LEARNING RESOURCES:

1	https://youtu.be/jteRvnNiD6w?si=HmAS7a_gng6hYcL_
2	https://youtu.be/WZwdo20QgP8?si=2oTevNPCiGzTPi0-
3	https://youtu.be/05e3v0xGA9k?si=SHa2vsQG39RpDPtZ
4	https://youtu.be/bxYdHw4rvec?si=Eryg0PF72BPhbRBH
5	https://youtu.be/MRfbeJvJZ0k?si=YpAYFFEpLdV8FIrX
6	https://youtu.be/BS_BSDZp6HA?si=D_QdZn1Zr6X3C95p

CO – PO – PSO MAPPING															
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO 10	PO 11	PO 12	PS O1	PS O2	PSO 3
CO1	-	-	-	-	-	-	3	3	-	2	-	3	-	-	-
CO2	-	-	-	-	-	-	3	3	-	2	-	3	-	-	-
CO3	-	-	-	-	-	-	3	3	-	2	-	3	-	-	-
CO4	-	-	-	-	-	-	3	3	-	2	-	3	-	-	-
CO5	-	-	-	-	-	-	3	3	-	2	-	3	-	-	-
AVG	-	-	-	-	-	-	3	3	-	2	-	3	-	-	-

<table>
<tr><td>R 2024</td><td colspan="5">DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING</td><td>SEMESTER:II</td></tr>
<tr><td rowspan="2">24ES208</td><td rowspan="2">PYTHON PROGRAMMING</td><td>L</td><td>T</td><td>P</td><td>C</td><td rowspan="2">ES</td></tr>
<tr><td>3</td><td>0</td><td>0</td><td>3</td></tr>
<tr><td colspan="7">Common to AI&DS, CSE and IT</td></tr>
</table>

COURSE OBJECTIVES:

The objectives of learning this course are:

- To understand the basics of algorithmic problem solving.
- To learn to solve problems using Python conditionals and loops.
- To define Python functions and use function calls to solve problems.
- To use Python data structures - lists, tuples, dictionaries to represent complex data.
- To do input/output with files in Python

COURSE OUTCOMES:

At the end of this course, students able to
CO1: Develop algorithmic solutions to simple computational problems and execute simple Python programs.
CO2: Write simple Python programs using conditionals and loops for solving problems.
CO3: Decompose a Python program into functions.
CO4: Represent compound data using Python lists, tuples, dictionaries etc.
CO5: Read and write data from/to files in Python programs.

UNIT: I	**COMPUTATIONAL THINKING AND PROBLEM SOLVING**	**9**

Fundamentals of Computing – Identification of Computational Problems -Algorithms, building blocks of algorithms (statements, state, control flow, functions), notation (pseudo code, flow chart, programming language), algorithmic problem solving, simple strategies for developing algorithms (iteration, recursion). Illustrative problems: Flowchart to find minimum in a list, Flowchart to insert a card in a list of sorted cards, Pseudo code to find an integer number in a range, Pseudo code to find the position of the largest element in an list of n numbers, Towers of Hanoi. Features of Python (Readability, Simplicity, Large ecosystem). Evolution (From 1991 to 2025, Version 0.9.0 to 3.12.3). Installation of Python in Windows.

Pedagogical Tools	Black board, chalk, Group Discussion, Role Play, Youtube Videos,Nptel videos.

UNIT: II	**DATA TYPES, EXPRESSIONS, STATEMENTS**	**9**

Python interpreter and interactive mode, debugging; values and types: int, float, boolean, string, and list; variables, expressions, statements, packing and unpacking arguments, precedence of operators, comments; Illustrative programs: swap the values of two variables, circulate the values of n variables, distance between two points, reverse the string.

Pedagogical Tools	Black board, chalk, Group Discussion, Role Play, Youtube Videos,Nptel videos.

UNIT: III	**CONTROL FLOW, FUNCTIONS, STRINGS**	**9**

Conditionals: Boolean values and operators, conditional (if), alternative (if-else),chained conditional (if-elif-else);Iteration: state, while, for, break, continue, pass; Fruitful functions: return values, parameters, local and global scope, function composition, recursion; Strings: string slices, immutability, string functions and methods, string module; Lists as arrays. Illustrative programs: square root, gcd, exponentiation, sum an array of numbers, factorial, fibonacci series, palindrome, linear search, binary search.

Pedagogical Tools	Black board, chalk, Group Discussion, Role Play, Youtube Videos,Nptel videos.	
UNIT: IV	**LISTS, TUPLES, DICTIONARIES**	**9**
Lists: list operations, list slices, list methods, list loop, mutability, aliasing, cloning lists, list parameters; Tuples: tuple assignment, tuple as return value; Dictionaries: operations and methods; advanced list processing - list comprehension; Illustrative programs: Bubble sorting, Insertion, selection, merge sort, histogram, Add Two Matrices, Transpose a Matrix, Students marks statement, Retail bill preparation.		
Pedagogical Tools	Black board, chalk, Group Discussion, Role Play, Youtube Videos,Nptel videos.	
UNIT: V	**FILES, MODULES, PACKAGES**	**9**
Files and exceptions: text files, reading and writing files, format operator; command line arguments, errors and exceptions, handling exceptions, modules (numpy, pandas, scipy, matplotlib, statmodels), packages; Illustrative programs: word count, copy file, check voting eligibility, count the number of each vowel in a string, random number generation, time series analysis, Marks range validation (0-100).		
Pedagogical Tools	Black board, chalk, Group Discussion, Role Play, Youtube Videos,Nptel videos.	
		Total Periods : 45

TEXT BOOKS:

Sl. No	Authors	Title of the Book	Publisher	Year of publication
1	Allen B. Downey	Think Python: How to Think like a Computer Scientist	O'Reilly Publishers	2016
2	Karl Beecher	Computational Thinking: A Beginner's Guide to Problem Solving and Programming	BCS Learning & Development Limited	2017

REFERENCE BOOKS:

Sl. No	Authors	Title of the Book	Publisher	Year of publication
1	Paul Deitel and Harvey Deitel	Python for Programmers	Pearson Education	2021
2	G Venkatesh and Madhavan Mukund	Computational Thinking: A Primer for Programmers and Data Scientists	Notion Press	2021
3	John V Guttag	Introduction to Computation and Programming Using Python: With Applications to Computational Modeling and Understanding Data	MIT Press	2021

WEB LEARNING RESOURCES:

1. https://www.python.org/
2. https://www.geeksforgeeks.org/python-programming-language-tutorial/

3. https://www.w3schools.com/python/

CO PO PSO MAPPING

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	3	3	3	3	2	-	-	-	-	-	2	2	3	3	-
CO2	3	3	3	3	2	-	-	-	-	-	2	2	3	-	-
CO3	2	2	-	2	2	-	-	-	-	-	1	-	3	-	-
CO4	1	2	-	-	1	-	-	-	-	-	1	-	2	-	-

<table>
<tr><td>R 2024</td><td colspan="5">SCIENCE & HUMANITIES</td><td>SEMESTER: II</td></tr>
<tr><td rowspan="2">24BS204</td><td rowspan="2">PHYSICS FOR ENGINEERS</td><td>L</td><td>T</td><td>P</td><td>C</td><td rowspan="2">BS</td></tr>
<tr><td>3</td><td>0</td><td>2</td><td>4</td></tr>
<tr><td colspan="7">COMMON TO: AI & DS, CSE, ECE and IT</td></tr>
</table>

COURSE OBJECTIVES:

The objectives of learning this course are to:

- ✔ Achieve an understanding of rotational dynamics of multi-particles
- ✔ Acquire the knowledge of transfer of heat in conductors and insulators
- ✔ Introduce the basics of oscillations, optics and lasers
- ✔ Equip the students to understand the importance of quantum physics
- ✔ Introduce and classify crystal structures of materials

COURSE OUTCOMES:

At the end of this course, students are able to

CO1: Understand and analyze the rotational dynamics of multi-particles

CO2: Apply the concepts of heat transfer in various applications.

CO3: Demonstrate a strong foundational knowledge in oscillations, optics and lasers

CO4: Recognize the basics of quantum physics.

CO5: Differentiate crystal structures of materials

UNIT: I	MECHANICS	9

Multi-particle dynamics: Center of mass (C.M) – CM of continuous bodies – motion of the CM – kinetic energy of system of particles. Rotation of rigid bodies: Rotational kinematics – rotational kinetic energy and moment of inertia - theorems of M .I –moment of inertia of continuous bodies – M.I of a diatomic molecule - torque – rotational dynamics of rigid bodies – conservation of angular momentum – rotational energy state of a rigid diatomic molecule - gyroscope - torsional pendulum – double pendulum –Introduction to nonlinear oscillations.

Pedagogical Tools	Chalk & board, PPT, NPTEL videos and Youtube videos

UNIT: II	THERMAL PHYSICS	9

Transfer of heat energy – thermal expansion of solids and liquids – expansion joints - bimetallic strips - thermal conduction, convection and radiation –rectilinear heat flow – thermal conductivity - Forbe's and Lee's disc method: theory and experiment-conduction through compound media (series and parallel)–thermal insulation – applications: heat exchangers, refrigerators, ovens and solar water heaters.

Pedagogical Tools	Chalk & board, PPT, NPTEL videos and Youtube videos

UNIT: III	OSCILLATIONS, OPTICS AND LASERS	9

Simple harmonic motion - resonance –analogy between electrical and mechanical oscillating systems - waves on a string - standing waves - traveling waves - Energy transfer of a wave - sound waves - Doppler effect. Reflection and refraction of light waves - total internal reflection - interference –Michelson interferometer –Theory of air wedge and experiment. Theory of laser - characteristics - Spontaneous and stimulated emission - Einstein's coefficients - population inversion - Nd-YAG laser, CO_2 laser, semiconductor laser –Basic applications of lasers in industry.

Pedagogical Tools	Chalk & board, PPT, NPTEL videos and Youtube videos

UNIT: IV	BASIC QUANTUM MECHANICS	9

Photons and light waves - Electrons and matter waves – Compton effect - The Schrodinger equation (Time dependent and time independent forms) - meaning of wave function - Normalization –Free particle - particle in a infinite potential well: 1D,2D and 3D Boxes- Normalization, probabilities and the correspondence principle

Pedagogical Tools	Chalk & board, PPT, NPTEL videos and Youtube videos

UNIT: V	CRYSTAL STRUCTURE	9
Introduction – Classification of solids –Space lattice –Basis-Lattice parameter – Unit cell – Crystal system –Miller indices –d-spacing in cubic lattice - Calculation of number of atoms per unit cell – Atomic radius-Coordination number – Packing factor for SC, BCC, FCC and HCP structures – crystal imperfection – Burger vector.		
Pedagogical Tools	Chalk & board, PPT, NPTEL videos and Youtube videos	
		Total Periods: 45

Practical Exercises: (Any six experiments to be conducted)	**Total Periods: 30**
1. Non-uniform bending - Determination of Young's modulus	
2. Uniform bending – Determination of Young's modulus	
3.Torsional pendulum - Determination of rigidity modulus of wire and moment of inertia of regular and irregular objects.	
4. Laser- Determination of the wave length of the laser using grating	
5. Optical fibre -Determination of numerical aperture (NA) and acceptance angle (AA)	
6. Air wedge - Determination of thickness of a thin sheet/wire	
7. Ultrasonic interferometer – determination of the velocity of sound and compressibility of liquids	
8. Acoustic grating- Determination of velocity of ultrasonic waves in liquids.	
9. Simple harmonic oscillations of cantilever.	
	Total Periods: 75

TEXT BOOKS:

Sl.No	Authors	Title of the Book	Publisher	Year of publication
1	D.Kleppner and R.Kolenkow	An Introduction to Mechanics	McGraw Hill Education (Indian Edition)	2017
2	Gaur,R.K.andGupta,S.L	Engineering Physics	DhanpatRai Publishers	2018
3	D.Halliday, R.Resnick and J.Walker	Principles of Physics	Wiley (Indian Edition)	2015
4	Arthur Beiser, ShobhitMahajan, S.RaiChoudhury	Concepts of Modern Physics	McGraw-Hill (Indian Edition)	2017
5	M.Arumugam	Engineering Physics	Anuradha publications	2010
6	Gaur,R.K.andGupta,S.L	Engineering Physics	DhanpatRai Publishers	2018

REFERENCE BOOKS:

1	R.Wolfson	Essential University Physics. Volume 1 & 2	Pearson Education (Indian Edition)	2020
2	K.Thyagarajan and A.Ghatak	Lasers: Fundamentals and Applications	Laxmi Publications, (Indian Edition)	2019
3	R.K.Rajput	Thermal Engineering	Laxmi Publications,	2011
4	S.O.Pillai,	Solid State Physics	New Age International, (Multicolour Edition)	2018

WEB LEARNING RESOURCES:

1. https://youtu.be/fDJeVR0o w?list=PLyQSN7X0ro203puVhQsmCj9qhlFQ-As8e (Rotating Objects, Moment of Inertia, Rotational KE)

2. https://archive.nptel.ac.in/courses/104/104/104104085/ (Lasers)

3. https://www.youtube.com/playlist?list=PL1gyM10tgL1hK9666oGndGlWDQdpQzkY9 (NPTEL: Heat transfer lectures by Dr.Gangesh A. Viswanathan, IITB)

4 https://archive.nptel.ac.in/courses/115/101/115101107/ (Quantum mechanics)
5 https://youtu.be/5EiZjZjG-IY (NPTEL lectures: Crystal Structure - 2 (Unit Cell, Lattice, Crystal)
6. https://www.youtube.com/watch?v=mx2P1_M-7UA&list=PLFE3074A4CB751B2B&index=9 (Rotations, Part I: Dynamics of Rigid Bodies)
7. https://www.youtube.com/watch?v=UzrZxpup3rc&list=PLFE3074A4CB751B2B&index=10 (Rotations, Part II: Parallel Axis Theorem)
8. https://youtu.be/7Bj3N1E7vZk?list=PLZOZfX_TaWAHZOgn8CRjpqREIp5Dd-GaY (Introduction to heat transfer, conduction, convection, and radiation)
9. https://youtu.be/dRpyfm66GxM (Particle in an Infinite Potential Well (QUANTUM MECHANICS)

CO – PO – PSO MAPPING

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO 10	PO 11	PO 12	PS O1	PS O2	PS O3
CO1	3	3	2	1	1	1	-	-	-	-	-	-	-	-	-
CO2	3	-	1	1	-	-	-	-	-	-	-	1	-	-	-
CO3	3	3	2	1	2	1	-	-	-	-	-	-	-	-	-
CO4	3	3	1	1	2	1	-	-	-	-	-	-	-	-	-
CO5	3	1	-	-	-	-	-	-	-		-	-	-	-	-
AVG	3	2.5	1.5	1	1.6	1						1			

<table>
<tr><td>R 2024</td><td colspan="5">DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING</td><td>SEMESTER:II</td></tr>
<tr><td rowspan="2">24ES210</td><td rowspan="2">DATA STRUCTURES AND ALGORITHMS</td><td>L</td><td>T</td><td>P</td><td>C</td><td rowspan="2">ES</td></tr>
<tr><td>3</td><td>0</td><td>2</td><td>4</td></tr>
<tr><td colspan="7">Common to CSE, IT AND AI&DS Departments</td></tr>
</table>

COURSE OBJECTIVES:

The objectives of learning this course are:

- To understand the concepts of ADTs
- To design linear data structures – lists, stacks, and queues
- To understand sorting, searching and hashing algorithms
- To apply Tree and Graph structures

COURSE OUTCOMES:

At the end of this course, students are able to:

CO1: Explain abstract data types

CO2: Design, implement, and analyse linear data structures, such as lists, queues, and stacks,according to the needs of different applications.

CO3: Design, implement, and analyse efficient tree structures to meet requirements such as searching, indexing, and sorting.

CO4: Model problems as graph problems and implement efficient graph algorithms to solve them.

CO5: Apply Graph Structures in real-world application

UNIT: I	ABSTRACT DATA TYPES	9
Abstract Data Types (ADTs) – ADTs and classes – introduction to OOP – classes in Python – inheritance – namespaces – shallow and deep copying, Introduction to analysis of algorithms – asymptotic notations – recursion – analyzing recursive algorithms.		
Pedagogical Tools	Black board, chalk, Group Discussion, Role Play, Youtube Videos,Nptel videos.	
UNIT: II	**LINEAR STRUCTURES**	**9**
List ADT – array-based implementations – linked list implementations – singly linked lists – circularly linked lists – doubly linked lists – applications of lists – Stack ADT – Queue ADT – double ended queues.		
Pedagogical Tools	Black board, chalk, Group Discussion, Role Play, Youtube Videos,Nptel videos.	
UNIT: III	**LINEAR STRUCTURES**	**9**
Bubble sort – selection sort – insertion sort – merge sort – quick sort – linear search – binary search – hashing – hash functions – collision handling – load factors, rehashing, and efficiency.		
Pedagogical Tools	Black board, chalk, Group Discussion, Role Play, Youtube Videos,Nptel videos.	
UNIT: IV	**TREE STRUCTURES**	**9**
Tree ADT – Binary Tree ADT – tree traversals – binary search trees – AVL trees – heaps – multiway search trees.		
Pedagogical Tools	Black board, chalk, Group Discussion, Role Play, Youtube Videos,Nptel videos.	
UNIT: V	**GRAPH STRUCTURES**	**9**
Graph ADT – representations of graph – graph traversals – DAG – topological ordering – shortest paths – minimum spanning trees.		

Pedagogical Tools	Black board, chalk, Group Discussion, Role Play, Youtube Videos,Nptel videos.
	45 Periods

PRACTICAL EXERCISES **30 Periods**

1. Implement simple ADTs as Python classes
2. Implement List ADT using Python arrays
3. Linked list implementations of List
4. Implementation of Stack and Queue ADTs
5. Implementation of sorting and searching algorithms
6. Tree representation and traversal algorithms
7. Implementation of Heaps
8. Implementation of single source shortest path algorithm
9. Implementation of minimum spanning tree algorithms
10. Mini Project
    - Creating a To-do list.
    - Building a Phonebook.
    - Build a simple calculator.
    - Students grade checker.
    - Plagiarism detection system.
    - Banking management system.
    - Travel planner using Graph.
    - Cash flow minimizer.

Total: 75 Periods

TEXT BOOKS:

Sl. No	Authors	Title of the Book	Publisher	Year of publication
1	Michael T. Goodrich, Roberto Tamassia, and Michael H. Goldwasser	Data Structures and Algorithms in Python	Wiley	2021
2	**Lee**, Kent D., **Hubbard**, Steve	Data Structures and Algorithms with Python	Springer Edition	2015
3	Narasimha Karumanchi	Data Structures and Algorithmic Thinking with Python	Careermonk	2015

REFERENCE BOOKS:

Sl. No	Authors	Title of the Book	Publisher	Year of publication
1	Rance D. Necaise	Data Structures and Algorithms Using Python	John Wiley & Sons	2011
2	Thomas H. Cormen, Charles E. Leiserson, Ronald L. Rivest, and Clifford Stein	Introduction to Algorithms	PHI Learning	2010
3	Mark Allen Weiss	Data Structures and Algorithm Analysis in C++	Pearson Education	2014

CO PO PSO MAPPING

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PS01	PS02	PS03
CO1	2	2	2	1	2	-	-	-	-	-	-	2	2	-	-

CO2	2	3	2	1	2	-	-	-	-	-	-	2	2	-	-
CO3	3	2	2	1	3	-	-	-	-	-	-	3	2	-	-
CO4	2	3	3	1	2	-	-	-	-	-	-	2	2	-	-
CO5	2	3	3	1	2	-	-	-	-	-	-	2	2	-	-
AVG	3	2	2	2	2	-	-	-	2	2	2	2	2	2	2

R 2024	DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING					SEMESTER:01
24ES209	PYTHON PROGRAMMING LABORATORY	L	T	P	C	ES
		0	0	4	2	
Common to CSE,IT AND AI&DS Departments						

COURSE OBJECTIVES:

The objectives of learning this course are:

- To understand the problem solving approaches.
- To learn the basic programming constructs in Python.
- To practice various computing strategies for Python-based solutions to real world problems.
- To use Python data structures - lists, tuples, dictionaries.
- To do input/output with files in Python.

COURSE OUTCOMES:

At the end of this course, students able to

CO1: Develop algorithmic solutions to simple computational problems

CO2: Implement programs in Python using conditionals and loops for solving problems.

CO3: Deploy functions to decompose a Python program.

CO4: Process compound data using Python data structures.

CO5: Utilize Python packages in developing software applications.

LIST OF EXPERIMENTS:

1. Identification and solving of simple real life or scientific or technical problems, and developing flow charts for the same. (Electricity Billing, Retail shop billing, Sin series, weight of a motorbike, Weight of a steel bar, compute Electrical Current in Three Phase AC Circuit, etc.)
2. Python programming using simple statements and expressions (exchange the values of two variables, circulate the values of n variables, distance between two points).
3. Scientific problems using Conditionals and Iterative loops. (Number series, Number Patterns, pyramid pattern)
4. Implementing real-time/technical applications using Lists, Tuples. (Items present in a library/Components of a car/ Materials required for construction of a building –operations of list & tuples)
5. Implementing real-time/technical applications using Sets, Dictionaries. (Language, components of an automobile, Elements of a civil structure, etc.- operations of Sets & Dictionaries)
6. Implementing programs using Functions. (Factorial, largest number in a list, area of shape)
7. Implementing programs using Strings. (reverse, palindrome, character count, replacing characters)
8. Implementing programs using written modules and Python Standard Libraries (pandas, numpy. Matplotlib, scipy)
9. Implementing real-time/technical applications using File handling. (copy from one file to another, word count, longest word)
10. Implementing real-time/technical applications using Exception handling. (divide by zero error, voter’s age validity, student mark range validation)
11. Exploring Pygame tool.
12. Mini Project - Developing a game activity using Pygame like bouncing ball, car race, Cricket alerts etc.

Total Periods : 60

LIST OF COMPONENTS REQUIRED: (For a Batch of 30 Students)

1. INTEL based desktop PC with min. 8GB RAM and 500 GB HDD, 17” or higher TFT Monitor, Keyboard and mouse. – 30 Nos
2. Windows 10 or higher operating system / Linux Ubuntu 20 or higher. – 30 Nos
3. Python 3.9 or above – 30 Nos

CO PO PSO MAPPING

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PS01	PS02	PS03
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO 10	PO11	PO12	PSO1	PSO2	PSO3
CO1	2	2	2	1	2	1	1	1	2	-	3	2	2	2	-
CO2	2	3	2	1	2	1	1	1	2	-	3	2	2	2	-
CO3	3	2	2	1	3	1	1	1	2	-	3	3	2	2	-
CO4	2	3	3	1	2	1	2	1	2	-	3	2	2	3	-
CO5	2	3	3	1	2	1	-	-	2	1	2	2	2	2	-
AVG	2	3	2	1	2	1	1	1	2	1	3	2	2	2	-

R 2024	MECHANICAL ENGINEERING					SEMESTER: II
24 ES 205	ENGINEERING DRAWING	L	T	P	C	PC
		0	0	4	2	

COMMON TO : AI&DS, BME, CSE, ECE, EEE and IT

COURSE OBJECTIVES:

The main objectives of this course are to:

- To learn conventions and use of drawing tools in making engineering drawings
- To draw orthographic projection of points and lines
- To understand the projection of planes and simple solids
- To teach the section of solids and obtain the development of surfaces of given solids
- To deliver how to draw isometric and perspective projections of the given solids

COURSE OUTCOMES:

Upon completion of the course, the student are able to

CO1: Recognize the conventions and construct basic engineering curves.

CO2: Draw the projection of points and lines.

CO3: Sketch the projection of planes and simple solids.

CO4: Produce the projection section of solids and development of surfaces of given solids

CO5: Develop the isometric projection and Perspective projections of the given objects

PRACTICAL EXERCISES:

1. Fundamental of drawing: Importance of graphics in engineering applications–Use of drafting instruments–BIS conventions and specifications – Size, layout and folding of drawing sheets – Lettering and dimensioning. (Not for examination)
2. Fundamental of drawing: Importance of graphics in engineering applications–Use of drafting instruments–BIS conventions and specifications – Size, layout and folding of drawing sheets – Lettering and dimensioning. (Not for examination)
3. Projection of straight lines (only First angle projection) inclined to both the principal planes - Determination of true l lengths and true inclinations by rotating line method.
4. Projection of polygonal plane surface inclined to both the principal planes by rotating object method (Pentagonal and Hexagonal plane surface)
5. Projection of Circular plane inclined to both the principal planes by rotating object method.
6. Projection of simple prisms (Hexagon and pentagon) when the axis is inclined to one of the principal planes.
7. Projection of simple prisms (Hexagon and pentagon) when the axis is inclined to one of the principal planes.
8. Projection of simple pyramids (Hexagon and pentagon), cylinder and cone when the axis is inclined to one of the principal planes.
9. Projection of cylinder and cone when the axis is inclined to one of the principal planes.
10. Projection of sectioned solids in simple vertical position when the cutting plane is inclined to the one of the principal planes and perpendicular to the other – obtaining true shape of section (Prism or Pyramid)
11. Development of lateral surfaces of simple and sectioned solids (Prism or Pyramid)
12. Draw the isometric view of frustum of solids like Prism or Pyramid of pentagonal or hexagonal base.
13. Perspective projection of simple solids-Prisms, pyramids and cylinders by visual ray method.

Total Periods : 60

CO PO PSO MAPPING:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PS01	PS02	PS03
CO1	3	3	3	3	2	-	-	-	-	3	-	2	1	3	2
CO2	3	3	3	3	2	-	-	-	-	3	-	2	1	3	2
CO3	3	3	3	3	2	-	-	-	-	3	-	2	1	3	2
CO4	3	3	3	2	2	-	-	-	-	3	-	2	1	3	2
CO5	3	3	3	2	2	-	-	-	-	3	-	2	1	3	2
AVG	3	3	3	2	2	-	-	-	-	3	-	2	1	3	2

R 2024	CAREER DEVELOPMENT AND PLACEMENT CELL M.A.M.SCHOOL OF ENGINEERING					SEMESTER:II
24TP201	Aptitude and Communication Skills - I	L	T	P	C	EEC
		0	0	2	1	

COURSE OBJECTIVES:

The main objectives of this course are to:

- To Learn and Practice Vedic Mathematics Principles and Techniques
- To Understand the Components of Effective Communication
- To understand the components of Presentation Skills and Delivery Techniques that are needed for Individual & Group Presentations.
- To learn about personal grooming, body language and Dress code.

COURSE OUTCOMES:

At the end of this course, students are able to:

CO1: Effectively applying the Vedic Mathematics Techniques to solve the Mathematical Aptitude Questions.

CO2: Learn and Practice the ways of Effective Communication and hence to excel in Public Speaking.

CO3: Present their Ideas in an professional way by learning the Presentation Skills and Delivery Techniques.

CO4: Effectively apply the body language and show case them with better dress code and grooming.

LIST OF ACTIVITIES/EXCERCISES:

1. Squares ending with 5 and 55.
2. Multiplication of Numbers by 5, 25, 50, 125, 9, 99, 999, 9999.
3. Multiplication of Two Numbers where Sum of unit digit is 10
4. Multiplication of Two Numbers where Sum of unit digit is 10, 1000 others digits same
5. Multiplication of Two numbers both having '5' at Unit digits.
6. Multiples of 11, 111 & 22, 33, 44, 55 etc.,
7. Squaring of numbers using Base 10, 100, 1000, 50, 500, 5000.
8. Multiplication of numbers more than or below the Base 10, 100, 1000, 50, 500, 5000.
9. Squares ending with 555.
10. Dividing of 9, 19, 29, 39, 49.
11. Square Root & Cube Root, Decimals, Fractions.
12. Components of Effective Communication and Communication styles of others.
13. Barriers of Communication.
14. Dealing with emotions while communicating
15. Just a Minute (JAM) Session
16. Delivery Techniques & Visual Effects / Individual & Group Presentations
17. SWOT Analysis
18. Personality Enhancement & Body Language.
19. Hand Shaking & Dress Code.
20. Personal Grooming.

Total Periods : 30

CO PO PSO MAPPING

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PS01	PS02	PS03
CO1	-	1	1	-	-	-	-	-	-	-	-	-	-	-	-
CO2	-	-	-	-	-	-	-	-	3	3	-	-	-	-	-
CO3	-	-	-	-	2	-	-	-	3	3	-	-	-	-	-
CO4	-	-	-	-	-	-	-	-	3	3	-	-	-	-	-
AVG	-	1	1	-	2	-	-	-	3	3	-	-	-	-	-

R 2024	DEPARTMENT OF ARTIFICIAL INTELLIGENCE & DATA SCIENCE					SEMESTER:III
24AD301	FUNDAMENTALS OF AI	L	T	P	C	PC
		3	0	0	3	

COURSE OBJECTIVES:

The main objectives of this course are to:

- Learn the fundamentals of AI Agents and Environments.
- Understand how to define the problem for AI agent and how to do search?
- Learn how to apply AI in Game theory.
- Introduce the concepts of Knowledge based agents.
- Apply probabilistic reasoning in AI

COURSE OUTCOMES:

At the end of this course, students are able to:
CO1: Explain intelligent agent frameworks
CO2: Apply problem solving techniques
CO3: Apply game playing and CSP techniques
CO4: Perform logical reasoning
CO5: Perform probabilistic reasoning under uncertainty.

UNIT: I	AI AGENTS AND ENVIRONMENTS	9

AI –Definition and History of Evolution – Turing Test - Applications of AI across Industries – Intelligent agents : Structure, Agent Function, Agent Program, Vaccum Cleaner World Example – Concept of Rationality - Environment : PEAS Description and Examples, Types of Environments, Types of Agents: Simple reflex agents, Model-based reflex agents; Goal-based agents and Utility-based agents.

UNIT: II	PROBLEM SOLVING	9

Formulation of Problem – Example: 8 Puzzle, 8 Queen – Uninformed Searching Strategies: Breadth First Search, Uniform Cost Search, Depth First Search, Depth Limited Search, Iterative Deepening, Bidirectional Search. Informed (Heuristic) search strategies: Greedy Best First Search, A* Search - Local search: Hill Climbing - Search in partially observable environments.

UNIT: III	GAME PLAYING AND CSP	9

Game theory – optimal decisions in games – alpha-beta search – monte-carlo tree search – stochastic games – partially observable games. Constraint satisfaction problems – constraint propagation – backtracking search for CSP – local search for CSP – structure of CSP.

UNIT: IV	LOGICAL REASONING	9

Knowledge-based agents – propositional logic – propositional theorem proving – propositional model checking – agents based on propositional logic. First-order logic – syntax and semantics – knowledge representation and engineering – inferences in first-order logic – forward chaining – backward chaining – resolution.

UNIT: V	PROBABILISTIC REASONING	9

Acting under uncertainty – Bayesian inference – naïve Bayes models. Probabilistic reasoning – Bayesian networks – exact inference in BN – approximate inference in BN – causal networks.

Total Periods : 45

TEXT BOOKS:

Sl. No	Authors	Title of the Book	Publisher	Year of publication
1	Stuart Russell and Peter Norvig	Artificial Intelligence – A Modern Approach	Pearson Education	Fourth Edition, 2021

REFERENCE BOOKS:

Sl. No	Authors	Title of the Book	Publisher	Year of publication
1	Dan W. Patterson	Introduction to AI and ES	Pearson Education	2007
2	Kevin Night, Elaine Rich, Nair B.	Artificial Intelligence	McGraw Hill	2008
3	Patrick H. Winston	Artificial Intelligence	Pearson Education	2006
4	Deepak Khemani	Artificial Intelligence	Tata McGraw Hill Education	2013

CO PO PSO MAPPING

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PS01	PS02	PS03
CO1	3	1	3	3	-	-	-	-	2	3	3	1	2	1	1
CO2	2	2	1	1	1	-	-	-	2	2	3	1	3	2	2
CO3	2	1	2	1	-	-	-	-	2	1	1	3	1	2	1
CO4	2	1	2	2	-	-	-	-	2	1	2	2	1	3	3
CO5	3	2	2	1	1	-	-	-	3	2	1	2	2	2	1
AVG	2	1	2	2	1	-	-	-	2	2	2	2	2	2	2

<table>
<tr><td>R 2024</td><td colspan="4">DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING</td><td>SEMESTER:III</td></tr>
<tr><td>24CS302</td><td colspan="3">OBJECT ORIENTED PROGRAMMING</td><td>L T P C
3 0 0 3</td><td>PC</td></tr>
<tr><td colspan="6">Common to CSE, AI&DS, IT, ECE AND BME Departments</td></tr>
<tr><td colspan="6">COURSE OBJECTIVES:</td></tr>
<tr><td colspan="6">The main objectives of this course are to:
• To understand Object Oriented Programming concepts and basics of Java programming language
• To know the principles of packages, inheritance and interfaces
• To develop a java application with threads and generics classes
• To define exceptions and use I/O streams
• To design and build Graphical User Interface Application using JAVAFX</td></tr>
<tr><td colspan="6">COURSE OUTCOMES:</td></tr>
<tr><td colspan="6">At the end of this course, students are able to:
CO1:Apply the concepts of classes and objects to solve simple problems
CO2:Develop programs using inheritance, packages and interfaces
CO3:Make use of exception handling mechanisms and multithreaded model to solve real world problems
CO4:Build Java applications with I/O packages, string classes, Collections and generics concepts
CO5:Integrate the concepts of event handling and JavaFX components and controls for developing GUI based applications</td></tr>
<tr><td>UNIT: I</td><td colspan="4">INTRODUCTION TO OOP AND JAVA</td><td>9</td></tr>
<tr><td colspan="6">Creation of JAVA – Evolution, Features – Basic Concepts of OOPs – Encapsulation, Inheritance and Polymorphism – Installation of Java, Entering and Compiling the simple JAVA Program – Primitive Data types, Literals – Variables, Type Conversion and Casting – Arrays : One Dimensional and Multi Dimensional - Operators</td></tr>
<tr><td>UNIT: II</td><td colspan="4">INHERITANCE, PACKAGES AND INTERFACES</td><td>9</td></tr>
<tr><td colspan="6">Control Statements : If, Nested If, If-else-If Ladder, switch, Nested Switch, while, do-while, for, jump, continue and return - Class Fundamentals: Programming Structures in Java – Defining classes in Java – Constructors-Methods -Access specifiers - Static members- Java Doc comments</td></tr>
<tr><td>UNIT: III</td><td colspan="4">EXCEPTION HANDLING AND MULTITHREADING</td><td>9</td></tr>
<tr><td colspan="6">Exception Handling basics – Multiple catch Clauses – Nested try Statements – Java's Built-in Exceptions – User defined Exception. Multithreaded Programming: Java Thread Model–Creating a Thread and Multiple Threads – Priorities – Synchronization – Inter Thread Communication- Suspending –Resuming, and Stopping Threads –Multithreading. Wrappers – Auto boxing.</td></tr>
<tr><td>UNIT: IV</td><td colspan="4">I/O, GENERICS, STRING HANDLING</td><td>9</td></tr>
<tr><td colspan="6">I/O Basics – Reading and Writing Console I/O – Reading and Writing Files. Generics: Generic Programming – Generic classes – Generic Methods – Bounded Types – Restrictions and Limitations. Strings: Basic String class, methods and String Buffer Class.</td></tr>
<tr><td>UNIT: V</td><td colspan="4">JAVAFX EVENT HANDLING, CONTROLS AND COMPONENTS</td><td>9</td></tr>
<tr><td colspan="6">JAVAFX Events and Controls: Event Basics – Handling Key and Mouse Events. Controls: Checkbox, ToggleButton – RadioButtons – ListView – ComboBox – ChoiceBox – Text Controls – ScrollPane. Layouts – FlowPane – HBox and VBox – BorderPane – StackPane – GridPane. Menus – Basics – Menu – Menu bars – MenuItem.</td></tr>
<tr><td colspan="6" align="right">Total Periods : 45</td></tr>
<tr><td colspan="6">TEXT BOOKS:</td></tr>
<tr><td>Sl. No</td><td>Authors</td><td>Title of the Book</td><td>Publisher</td><td colspan="2">Year of publication</td></tr>
</table>

1	Herbert Schildt	Java: The Complete Reference (11th Edition)	McGraw Hill Education, New Delhi	2019
2	Herbert Schildt	Introducing JavaFX 8 Programming (1st Edition)	McGraw Hill Education, New Delhi	2015

REFERENCE BOOKS:

Sl. No	Authors	Title of the Book	Publisher	Year of publication
1	Cay S. Horstmann	Core Java Fundamentals (Volume 1, 11th Edition)	Prentice Hall	2018

CO PO PSO MAPPING

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PS01	PS02	PS03
CO1	1	1	3	1	3	-	-	-	3	2	2	2	3	1	2
CO2	2	1	3	2	1	-	-	-	2	1	1	3	3	3	2
CO3	3	3	1	2	2	-	-	-	3	2	1	2	3	1	3
CO4	3	1	2	2	2	-	-	-	1	2	1	3	3	1	1
CO5	1	1	2	3	2	-	-	-	3	2	1	2	3	3	3
AVG	2	1	2	2	2	-	-	-	2	2	1	2	3	2	2

R 2024	DEPARTMENT OF INFORMATION TECHNOLOGY					SEMESTER:III
24IT301	DATA SCIENCE USING R	L	T	P	C	PC
		3	0	2	4	

COURSE OBJECTIVES:

The main objectives of this course are to:

- To understand the basics in R programming in terms of constructs, control statements, string functions
- To learn to apply R programming for Text processing
- To understand the use of R Big Data analytics
- To able to appreciate and apply the R programming from a statistical perspective

COURSE OUTCOMES:

At the end of this course, students are able to:

CO1: Create artful graphs to visualize complex data sets and functions
CO2: Write more efficient code using parallel R and vectorization
CO3: Interface R with C/C++ and Python for increased speed or functionality
CO4: Find new packages for text analysis, image manipulation, and perform statistical analysis of the same
CO5: Develop interfacing R to other Languages

UNIT: I	INTRODUCING TO R	9

Introducing to R – R Data Structures – Help functions in R – Vectors – Scalars – Declarations – recycling – Common Vector operations – Using all and any – Vectorized operations – NA and NULL values – Filtering – Vectorised if-then else – Vector Equality – Vector Element names

UNIT: II	MATRICES, ARRAYS AND LISTS	9

Matrices, Arrays And Lists Creating matrices – Matrix operations – Applying Functions to Matrix Rows and Columns – Adding and deleting rows and columns – Vector/Matrix Distinction – Avoiding Dimension Reduction – Higher Dimensional arrays – lists – Creating lists – General list operations – Accessing list components and values – applying functions to lists – recursive lists.

UNIT: III	DATA FRAMES	9

Creating Data Frames – Matrix-like operations in frames – Merging Data Frames – Applying functions to Data frames – Factors and Tables – factors and levels – Common functions used with factors – Working with tables - Other factors and table related functions

UNIT: IV	CONTROL STATEMENTS, FUNCTIONS, R GRAPHS	9

Control statements – Arithmetic and Boolean operators and values – Default values for arguments - Returning Boolean values – functions are objects – Environment and Scope issues –Writing Upstairs - Recursion – Replacement functions – Tools for composing function code – Math and Simulations in R Creating Graphs – Customizing Graphs – Saving graphs to files – Creating three-dimensional plots

UNIT: V	INTERFACING	9

Interfacing R to other languages – Parallel R – Basic Statistics – Linear Model – Generalized Linear models – Non-linear models – Time Series and Auto-correlation – Clustering

Periods : 45

PRACTICAL EXCERCISES
1. Arithmetic Operations using R Language **2.** Define and use Functions, Vectors, and Data Frames **3.** Matrix operations for data science. **4.** Descriptive Statistics using R **5.** Data Visualization using Plots **6.** Data handling and Cleaning
Periods : 30
Total Periods : 75

TEXT BOOKS:

Sl. No	Authors	Title of the Book	Publisher	Year of publication
1	Norman Mat off	"The Art of R Programming: A Tour of Statistical Software Design"	No Starch Press	2011

REFERENCE BOOKS:

Sl. No	Authors	Title of the Book	Publisher	Year of publication
1	Jared P. Lander	R for Everyone: Advanced Analytics and Graphics	Addison-Wesley Data & Analytics Series	2013
2	Mark Gardener	Beginning R – The Statistical Programming Language	Wiley	2013
3	Robert Knell	Introductory R: A Beginner's Guide to Data Visualization, Statistical Analysis and Programming in R	Amazon Digital South Asia Services Inc.	2013

CO PO PSO MAPPING

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PS01	PS02	PS03
CO1	3	2	1	-	2	-	-	-	-	-	-	1	-	-	-
CO2	3	3	2	-	2	-	-	-	-	-	-	1	-	-	-
CO3	3	3	3	-	2	1	-	-	1	-	-	2	-	-	-
CO4	2	3	2	-	3	2	-	-	2	-	-	2	-	-	-
CO5	3	3	3	-	3	2	-	-	1	-	-	3	-	-	-
AVG	2.8	2.8	2.2	-	2.4	1	-	-	0.8	-	-	1.8	-	-	-

<table>
<tr><td>R 2024</td><td colspan="4">DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING</td><td>SEMESTER:III</td></tr>
<tr><td rowspan="2">24CS304</td><td rowspan="2">DESIGN AND ANALYSIS OF ALGORITHMS</td><td>L</td><td>T</td><td>P</td><td>C</td></tr>
<tr><td>3</td><td>0</td><td>0</td><td>3</td></tr>
</table>

PC

Common to CSE and AI&DS & IT Departments

COURSE OBJECTIVES:

The main objectives of this course are to:

- To critically analyze the efficiency of alternative algorithmic solutions for the same problem
- To illustrate brute force and divide and conquer design techniques.
- To explain dynamic programming and greedy techniques for solving various problems.
- To apply iterative improvement technique to solve optimization problems
- To examine the limitations of algorithmic power and handling it in different problems.

COURSE OUTCOMES:

At the end of this course, students are able to:

CO1: Analyze the efficiency of recursive and non-recursive algorithms mathematically

CO2: Analyze the efficiency of brute force, divide and conquer, decrease and conquer, Transform and conquer algorithmic techniques

CO3: Implement and analyze the problems using dynamic programming and greedy algorithmic techniques.

CO4: Solve the problems using iterative improvement techniques for optimization.

CO5: Compute the limitations of algorithmic power and solve the problems using backtracking and branch and bound techniques.

UNIT: I	INTRODUCTION	8

Notion of an Algorithm – Fundamentals of Algorithmic Problem Solving – Important Problem Types –Fundamentals of the Analysis of Algorithm Efficiency – Analysis Framework - Asymptotic Notations and their properties – Empirical analysis - Mathematical analysis of Recursive and Non-recursive algorithms – Visualization.

UNIT: II	BRUTE FORCE AND DIVIDE AND CONQUER	10

Brute Force – String Matching - Exhaustive Search - Traveling Salesman Problem - Knapsack Problem - Assignment problem. Divide and Conquer Methodology – Multiplication of Large Integers and Strassen's Matrix Multiplication – Closest-Pair and Convex - Hull Problems. Decrease and Conquer: - Topological Sorting – Transform and Conquer: Presorting – Heaps and Heap Sort.

UNIT: III	DYNAMIC PROGRAMMING AND GREEDY TECHNIQUE	10

Dynamic programming – Principle of optimality - Coin changing problem – Warshall's and Floyd's algorithms – Optimal Binary Search Trees - Multi stage graph - Knapsack Problem and Memory functions. Greedy Technique – Dijkstra's algorithm - Huffman Trees and codes - 0/1 Knapsack problem.

UNIT: IV	ITERATIVE IMPROVEMENT	8

The Simplex Method-The Maximum-Flow Problem – Maximum Matching in Bipartite Graphs- The Stable marriage Problem.

UNIT: V	LIMITATIONS OF ALGORITHM POWER	9

Lower - Bound Arguments - P, NP, NP- Complete and NP Hard Problems. Backtracking – N-Queen problem - Hamiltonian Circuit Problem – Subset Sum Problem. Branch and Bound – LIFO Search and FIFO search - Assignment problem – Knapsack Problem – Traveling Salesman Problem - Approximation Algorithms for NP-Hard Problems – Traveling Salesman problem – Knapsack problem.

Total Periods : 45

TEXT BOOKS:

Sl. No	Authors	Title of the Book	Publisher	Year of publication
1	Anany Levitin	Introduction to the Design and Analysis of Algorithms, Third Edition	Pearson Education	2012

REFERENCE BOOKS:

Sl. No	Authors	Title of the Book	Publisher	Year of publication
1	Ellis Horowitz, Sartaj Sahni and Sanguthevar Rajasekaran	Computer Algorithms/ C++, Second Edition	Universities Press	2019
2	Thomas H. Cormen, Charles E. Leiserson, Ronald L. Rivest and Clifford Stein	Introduction to Algorithms, Third Edition	PHI Learning Private Limited	2012
3	S. Sridhar	Design and Analysis of Algorithms	Oxford University Press	2014
4	Alfred V. Aho, John E. Hopcroft and Jeffrey D. Ullman	Data Structures and Algorithms	Pearson Education	Reprint 2006

CO PO PSO MAPPING

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PS01	PS02	PS03
CO1	3	3	3	1	1	-	-	-	1	1	2	2	3	2	1
CO2	2	1	1	3	2	-	-	-	2	2	1	2	2	2	2
CO3	3	2	1	2	2	-	-	-	2	1	1	2	1	3	3
CO4	3	2	3	2	2	-	-	-	3	3	3	2	2	1	2
CO5	3	1	2	3	3	-	-	-	2	2	2	2	3	1	3
AVG	3	2	2	2	2	-	-	-	2	2	2	2	2	2	2

R 2024	DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING					SEMESTER:III
24CS305	DIGITAL PRINCIPLES AND SYSTEM DESIGN	L	T	P	C	PC
		3	0	2	4	

COURSE OBJECTIVES:

The main objectives of this course are to:

- To analyze and design combinational circuits.
- To analyze and design sequential circuits
- To understand the basic structure and operation of a digital computer.
- To study the design of data path unit, control unit for processor and to familiarize with hazards
- To understand the concept of various memories and I/O interfacing

COURSE OUTCOMES:

At the end of this course, students are able to:

CO1: Design various combinational digital circuits using logic gates
CO2: Design sequential circuits and analyze the design procedures
CO3: State the fundamentals of computer systems and analyze the execution of an instruction
CO4: Analyze different types of control design and identify hazards
CO5: Identify the characteristics of various memory systems and I/O communication

UNIT: I	COMBINATIONAL LOGIC	9

Combinational Circuits – Karnaugh Map - Analysis and Design Procedures – Binary Adder – Subtractor – Decimal Adder - Magnitude Comparator – Decoder – Encoder – Multiplexers - Demultiplexers

UNIT: II	SYNCHRONOUS SEQUENTIAL LOGIC	9

Introduction to Sequential Circuits – Flip-Flops – operation and excitation tables, Triggering of FF, Analysis and design of clocked sequential circuits – Design – Moore/Mealy models, state minimization, state assignment, circuit implementation - Registers – Counters.

UNIT: III	COMPUTER FUNDAMENTALS	9

Functional Units of a Digital Computer: Von Neumann Architecture – Operation and Operands of Computer Hardware Instruction – Instruction Set Architecture (ISA): Memory Location, Address and Operation – Instruction and Instruction Sequencing – Addressing Modes, Encoding of Machine Instruction – Interaction between Assembly and High Level Language.

UNIT: IV	PROCESSOR	9

Instruction Execution – Building a Data Path – Designing a Control Unit – Hardwired Control, Microprogrammed Control – Pipelining – Data Hazard – Control Hazards.

UNIT: V	MEMORY AND I/O	9

Memory Concepts and Hierarchy – Memory Management – Cache Memories: Mapping and Replacement Techniques – Virtual Memory – DMA – I/O – Accessing I/O: Parallel and Serial Interface – Interrupt I/O – Interconnection Standards: USB, SATA

Total Periods : 45

PRACTICAL EXERCISES
1. Verification of Boolean theorems using logic gates. 2. Design and implementation of combinational circuits using gates for arbitrary functions. 3. Implementation of 4-bit binary adder/subtractor circuits. 4. Implementation of code converters. 5. Implementation of BCD adder, encoder and decoder circuits 6. Implementation of functions using Multiplexers. 7. Implementation of the synchronous counters 8. Implementation of a Universal Shift register. 9. Simulator based study of Computer Architecture
Periods : 30
Total Periods : 75

TEXT BOOKS:

Sl. No	Authors	Title of the Book	Publisher	Year of publication
1	. M. Morris Mano, Michael D. Ciletti,	Digital Design : With an Introduction to the Verilog HDL, VHDL, and System Verilog	Pearson Education	2018
2	David A. Patterson, John L. Hennessy	Computer Organization and Design, The Hardware/Software Interface", Sixth Edition	Morgan Kaufmann/Elsevier	2020

REFERENCE BOOKS:

Sl. No	Authors	Title of the Book	Publisher	Year of publication
1	Carl Hamacher, Zvonko Vranesic, Safwat Zaky, Naraig Manjikian,	Computer Organization and Embedded Systems", Sixth Edition	Tata McGraw-Hill	2012
2	William Stallings	Computer Organization and Architecture – Designing for Performance	Pearson Education	2016
3	M. Morris Mano	Digital Logic and Computer Design	Pearson Education	2016

CO PO PSO MAPPING

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PS01	PS02	PS03
CO1	3	3	3	3	3	2	1	1	1	1	2	3	2	3	3
CO2	3	3	3	3	2	1	1	1	1	1	2	3	1	2	2
CO3	3	3	3	3	2	2	1	1	1	1	2	3	2	3	1
CO4	3	3	3	3	1	1	1	1	1	1	1	2	1	3	1
CO5	3	3	3	3	1	2	1	1	1	1	1	2	1	2	1
AVG	3	3	3	3	1.8	1.6	1	1	1	1	1,6	2.6	1.4	2.6	1.6

R 2024	DEPARTMENT OF ARTIFICIAL INTELLIGENCE AND DATA SCIENCE					SEMESTER:III
24 AD303	AI LAB	L	T	P	C	PC
		0	0	4	2	

COURSE OBJECTIVES:

The main objectives of this course are to:

- To design and implement search strategies
- To implement game playing techniques
- To implement CSP techniques
- To develop systems with logical reasoning
- To develop systems with probabilistic reasoning

COURSE OUTCOMES:

At the end of this course, students are able to:

CO1: Design and implement search strategies
CO2: Implement game playing and CSP techniques
CO3: Develop logical reasoning systems
CO4: Develop probabilistic reasoning systems

LIST OF EXPERIMENTS:

1. Implement basic search strategies – 8-Puzzle, 8 - Queens problem, Cryptarithmetic.
2. Implement A* and memory bounded A* algorithms
3. Implement Minimax algorithm for game playing (Alpha-Beta pruning)
4. Solve constraint satisfaction problems
5. Implement propositional model checking algorithms
6. Implement forward chaining, backward chaining, and resolution strategies
7. Build naïve Bayes models
8. Implement Bayesian networks and perform inferences
9. Classification Algorithms
10. Mini-Project

Total Periods : 45

CO PO PSO MAPPING

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PS01	PS02	PS03
CO1	2	1	3	3	-	-	-	-	1	1	2	1	3	2	1
CO2	1	2	3	3	2	-	-	-	3	2	3	3	3	3	2
CO3	3	1	3	3	1	-	-	-	1	3	1	2	1	1	3
CO4	2	1	1	1	1	-	-	-	2	3	1	2	2	2	1
CO5	3	1	1	1	1	-	-	-	1	3	3	3	3	3	2
AVG	2	1	2	2	1	-	-	-	2	2	2	2	2	2	2

<table>
<tr><td rowspan="1">R 2024</td><td colspan="5">DEPARTMENT OF ARTIFICIAL INTELLIGENCE AND DATA SCIENCE</td><td>SEMESTER:III</td></tr>
<tr><td rowspan="2">24 CS307</td><td rowspan="2">OBJECT ORIENTED PROGRAMMING LAB</td><td>L</td><td>T</td><td>P</td><td>C</td><td rowspan="2">PC</td></tr>
<tr><td>0</td><td>0</td><td>4</td><td>2</td></tr>
</table>

COURSE OBJECTIVES:

The main objectives of this course are to:

•To build software development skills using java programming for real-world applications.

•To understand and apply the concepts of classes, packages, interfaces, inheritance, exception handling and file processing.

•To develop applications using generic programming and event handling

COURSE OUTCOMES:

At the end of this course, students are able to:

CO1 : Design and develop java programs using object oriented programming concepts

CO2 : Develop simple applications using object oriented concepts such as package, exceptions

CO3: Implement multithreading, and generics concepts

CO4 : Create GUIs and event driven programming applications for real world problems

CO5: Implement and deploy web applications using Java

LIST OF EXPERIMENTS:

1. Solve problems by using sequential search, binary search, and quadratic sorting algorithms (selection, insertion)
2. Develop stack and queue data structures using classes and objects.
3. Develop a java application with an Employee class with Emp_name, Emp_id, Address, Mail_id, Mobile_no as members. Inherit the classes, Programmer, Assistant Professor, Associate Professor and Professor from employee class. Add Basic Pay (BP) as the member of all the inherited classes with 97% of BP as DA, 10 % of BP as HRA, 12% of
BP as PF, 0.1% of BP for staff club funds. Generate pay slips for the employees with their gross and net salary.
4. Write a Java Program to create an abstract class named Shape that contains two integers and an empty method named printArea(). Provide three classes named Rectangle, Triangle and Circle such that each one of the classes extends the class Shape. Each one of the classes contains only the method printArea( ) that prints the area of the given shape.
5. Solve the above problem using an interface.
6. Implement exception handling and creation of user defined exceptions.
7. Write a java program that implements a multi-threaded application that has three threads. First thread generates a random integer every 1 second and if the value is even, the second thread computes the square of the number and prints. If the value is odd, the third thread will print the value of the cube of the number.
8. Write a program to perform file operations.
9. Develop applications to demonstrate the features of generics classes.
10. Develop applications using JavaFX controls, layouts and menus.
11. Develop a mini project for any application using Java concepts.

Total Periods : 45

CO PO PSO MAPPING

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PS01	PS02	PS03
CO1	2	1	2	1	-	-	-	-	1	2	2	2	1	2	3
CO2	2	1	3	1	-	-	-	-	2	3	3	2	1	3	1
CO3	2	2	1	2	1	-	-	-	1	2	1	3	2	3	2
CO4	2	2	1	3	-	-	-	-	3	1	1	1	2	1	2
CO5	1	3	3	1	3	-	-	-	1	1	1	1	2	1	2
AVG	2	2	2	2	2	-	-	-	2	2	2	2	2	2	2

R 2024	ELECTRONICS AND COMMUNICATION ENGINEERING					SEMESTER: 04
24ECE17	MICROPROCESSORS AND MICROCONTROLLERS	L	T	P	C	PC
		3	0	0	3	

COURSE OBJECTIVES:

- To study the addressing modes & instruction set of 8085 &8051
- To develop skills in simple program writing in assembly languages.
- To introduce commonly used peripheral/interfacing ICs.
- To study and understand typical applications of micro-processors.
- To study and understand the typical applications of micro-controllers

COURSE OUTCOMES:

Upon successful completion of the course, students will be able to:

CO1 :To write assembly language program for microprocessor and microcontroller

CO2 :To design and implement interfacing of peripheral with microprocessor and microcontroller

CO3 :To analyze, comprehend, design and simulate microprocessor based systems used for control and monitoring

CO4 :To analyze, comprehend, design and simulate microcontroller based systems used for control and monitoring.

CO5 : To understand and appreciate advanced architecture evolving microprocessor field

UNIT: I	INTRODUCTION TO 8085 ARCHITECTURE	9

Functional block diagram – Memory interfacing–I/O ports and data transfer concepts – Timing Diagram – Interrupt structure

UNIT: II	8085 INSTRUCTION SET AND PROGRAMMING	9

Instruction format and addressing modes – Assembly language format – Data transfer, data manipulation & control instructions – Programming: Loop structure with counting & Indexing - Look up table - Subroutine instructions, stack.

UNIT: III	INTERFACING BASICS AND ICS	9

Study of Architecture and programming of ICs: 8255 PPI, 8259PIC, 8251USART, 8279 Keyboard display controller and 8254 Timer/Counter – Interfacing with 8085 -A/D and D/A converter interfacing.

UNIT: IV	INTRODUCTION TO 8051 MICROCONTROLLER	9

Functional block diagram - Instruction format and addressing modes – Interrupt structure – Timer – I/O ports – Serial communication, Simple programming –keyboard and display interface –Temperature control system –stepper motor control - Usage of IDE for assembly language programming.

UNIT: V	INTRODUCTIONTO RISC BASED ARCHITECTURE	9

PIC16 /18 architecture, Memory organization – Addressing modes – Instruction set - Programming techniques – Timers – I/O ports – Interrupt programming.

Pedagogical Tools : Chalk & Board, PPT, NPTEL video, you tube video

Total :45

TEXT BOOKS:

Sl.No	Authors	Title of the Book	Publisher	Year of publication
1	Ramesh S. Gaonkar	Microprocessor Architecture Programming and Application	Penram International (P) Ltd., Mumbai	2013
2	Muhammad Ali Mazidi, Janice Gillispie Mazidi	The 8051 Microcontroller and Embedded Systems	Pearson Education	2011

3	Muhammad Ali Mazidi, Janice Gillispie Mazidi	The PIC Microcontroller and Embedded Systems	—	2010
REFERENCE BOOKS:				
1	Douglas V. Hall	Micro-processors & Interfacing	Tata McGraw Hill	2017
2	Krishna Kant	Micro-processors & Micro-controllers	Prentice Hall of India	2007
3	Mike Predko	8051 Micro-controllers	McGraw Hill	2009
4	Kenneth Ayala	The 8051 Microcontroller	Thomson	2004

CO – PO – PSO MAPPING:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	3	3	1	3	-	-	-	-	-	-	-	-	1	1	1
CO2	3	3	3	2	-	-	-	-	-	-	-	-	1	1	1
CO3	3	3	3	3	-	-	-	-	-	-	-	-	1	1	1
CO4	3	3	2	3	-	-	-	-	-	-	-	-	1	1	1
CO5	3	2	2	**3**	-	-	-	-	-	-	-	-	1	1	1
CO	3	3	3	3	-	-	-	-	-	-	-	-	1	1	1

1 - low, 2 - medium, 3 - high, '-' - no correlation

R 2024	DEPARTMENT OF ARTIFICIAL INTELLIGENCE AND DATA SCIENCE					SEMESTER:IV
24CS402	DATABASE MANAGEMENT SYSTEMS	L	T	P	C	PC
		3	0	0	3	

COURSE OBJECTIVES:

The main objectives of this course are to:

•To learn the fundamentals of data models, relational algebra and SQL

•To represent a database system using ER diagrams and to learn normalization techniques

•To understand the fundamental concepts of transaction, concurrency and recovery processing

•To understand the internal storage structures using different file and indexing techniques which will help in physical DB design

•To have an introductory knowledge about the Distributed databases, NOSQL and database security

COURSE OUTCOMES:

At the end of this course, students are able to:

CO1: Construct SQL Queries using relational algebra

CO2: Design database using ER model and normalize the database

CO3: Construct queries to handle transaction processing and maintain consistency of the database

CO4: Compare and contrast various indexing strategies and apply the knowledge to tune the performance of the database

CO5: Appraise how advanced databases differ from Relational Databases and find a suitable database for the given requirement.

UNIT: I	RELATIONAL DATABASES	10

Purpose of Database System – Views of data – Data Models – Database System Architecture – Introduction to relational databases – Relational Model – Keys – Relational Algebra – SQL fundamentals – Advanced SQL features – Embedded SQL– Dynamic SQL

UNIT: II	DATABASE DESIGN	8

Entity-Relationship model – E-R Diagrams – Enhanced-ER Model – ER-to-Relational Mapping – Functional Dependencies – Non-loss Decomposition – First, Second, Third Normal Forms, Dependency Preservation – Boyce/Codd Normal Form – Multi-valued Dependencies and Fourth Normal Form – Join Dependencies and Fifth Normal Form

UNIT: III	TRANSACTIONS	9

Transaction Concepts – ACID Properties – Schedules – Serializability – Transaction support in SQL – Need for Concurrency – Concurrency control –Two Phase Locking- Timestamp – Multiversion – Validation and Snapshot isolation– Multiple Granularity locking – Deadlock Handling – Recovery Concepts – Recovery based on deferred and immediate update – Shadow paging – ARIES Algorithm

UNIT: IV	IMPLEMENTATION TECHNIQUES	9

RAID – File Organization – Organization of Records in Files – Data dictionary Storage – Column Oriented Storage– Indexing and Hashing –Ordered Indices – B+ tree Index Files – B tree Index Files – Static Hashing – Dynamic Hashing – Query Processing Overview – Algorithms for Selection, Sorting and join operations – Query optimization using Heuristics - Cost Estimation.

UNIT: V	ADVANCED TOPICS	9

Overview of No SQL Databases – Types – Comparison of Relational Database and No SQL Database – Mongo DB, Apache Cassandra – HBASE – Challenges and Applications of No SQL.

Total Periods : 45

TEXT BOOKS:

Sl. No	Authors	Title of the Book	Publisher	Year of publication

1	Abraham Silberschatz, Henry F. Korth, S. Sudharshan	Database System Concepts, Seventh Edition	McGraw Hill	2020
2	Ramez Elmasri, Shamkant B. Navathe	Fundamentals of Database Systems, Seventh Edition	Pearson Education	2017

REFERENCE BOOKS:

Sl. No	Authors	Title of the Book	Publisher	Year of publication
1	J. Date, A. Kannan, S. Swaminathan	An Introduction to Database Systems, Eighth Edition	Pearson Education	2006

CO PO PSO MAPPING

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PS01	PS02	PS03
CO1	2	2	3	2	1	-	-	-	2	1	1	1	2	1	3
CO2	3	1	1	1	1	-	-	-	2	3	3	3	3	1	2
CO3	3	2	3	2	1	-	-	-	2	1	1	2	2	3	3
CO4	1	2	3	2	-	-	-	-	3	2	3	3	1	2	3
CO5	1	1	3	3	2	-	-	-	1	3	3	1	2	2	2
AVG	2	2	3	2	1	-	-	-	2	2	2	2	2	2	3

R 2024	DEPARTMENT OF INFORMATION TECHNOLOGY					SEMESTER:IV
24IT401	COMPUTER ARCHITECTURE	L	T	P	C	PC
		3	0	0	3	

COURSE OBJECTIVES:

The main objectives of this course are to:

- To learn the basic structure and operations of a computer.
- To learn the arithmetic and logic unit and implementation of fixed-point and floating point arithmetic unit.
- To learn the basics of pipelined execution.
- To understand parallelism and multi-core processors.
- To understand the memory hierarchies, cache memories and virtual memories.
- To learn the different ways of communication with I/O devices.

COURSE OUTCOMES:

At the end of this course, students are able to:
CO1: Understand the basics structure of computers, operations and instructions.
CO2: Design arithmetic and logic unit.
CO3: Understand pipelined execution and design control unit.
CO4: Understand parallel processing architectures.
CO5: Understand the various memory systems and I/O communication

UNIT: I	BASIC STRUCTURE OF A COMPUTER SYSTEM	9

Functional Units - Basic Operational Concepts - Performance - Instructions: Language of the Computer - Operations, Operands - Instruction representation - Logical operations - decision making - MIPS Addressing.

UNIT: II	ARITHMETIC FOR COMPUTERS	9

Addition and Subtraction - Multiplication - Division - Floating Point Representation - Floating Point Operations - Sub word Parallelism

UNIT: III	PROCESSOR AND CONTROL UNIT	9

A Basic MIPS implementation – Building a Data path – Control Implementation Scheme – Pipelining – Pipelined data path and control – Handling Data Hazards & Control Hazards – Exceptions.

UNIT: IV	PARALLELISIM	9

Parallel processing challenges – Flynn's classification – SISD, MIMD, SIMD, SPMD, and Vector Architectures - Hardware multithreading – Multi-core processors and other Shared Memory Multiprocessors - Introduction to Graphics Processing Units, Clusters, Warehouse Scale Computers and other Message-Passing Multiprocessors.

UNIT: V	MEMORY & I/O SYSTEMS	9

Memory Hierarchy - memory technologies – cache memory – measuring and improving cache performance - virtual memory, TLB's - Accessing I/O Devices - Interrupts - Direct Memory Access - Bus structure – Bus operation - Arbitration - Interface circuits - USB.
Total Periods : 45

TEXT BOOKS:

Sl. No	Authors	Title of the Book	Publisher	Year of publication
1	David A. Patterson and John L. Hennessy	Computer Organization and Design: The Hardware/Software Interface, Fifth Edition, Morgan Kaufmann	Elsevier Publishers	2014
2	Carl Hamacher, Zvonko Vranesic, Safwat Zaky and Naraig Manjikian	Computer Organization and Embedded Systems	Sixth Edition, Tata McGraw Hill	2012

REFERENCE BOOKS:

Sl. No	Authors	Title of the Book	Publisher	Year of publication
1	William Stallings	Computer Organization and Architecture – Designing for Performance, Eighth Edition	Pearson Education,	2010
2	John P. Hayes	Computer Architecture and Organization, Third Edition	Tata McGraw Hill,	2012
3	John L. Hennessey and David A. Patterson	Computer Architecture - A Quantitative Approach‖, Morgan Kaufmann	Elsevier Publishers, Fifth Edition	2012

CO PO PSO MAPPING

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PS01	PS02	PS03
CO1	3	2	1	-	-	-	-	-	-	-	-	1	-	-	-
CO2	3	3	2	1	-	-	-	-	-	-	-	-	-	-	-
CO3	3	2	3	2	2	-	-	-	-	-	-	-	-	-	-
CO4	2	3	3	3	2	-	-	-	-	-	-	-	-	-	-
CO5	3	3	2	2	2	-	-	-	-	-	-	2	-	-	-
AVG	2.8	2.6	2.2	1.6	1.2	-	-	-	-	-	-	0.6	-	-	-

<table>
<tr><td>R 2024</td><td colspan="5">CHEMISTRY</td><td>IV</td></tr>
<tr><td rowspan="2">24MC401</td><td rowspan="2">ENVIRONMENTAL SCIENCES</td><td>L</td><td>T</td><td>P</td><td>C</td><td rowspan="2">BS</td></tr>
<tr><td>2</td><td>0</td><td>0</td><td>0</td></tr>
<tr><td colspan="7">COMMON TO ALL DEPARTMENTS</td></tr>
</table>

COURSE OBJECTIVES:

The objectives of learning this course are:

- To introduce the basic concepts of environment and ecosystems
- To introduce the basic concepts biodiversity and emphasize on the biodiversity of India and its conservation
- To impart knowledge on the causes, effects and control or prevention measures of environmental pollution and natural disasters.
- To facilitate the understanding of global and Indian scenario of renewable and non-renewable resources, causes of their degradation and measures to preserve them.
- To familiarize the concept of sustainable development goals and appreciate the interdependence of economic and social aspects of sustainability, recognize and analyze climate changes, concept of carbon credit and the challenges of environmental management.

COURSE OUTCOMES:

At the end of this course, students are able to:

CO1: To recognize and understand the functions of environment and ecosystem.

CO2: To develop an understanding of the roles and functions of the biodiversity and to emphasize the need for their conservation.

CO3: To identify the causes, effects of environmental pollution and natural disasters and contribute to the preventive measures in the society.

CO4: To identify and apply the understanding of renewable and non-renewable resources and contribute to the sustainable measures to preserve them for future generations

CO5: To recognize the different goals of sustainable development and apply them for suitable technological advancement and societal development.

UNIT: I	**ENVIRONMENT AND ECOSYSYTEM**	**6**

Definition, scope and importance of environment – need for public awareness - concept of an ecosystem – structure and function of an ecosystem – energy flow in the ecosystem – ecological succession - food chains, food webs and ecological pyramids - Introduction, types, characteristic features, structure and function of the (a) Terrestrial ecosystem (b) Aquatic ecosystem

UNIT: II	**BIODIVERSITY**	**6**

Definition, importance of bio-diversity-types of biodiversity: genetic, species and ecosystem diversity-values of biodiversity, India as a mega-diversity nation- hotspots of biodiversity-threats to biodiversity – endangered and endemic species of India – conservation of biodiversity: In-situ and ex-situ.

UNIT: III	**ENVIRONMENTALPOLLUTION**	**6**

Causes, Effects and Preventive measures of Water, Soil, Air, Marine and Noise Pollution. Solid, Biological, Hazardous and E-Waste management. Case studies on Occupational Health and Safety Management system (OHASMS). Environmental protection, Environment (Protection) Amendment Rules, 2024

UNIT: IV	**RENEWABLE SOURCES OF ENERGY**	**6**

Energy conservation, New Energy Sources: Need of new sources. Different types new energy sources. Wind Energy, Ocean energy resources, Tidal energy conversion, Concept, origin and power plants of geothermal energy.

UNIT: V	SUSTAINABILITY AND MANAGEMENT			
Sustainability- concept, need for sustainability- from unsustainability to sustainability-millennium development goals - Sustainable Development Goals, Climate change - environmental issues and possible solutions-case studies on climate change.				
Tools Required:	Board & Chalk, PPT, NPTEL Videos, Role play			
				Total Periods :30
TEXT BOOKS:				
Sl.No	Authors	Title of the Book	Publisher	Year of publication
1	Anubha Kaushik and C. P. Kaushik	Perspectives in Environmental Studies	6th Edition, New Age International Publishers	2018
2	Benny Joseph	Environmental Science And Engineering	Tata Mc Graw-Hill, NewDelhi,	2016
3	Gilbert M. Masters	Introduction to Environmental Engineering and Science	2nd edition, Pearson Education	2004
4	Allen, D. T. and Shonnard, D. R.	Sustainability Engineering: Concepts, Design and Case Studies	Prentice Hall	2012
5	Bradley. A.S; Adebayo, A.O., Maria, P.	Engineering applications in sustainable design and development	Cengage learning	2022
6		Environment Impact Assessment Guidelines	Notification of Government of India	2006
REFERENCE BOOKS:				
1	Rajagopalan, R	Environmental Studies-From Crisis to Cure'	Prentice hall of India PVT. LTD, New Delhi	2015
2	Erach Bharucha	Textbook of Environmental Studies for Undergraduate Courses	Orient Blackswan Pvt. Ltd.	2013
WEB LEARNING RESOURCES:				
1	https://archive.nptel.ac.in/courses/127/106/127106004/			
2	https://archive.nptel.ac.in/courses/127/105/127105018/			
3	https://www.youtube.com/watch?v=705DHheuG6w&t=16s			
4	https://www.youtube.com/watch?v=CYJECfoP-3E&t=863s			
5	https://www.youtube.com/watch?v=t7Q7y_xjR5E&t=14s			

CO PO PSO MAPPING:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2	PSO3
CO1	2	1	-	-	-	2	3	-	-	-	-	2	-	-
CO2	3	2	-	-	-	3	3	-	-	-	-	2	-	-
CO3	3	-	1	-	-	2	2	-	-	-	-	2	-	-
CO4	3	2	1	1	-	2	2	-	-	-	-	2	-	-
CO5	3	2	1	-	-	2	2	-	-	-	-	1	-	-
Avg	3	2	1	1	-	2	2	-	-	-	-	2	-	-

1-Low, 2-Medium,3-High

R 2024	DEPARTMENT OF ARTIFICIAL INTELLIGENCE AND DATA SCIENCE					SEMESTER:IV
24AD403	OPERATING SYSTEMS	L	T	P	C	PC
		3	0	2	4	

COURSE OBJECTIVES:

The main objectives of this course are to:

- To understand the basics and functions of operating systems.
- To understand Processes and Threads
- To analyze Scheduling algorithms and process synchronization.
- To understand the concept of Deadlocks.
- To analyze various memory management schemes.
- To be familiar with I/O management and File systems.
- To be familiar with the basics of virtual machines and Mobile OS like iOS and Android.

COURSE OUTCOMES:

At the end of this course, students are able to:
CO1: Analyze various scheduling algorithms and process synchronization.
CO2 : Explain deadlock, prevention and avoidance algorithms.
CO3 : Compare and contrast various memory management schemes.
CO4 : Explain the functionality of file systems I/O systems, and Virtualization
CO5 : Compare iOS and Android Operating Systems.

UNIT: I	INTRODUCTION	7

Computer System - Elements and organization; Operating System Overview - Objectives and Functions - Evolution of Operating System; Operating System Structures – Operating System Services - User Operating System Interface - System Calls – System Programs - Design and Implementation - Structuring methods.

UNIT: II	PROCESS MANAGEMENT	11

Processes - Process Concept - Process Scheduling - Operations on Processes - Inter-process Communication; CPU Scheduling - Scheduling criteria - Scheduling algorithms: Threads - Multithread Models – Threading issues; Process Synchronization - The critical-section problem - Synchronization hardware – Semaphores – Mutex - Classical problems of synchronization - Monitors; Deadlock - Methods for handling deadlocks, Deadlock prevention, Deadlock avoidance, Deadlock detection, Recovery from deadlock.

UNIT: III	MEMORY MANAGEMENT	10

Main Memory - Swapping - Contiguous Memory Allocation – Paging - Structure of the Page Table - Segmentation, Segmentation with paging; Virtual Memory - Demand Paging – Copy on Write - Page Replacement - Allocation of Frames –Thrashing.

UNIT: IV	STORAGE MANAGEMENT	10

Mass Storage system – Disk Structure - Disk Scheduling and Management; File-System Interface - File concept - Access methods - Directory Structure - Directory organization - File system mounting - File Sharing and Protection; File System Implementation - File System Structure - Directory implementation - Allocation Methods - Free Space Management; I/O Systems – I/O Hardware, Application I/O interface, Kernel I/O subsystem.

UNIT: V	VIRTUAL MACHINES AND MOBILE OS	7

Virtual Machines – History, Benefits and Features, Building Blocks, Types of Virtual Machines and their Implementations, Virtualization and Operating-System Components; Mobile OS - iOS and Android.

Total Periods : 45

PRACTICAL EXERCISES:

1. Installation of Operating system : Windows/ Linux
2. Illustrate UNIX commands and Shell Programming
3. Process Management using System Calls : Fork, Exec, Getpid, Exit, Wait, Close
4. Write C programs to implement the various CPU Scheduling Algorithms
5. Illustrate the inter process communication strategy
6. Implement mutual exclusion by Semaphores
7. Write a C program to avoid Deadlock using Banker's Algorithm
8. Write a C program to Implement Deadlock Detection Algorithm
9. Write C program to implement Threading
10. Implement the paging Technique using C program
1. Write C programs to implement the following Memory Allocation Methods

a. First Fit b. Worst Fit c. Best Fit

2. Write C programs to implement the various Page Replacement Algorithms
3. Write C programs to Implement the various File Organization Techniques
4. Implement the following File Allocation Strategies using C programs

a. Sequential b. Indexed c. Linked

5. Write C programs for the implementation of various disk scheduling algorithms

TEXT BOOKS:

Sl. No	Authors	Title of the Book	Publisher	Year of publication
1	Abraham Silberschatz, Peter Baer Galvin, Greg Gagne	Operating System Concepts, 9th Edition	John Wiley and Sons Inc.	2018
2	Andrew S. Tanenbaum	Modern Operating Systems, 4th Edition	Pearson	2016

REFERENCE BOOKS:

Sl. No	Authors	Title of the Book	Publisher	Year of publication
1	Ramaz Elmasri, A. Gil Carrick, David Levine	Operating Systems â€“ A Spiral Approach	Tata McGraw Hill Edition	2010
2	William Stallings	Operating Systems: Internals and Design Principles, 7th Edition	Prentice Hall	2018
3	Achyut S. Godbole, Atul Kahate	Operating Systems	McGraw Hill Education	2016

CO PO PSO MAPPING

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PS01	PS02	PS03
CO1	3	1	1	1	-	-	-	-	1	1	1	2	2	1	2
CO2	2	3	1	3	1	-	-	-	3	2	2	3	3	3	1
CO3	2	2	3	3	2	-	-	-	3	1	1	2	1	1	1
CO4	2	2	1	2	1	-	-	-	1	3	2	1	1	1	2
CO5	2	3	3	2	1	-	-	-	3	1	2	1	3	1	2
AVG	2	2	2	2	1	-	-	-	2	2	2	2	2	1	2

R2024	DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING					SEMESTER:IV
24CS405	JAVA PROGRAMMING	L	T	P	C	PC
		3	0	2	4	

COURSE OBJECTIVES:

- To Understand the fundamental concepts of Java, including JVM, program structure, classes, objects, constructors, and string handling.
- To Develop modular Java applications using packages, interfaces, and apply multithreading to build concurrent programs.
- To Gain the ability to design dynamic web applications using Java Server Pages (JSP) and understand their architecture, lifecycle, and components.
- To Learn to create desktop applications using JavaFX, incorporating UI controls, layouts, event handling, multimedia, and interactive elements.
- To Acquire skills to interact with relational databases using JDBC for performing CRUD operations and storing/retrieving files and images.

COURSE OUTCOMES:

At the end of this course, students are able to:
CO1: Apply object-oriented programming concepts to create Java programs using classes, objects, constructors, and strings.
CO2: Develop Java applications using packages, interfaces, and manage parallel execution using multithreading techniques.
CO3: Build and deploy JSP-based web applications capable of processing user inputs, interacting with databases, and dynamically generating web content.
CO4: Design and implement GUI applications using JavaFX components such as controls, panes, shapes, media elements, and event-driven programming.
CO5: Connect Java applications with databases using JDBC to perform insertion, retrieval, update, deletion, and manage file and image storage operations.

UNIT: I	INTRODUCTION TO JAVA BASIC CONCEPTS	9

Introduction, Java features, Java Virtual Machine, Java Program Structure, Command Line Arguments, Type Casting. Closer look at Methods and classes: Defining a Class, Creating Objects, Accessing class members, Constructors, Garbage Collection, String Handling.

UNIT: II	PACKAGES, INTERFACES & MULTI-THREADING PACKAGES	9

Built-in Packages, Creating User Defined Packages, Accessing a Package, Using a Package. Interfaces - Defining Interfaces, Extending Interfaces, Implementing Interfaces, Accessing Interface Variables.
Threading: Introduction, Thread creation, Extending the thread class, Life cycle, Methods and Priorities, Implementing thread using runnable interface.

UNIT: III	JAVA SERVER PAGES	9

JSP: Introduction, Architecture of JSP, Life Cycle of JSP, Scripting elements, Directive Elements, JSP Actions, Implicit objects, including HTML in JSP

UNIT: IV	JAVAFX BASICS, UI CONTROLS AND MULTIMEDIA	9

Basic structure of JAVAFX program, Panes, UI control and shapes, Property binding, the Color and the Font class, the Image and Image-View class, layout panes and shapes,Labeled and Label, button, Checkbox, RadioButton, Textfield, TextArea, Combo Box, ListView, Scrollbar, Slider, Video and Audio.

UNIT: V	INTERACTING WITH DATABASE	9

Introduction to JDBC, Essential JDBC classes, Connecting to database, Inserting data in database, Retrieving data from database, deleting data in database, updating data in database, store image in the database, to retrieve image from database, to store file in database, retrieve file from database

Periods : 45

PRACTICAL EXERCISES

1. Write a Java Program to demonstrate constructors.
2. Write a Java program to practice using String class and its methods.
3. Write a java program to demonstrate Packages.
4. Write a program to demonstrate use of implementing interfaces.
5. Write a java program to demonstrate threads.
6. Write a java to implementing thread using runnable interface.
7. Install Tomcat Server. Create a table which should contain at least the following fields: name, password, emailid, phone number Write a JSP to connect to that database and extract data from the tables and display them. Insert the details of the users who register with the web site, whenever a new user clicks the submit button in the registration page.
8. Write a JSP which does the following job: Insert the details of the 3 or 4 users who register with the web site by using registration form. Authenticate the user when he submits the login form using the user name and password from the database and display the message “successfully logged in”.
9. Write a JSP Program store and retrieve images and file from the database.
10. Write a GUI program that use button to move the message to the left and right and use the radio button to change the color for the message displayed.
11. Write a program that displays the color of a circle as red when the mouse button is pressed and as blue when the mouse button is released

Periods : 30

Total Periods : 75

TEXT BOOKS:

Sl. No	Authors	Title of the Book	Publisher	Year of publication
1	Herbert Schildt	Java: The Complete Reference, Seventh Edition	Tata McGraw Hill	2007
2	E.Balagurusamy	Programming with JAVA – A Primer, Fifth Edition	Tata McGraw Hill	2013
3	James Edward Keogh	J2EE: The Complete Reference	Tata McGraw Hill	2002
4	Maydene Fisher, Jon Ellis, Jonathan Bruce	JDBC, API Tutorial and Reference, Third Edition	Addison Wesley	2003

REFERENCE BOOKS:

Sl. No	Authors	Title of the Book	Publisher	Year of publication
1	Cay S.Horstmann, Gray Cornell	Core Java, Vol I – Fundamentals, Ninth Edition	Prentice Hall	2013
2	Sachin Malhotra & Saurabh Choudhary	Programming in JAVA, Second Edition	Oxford	2014
3	Bryan Basham, Kathy Sierra, Bert Bates	Headfirst Servlets & JSP, 2nd Edition.	O-reilly	2008

CO PO PSO MAPPING

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PS01	PS02	PS03
CO1	2	2	1	2	2	-	-	-	-	1	1	2	2	2	1
CO2	2	3	2	3	2	-	-	-	2	2	3	2	3	2	1
CO3	2	3	2	1	1	-	-	-	2	2	3	2	2	3	1
CO4	2	3	1	2	2	-	-	-	2	2	3	2	2	3	1
CO5	2	3	1	2	2	-	-	--	-	-	-	1	3	2	2
AVG	2	2	1	2	2	-	-	-	-	2	3	2	2	2	1

R 2024	ELECTRONICS AND COMMUNICATION ENGINEERING					SEMESTER: 04
24ECE18	MICROPROCESSOR AND MICROCONTROLLER LABORATORY	L	T	P	C	PC
		0	0	4	2	

COURSE OBJECTIVES:

The objectives of learning this course are:

- To perform simple arithmetic operations using assembly language program and study the addressing modes & instruction set of 8085 & 8051
- To develop skills in simple program writing in assembly languages
- To write an assembly language program to convert Analog input to Digital output and Digital input to Analog output.
- To perform interfacing experiments with µP808
- To perform interfacing experiments with µC8051.

COURSE OUTCOMES:

After completion of the course, the student will be able

CO1 : To write assembly language program for microprocessor

CO2 : To write assembly language program for microcontroller

CO3 : To design and implement interfacing of peripheral with microprocessor and microcontroller

CO4 : To analyze, comprehend, design and simulate microprocessor based systems used for control and monitoring

CO5 : To analyze, comprehend, design and simulate microcontroller based systems used for control and monitoring

LIST OF EXPERIMENTS

PROGRAMMING EXERCISES / EXPERIMENTS WITH µP8085:

1. Simple arithmetic operations: Multi precision addition / subtraction /multiplication / division.

2. Programming with control instructions: Increment / Decrement, Ascending / Descending order, Maximum / Minimum of numbers, Rotate instructions, Hex / ASCII / BCD code conversions.

3. Interface Experiments: A/D Interfacing. D/A Interfacing. Traffic light controller

4. Stepper motor controller interface.

5. Displaying a moving/ rolling message in the student trainer kit's output device.

PROGRAMMING EXERCISES / EXPERIMENTS WITH µC8051:

6. Simple arithmetic operations with 8051: Multi precision addition / subtraction / multiplication/ division.

7. Programming with control instructions: Increment / Decrement, Ascending / Descending. order, Maximum / Minimum of numbers, Rotate instructions, Hex / ASCII / BCD code conversions.

8. Interface Experiments: A/D Interfacing. D/A Interfacing. Traffic light controller

9. Stepper motor controller interface.

10. Displaying a moving/ rolling message in the student trainer kit's output device.

11. Programming PIC architecture with software tools

Total Hours : 60

CO-PO- PSO MAPPING:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	2	3	3	3	-	-	-	-	-	-	-	-	-	-	-
CO2	2	3	3	3	-	-	-	-	-	-	-	-	-	-	-
CO3	2	3	3	3	-	-	-	-	-	-	-	-	-	-	-
CO4	2	3	3	3	-	-	-	-	-	-	-	-	-	-	-
CO5	2	3	3	3	-	-	-	-	-	-	-	-	-	-	-

1 - low, 2 - medium, 3 - high, '-' - no correlation

R 2024	DEPARTMENT OF ARTIFICIAL INTELLIGENCE AND DATA SCIENCE					SEMESTER:IV
24CS407	DATABASE MANAGEMENT SYSTEMS LAB	L	T	P	C	PC
		0	0	4	2	

COURSE OBJECTIVES:

The main objectives of this course are to:

•To understand the database development life cycle

•To learn database design using conceptual modeling, Normalization

•To implement database using Data definition, Querying using SQL manipulation and SQL programming

•To implement database applications using IDE/RAD tools

•To learn querying Object-relational databases

COURSE OUTCOMES:

At the end of this course, students are able to:

CO1: Understand the database development life cycle

CO2: Design relational database using conceptual-to-relational mapping, Normalization

CO3: Apply SQL for creation, manipulation and retrieval of data

CO4: Develop a database applications for real-time problems

CO5: Design and query object-relational databases

LIST OF EXPERIMENTS:

1. Database Development Life cycle:

Problem definition and Requirement analysis Scope and Constraints

2. Database design using Conceptual modeling (ER-EER) – top-down approach Mapping conceptual to relational database and validate using Normalization

3. Implement the database using SQL Data definition with constraints, Views

4. Query the database using SQL Manipulation

5. Querying/Managing the database using SQL Programming

-Stored Procedures/Functions

-Constraints and security using Triggers

6. Database design using Normalization – bottom-up approach

7. Develop database applications using IDE/RAD tools (Eg., NetBeans,VisualStudio)

8. Database design using EER-to-ODB mapping / UML class diagrams

9. Object features of SQL-UDTs and sub-types, Tables using UDTs, Inheritance, Method definition

10. Querying the Object-relational database using Objet Query language

Total Periods : 45

CO PO PSO MAPPING

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PS01	PS02	PS03
CO1	3	1	3	3	-	-	-	-	1	1	1	3	2	2	1
CO2	2	2	1	3	1	-	-	-	3	2	3	1	1	1	2
CO3	2	1	3	1	-	-	-	-	3	3	1	1	2	1	1
CO4	2	2	3	1	-	-	-	-	2	3	2	1	2	1	2
CO5	3	3	1	3	1	-	-	-	1	3	2	3	3	3	2

AVG	2	2	2	2	1	-	-	-	2	2	2	2	2	2	2

R 2024	AI&DS					SEMESTER: V
24AD501	INTERNET OF THINGS	L	T	P	C	PC
		3	0	0	3	
Common to CSE, AIDS and IT						

COURSE OBJECTIVES:

- To learn the internal architecture and programming of an IOT.
- To introduce interfacing I/O devices to the processor.
- To introduce the evolution of the Internet of Things (IoT).
- To build a small low-cost embedded and IoT system using Arduino/Raspberry Pi/ open platform.
- To apply the concept of Internet of Things in real world scenario.

COURSE OUTCOMES:

CO1: learn the basics of IOT.
CO2: Analyse basic protocols of wireless and MAC.
CO3: Design simple embedded applications.
CO4: Compare the communication models in IOT
CO5: Design IoT applications using Arduino/Raspberry Pi /open platform.

UNIT: I	INTRODUCTION TO IOT	9

Introduction to IoT, Characteristics of IoT, Physical design of IoT, Logical design of IoT, Functional blocks of IoT, Communication models & APIs ,IoT & M2M Machine to Machine, Difference between IoT and M2M, Software define Network, Challenges in IoT(Design ,Development, Security)

UNIT: II	NETWORK AND COMMUNICATION ASPECTS:	9

Wireless medium access issues, MAC protocol survey, Survey routing protocols, Sensor deployment & Node discovery, Data aggregation & dissemination.

UNIT: III	WEB OF THINGS	9

Web of Things vs Internet of things, two pillars of web, Architecture and standardization of IoT, Unified multitier-WoT architecture, WoT portals and Business intelligence, Cloud of things: Grid/SOA and cloud computing, Cloud middleware, cloud standards

UNIT: IV	IOT COMMUNICATION AND OPEN PLATFORMS	9

IoT Communication Models and APIs – IoT Communication Protocols – Bluetooth – WiFi – ZigBee – GPS – GSM modules – Open Platform (like Raspberry Pi) – Architecture – Programming – Interfacing – Accessing GPIO Pins – Sending and Receiving Signals Using GPIO Pins – Connecting to the Cloud.

UNIT: V	APPLICATIONS DEVELOPMENT	9

Complete Design of Embedded Systems – Development of IoT Applications – Home Automation – Smart Agriculture – Smart Cities – Smart Healthcare.

Pedagogical Tools	Black board, chalk, Group Discussion, Role Play, Youtube Videos, Nptel videos.

Total Periods :45

TEXT BOOKS:

Sl.No	Authors	Title of the Book	Publisher	Year of publication
1	Muhammed Ali Mazidi, Janice Gillispie Mazidi, Rolin D. McKinlay	Design an IOT based system	Pearson Education, Second Edition	2014
2	Robert Barton, Patrick Grossetete, David Hanes, Jerome Henry, Gonzalo Salgueiro	IoT Fundamentals: Networking Technologies, Protocols, and Use Cases for the Internet of Things	CISCO Press	2017

REFERENCE BOOKS:

Sl.No	Authors	Title of the Book	Publisher	Year of publication
1	Michael J. Pont	Embedded C	Pearson Education	2007
2	Wayne Wolf	Computers as Components: Principles of Embedded Computer System Design	Elsevier	2006
3	Andrew N Sloss, D. Symes, C.Wright	Arm System Developer's Guide	Morgan Kauffman/ Elsevier	2006
4	Arshdeep Bahga, Vijay Madisetti	Internet of Things – A hands-on approach	Universities Press	2015

CO – PO – PSO MAPPING

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	3	2	2	1	-	1	1	-	-	-	-	2	-	-	-
CO2	3	1	2	1	-			-	-	-	-	2	-	-	-
CO3	3	1	-	-	-	-	-	-	-	-	-	-	-	-	-
CO4	3	1	1	-	-	1	2	-	-	-	-	-	-	-	-
CO5	3	1	1	-	-	1	2	-	-	-	-	-	-	-	-

R 2024	AI&DS					SEMESTER:V
24AD501	COMPUTER NETWORKS	L	T	P	C	PC
		3	0	0	3	

COMMON TO AIDS&IT

COURSE OBJECTIVES:

- Understand the fundamentals of data communication, computer networks, network architectures, and protocol layering concepts.
- Gain knowledge of the OSI reference model and TCP/IP protocol suite and their roles in network communication.
- Analyze the functions and protocols of the Data Link, Network, and Transport layers for reliable data transmission.
- Understand addressing, routing, congestion control, flow control, and error control mechanisms used in computer networks.

Explore Application Layer protocols and services, including web, file transfer, email, domain name resolution, and network management protocols, for effective end-to-end communication.

COURSE OUTCOMES:

At the end of this course, the students will be able to:

- CO1: Explain the fundamentals of data communication, network architectures, and protocol layering concepts based on the OSI and TCP/IP models.
- CO2: Analyze the services and protocols of the Transport Layer, including TCP and UDP, congestion control, flow control, and reliable data transfer mechanisms.
- CO3: Evaluate the operation of the Network Layer, including IP addressing, subnetting, routing algorithms, and packet forwarding mechanisms.
- CO4: Analyze the functions of the Data Link Layer, including framing, error detection, flow control, and Medium Access Control (MAC) protocols.

CO5: Analyze Application Layer protocols such as HTTP, FTP, SMTP, POP3, IMAP, DNS, and SNMP and their role in providing network services and end-to-end communication.

UNIT: I	**INTRODUCTION AND APPLICATION LAYER**	**9**

Data Communication - Networks - Network Types - Protocol Layering - TCP/IP Protocol suite - OSI Model - Introduction to Sockets - Application Layer protocols: HTTP - FTP - Email protocols (SMTP - POP3 - IMAP - MIME) - DNS - SNMP

UNIT: II	**TRANSPORT LAYER**	**9**

Introduction - Transport-Layer Protocols: UDP - TCP: Connection Management - Flow control - Congestion Control - Congestion avoidance (DECbit, RED) - SCTP - Quality of Service

UNIT: III	**NETWORK LAYER**	**9**

Switching: Packet Switching - Internet protocol - IPV4 - IP Addressing - Subnetting - IPV6, ARP, RARP, ICMP, DHCP

UNIT: IV	**ROUTING**	**9**

Routing and protocols: Unicast routing - Distance Vector Routing - RIP - Link State Routing - OSPF -Path-vector routing - BGP - Multicast Routing: DVMRP - PIM.

UNIT: V	**DATA LINK AND PHYSICAL LAYERS**	**9**

Data Link Layer - Framing - Flow control - Error control - Data-Link Layer Protocols - HDLC - PPP - Media Access Control - Ethernet Basics - CSMA/CD - Virtual LAN - Wireless LAN (802.11)- Physical Layer: Data and Signals - Performance - Transmission media- Switching - Circuit Switching.

Total Periods :45

Pedagogical Tools:	Black board, chalk, group discussion, role play, youtube videos, NPTEL videos

TEXT BOOKS:

Sl. No	Authors	Title of the Book	Publisher	Year of publication
1	James F. Kurose, Keith W. Ross	Computer Networking: A Top-Down Approach Featuring the Internet (8th Ed.)	Pearson Education	2021
2	Behrouz A. Forouzan	Data Communications and Networking with TCP/IP Protocol Suite (6th Ed.)	TMH	2022

REFERENCE BOOKS:				
Sl. No	**Authors**	**Title of the Book**	**Publisher**	**Year of Publication**
1	Larry L. Peterson, Bruce S. Davie	Computer Networks: A Systems Approach (5th Ed.)	Morgan Kaufmann Publishers Inc.	2012
2	William Stallings	Data and Computer Communications (10th Ed.)	Pearson Education	2013
3	Nader F. Mir	Computer and Communication Networks (2nd Ed.)	Prentice Hall	2014
4	Ying-Dar Lin, Ren-Hung Hwang, Fred Baker	Computer Networks: An Open Source Approach	McGraw Hill	2012

CO – PO – PSO MAPPING															
	PO1	**PO2**	**PO3**	**PO4**	**PO5**	**PO6**	**PO7**	**PO8**	**PO9**	**PO10**	**PO11**	**PO12**	**PSO1**	**PSO2**	**PSO3**
CO1	3	2	–	–	–	–	–	–	–	–	–	1	3	2	1
CO2	3	3	–	–	1	–	–	–	–	–	–	1	3	2	2
CO3	3	3	–	2	1	–	–	–	–	–	–	2	3	3	3
CO4	3	2	–	–	2	–	–	–	–	–	–	1	3	3	2
CO5	3	3	–	2	2	–	–	–	–	–	–	2	3	3	3

R 2024						SEMESTER: V
24AD502	BIG DATA ANALYTICS	L	T	P	C	PC
		3	0	2	4	

COMMON TO AIDS & IT

COURSE OBJECTIVES:

The objectives of learning this course are to:

- Understand the fundamentals of Big Data and its applications
- Learn MapReduce analytics using Hadoop and related tools
- Master Big Data stream processing algorithms
- Apply Big Data analytics to real-world case studies

COURSE OUTCOMES:

At the end of this course, students are able to:

CO1: Describe Big Data characteristics, technologies, and business applications
CO2: Install, configure, and run Hadoop and HDFS
CO3: Perform MapReduce analytics using Hadoop
CO4: Implement Big Data stream processing algorithms
CO5: Apply Big Data analytics to real-world problems in finance, healthcare, and social media

UNIT: I	INTRODUCTION TO BIG DATA	9

Introduction to Big Data – 5 V's (Volume, Velocity, Variety, Veracity, Value) – Big Data characteristics – convergence of key trends – unstructured data – industry examples – web analytics – Big Data applications in business, healthcare, social media – Big Data technologies overview – Cloud computing and Big Data – Mobile business intelligence – Data-driven decision making.

UNIT: II	HADOOP FUNDAMENTALS	9

Introduction to Hadoop – Hadoop ecosystem – data formats – analyzing data with Hadoop – scaling out – Hadoop streaming – design of Hadoop Distributed File System (HDFS) – HDFS concepts – Java interface – data flow – Hadoop I/O – data integrity – compression – serialization – Avro – file-based data structures – HDFS architecture – NameNode, DataNode, Secondary NameNode.

UNIT: III	MAPREDUCE AND HADOOP TOOLS	9

MapReduce paradigm – MapReduce workflows – unit tests with MRUnit – anatomy of MapReduce job run – Classic MapReduce vs YARN – job scheduling – shuffle and sort – task execution – MapReduce types – input formats – output formats. HBase – data model and implementations – HBase clients – HBase examples. Pig – Pig Latin – data model – developing and testing scripts. Hive – data types – file formats – HiveQL data definition and manipulation.

UNIT: IV	BIG DATA STREAM PROCESSING ALGORITHMS	9

Introduction to data streams – stream data model – sampling data in a stream – Bloom Filters – applications and implementation. Flajolet-Martin (FM) algorithm for counting distinct elements – Alon-Matias-Szegedy (AMS) algorithm for frequency moments. HyperLogLog algorithm – Count-Min Sketch algorithm. Reservoir sampling – sliding window techniques. Real-time stream processing with Apache Kafka – Apache Spark Streaming – Apache Flink fundamentals.

UNIT: V	REAL-TIME BIG DATA ANALYTICS CASE STUDIES	9

Financial Analytics: Stock market analysis – high-frequency trading analytics – fraud detection in banking – credit risk assessment – algorithmic trading using Big Data.

Healthcare Analytics: Patient health monitoring – disease outbreak prediction – genomics data analysis – personalized medicine.

Social Media Analytics: Twitter sentiment analysis – Facebook user behavior analysis – recommendation systems – influencer identification.

E-commerce Analytics: Customer segmentation – product recommendation – dynamic pricing – supply chain optimization.

IoT and Smart Cities: Smart traffic management – energy consumption optimization – predictive maintenance.

Total Periods: 45

Pedagogical Tools:	Blackboard, PowerPoint presentations, hands-on lab sessions, case study discussions, YouTube videos, NPTEL videos, Kaggle datasets, real-time Big Data projects

TEXT BOOKS:

Sl. No	Authors	Title of the Book	Publisher	Year
1	Tom White	Hadoop: The Definitive Guide, 4th Edition	O'Reilly	2015
2	Jure Leskovec, Anand Rajaraman, Jeffrey D. Ullman	Mining of Massive Datasets, 3rd Edition	Cambridge University Press	2020
3	Holden Karau, Andy Konwinski, Patrick Wendell, Matei Zaharia	Learning Spark: Lightning-Fast Data Analytics, 2nd Edition	O'Reilly	2020

REFERENCE BOOKS:

Sl. No	Authors	Title of the Book	Publisher	Year
1	Bill Chambers, Matei Zaharia	Spark: The Definitive Guide	O'Reilly	2018
2	Nathan Marz, James Warren	Big Data: Principles and Best Practices of Scalable Real-Time Data Systems	Manning	2015
3	Prajapati Vihang	Big Data Analytics with Java	Packt Publishing	2017
4	Rajiv Lal, Sameer Sachdev	Mastering Apache Kafka	Packt Publishing	2019

LIST OF EXPERIMENTS:

Total Periods: 30

1. Downloading and installing Hadoop; Understanding different Hadoop modes (Standalone, Pseudo-distributed, Fully-distributed). Startup scripts and configuration files.

2. Hadoop implementation of file management tasks: Adding files and directories, retrieving files, and deleting files using HDFS commands.

3. Implementation of Matrix Multiplication using Hadoop MapReduce.

4. Run a basic Word Count MapReduce program to understand the MapReduce paradigm.

5. Installation of Hive along with practice examples for data warehousing operations.

6. Installation of HBase with practice examples for NoSQL database operations.

7. Implementation of Bloom Filter algorithm for membership testing in Big Data.

8. Implementation of Flajolet-Martin algorithm for counting distinct elements in data streams.

9. Stock market data analysis using Hadoop: Price trend analysis, moving averages, and volatility calculation.

10. Twitter sentiment analysis: Collecting tweets, preprocessing, and analyzing sentiment using MapReduce.

Software Requirements: Hadoop 3.x, Java 8 or higher, Pig, Hive, HBase, Apache Kafka, Apache Spark

CO PO MAPPING

COs / POs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	2	–	–	1	–	–	–	–	–	–	1	2	1
CO2	2	2	1	2	3	–	–	–	–	–	–	1	3	2
CO3	2	3	2	2	3	–	–	–	–	–	–	1	3	3
CO4	2	3	3	2	3	–	–	–	–	–	–	1	3	3
CO5	3	3	2	2	2	1	–	–	1	1	1	1	3	3

R 2024	AIDS					SEMESTER: V
24AD504	**IOT Laboratory**	**L**	**T**	**P**	**C**	**PC**
		0	**0**	**4**	**2**	

COURSE OBJECTIVES:

The objectives of learning this course are:

- Understand microcontroller architecture and practice assembly and embedded C programming.
- Develop programs for data transfer, arithmetic, and logical operations.
- Gain hands-on experience with embedded platforms such as Arduino and Raspberry Pi.
- Interface sensors and communication modules for real-time applications.
- Design and implement IoT systems with cloud data logging.

COURSE OUTCOMES:

At the end of this course, students can be able to

CO1: Develop and test 8051 assembly programs using a simulator.
CO2: Implement data transfer and ALU operations in embedded platforms.
CO3: Build applications using Embedded C on Arduino.
CO4: Interface Raspberry Pi with sensors and establish device communication.
CO5: Design IoT systems with wireless communication and cloud integration.

PRACTICAL EXERCISES:

1. Write 8051 Assembly Language experiments using simulator.
2. Test data transfer between registers and memory.
3. Perform ALU operations.
4. Write Basic and arithmetic Programs Using Embedded C.
5. Introduction to Arduino platform and programming
6. Explore different communication methods with IoT devices (Zigbee, GSM, Bluetooth)
7. Introduction to Raspberry PI platform and python programming
8. Interfacing sensors with Raspberry PI
9. Communicate between Arduino and Raspberry PI using any wireless medium
10. Setup a cloud platform to log the data
11. Log Data using Raspberry PI and upload to the cloud platform
12. Design an IOT based system

Total Periods: 60

List of Components Required:

Hardware	Software
8051 Microcontroller kit	8051 Simulator (Keil / Proteus)
Arduino boards (UNO / Nano)	Arduino IDE
Raspberry Pi with SD card & power adapter	Python
Sensor modules (Temperature, Humidity, Gas, etc.)	IoT cloud platform (ThingSpeak / Blynk / AWS etc.)
Communication modules (Bluetooth, Zigbee, GSM, Wi-Fi)	
8051 Microcontroller kit	

CO – PO – PSO MAPPING

CO	PO												PSO		
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO 1	PSO 2	PSO 3
CO1	2	2	–	1	3	–	–	–	1	–	–	1	3	2	1
CO2	3	2	–	1	3	–	–	–	1	–	–	1	3	2	1
CO3	3	2	–	2	3	–	–	–	2	–	–	2	3	3	2
CO4	3	3	–	2	3	–	–	–	2	–	–	2	3	3	2
CO5	3	3	2	2	3	–	–	–	3	1	–	3	3	3	3

R 2024	AIDS					SEMESTER: V
24AD503	**COMPUTER NETWORKS LAB**	L	T	P	C	PC
		0	0	4	2	

COURSE OBJECTIVES:

The objectives of learning this course are:
1. To learn the basics of computer networks.
2. To understand how computers communicate with each other.
3. To use basic networking commands and tools.
4. To study network protocols and routing.
5. To analyze network performance and errors.

COURSE OUTCOMES:

At the end of this course, students can be able to
CO1: Understand basic networking concepts.
CO2: Use networking commands and tools.
CO3: Create simple client-server programs.
CO4: Analyze network packets and protocols.
CO5: Test and evaluate network performance.

PRACTICAL EXERCISES:

1. Learn to use commands like tcpdump, netstat, ifconfig, nslookup and traceroute. Capture ping and trace route PDUs using a network protocol analyzer and examine.
2. Write a HTTP web client program to download a web page using TCP sockets.
3. Applications using TCP sockets like:
4. Echo client and echo server
5. Chat
6. Simulation of DNS using UDP sockets.
7. Use a tool like Wireshark to capture packets and examine the packets
8. Write a code simulating ARP /RARP protocols.
9. Study of Network simulator (NS) and Simulation of Congestion Control Algorithms using NS.
10. Study of TCP/UDP performance using Simulation tool.
11. Simulation of Distance Vector/ Link State Routing algorithm.

Total Periods: 60

List of Components Required:

Desktop / Laptop Computer	1 Nos
Android Smartphone (Testing Device)	1 Each
Server System	1 Nos
Router	1 Nos
Network Switch	1 Nos
LAN Cable	1 Nos
Network Interface Card (NIC)	1 Nos
Wireshark Software	1 Nos
Cisco Packet Tracer :	1 Nos
Router	1 Nos

CO – PO – PSO MAPPING

CO	PO												PSO		
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO 1	PSO 2	PSO 3
CO1	3	3	2	1	-	-	1	-	2	-	-	1	3	2	3
CO2	3	3	2	1	-	-	1	-	2	-	-	1	3	2	3
CO3	3	3	2	1	-	1	1	-	2	-	-	1	3	2	3
CO4	3	3	2	1	-	1	1	-	2	-	-	1	3	2	3
CO5	3	3	2	1	-	1	1	-	2	-	-	1	3	2	3

R 2024	COMPUTER SCIENCE AND ENGINEERING					SEMESTER: VI
24HS101	PRINCIPLES OF MANAGEMENT	L	T	P	C	HS
		3	0	0	3	

COMMON TO ALL DEPARTMENTS

COURSE OBJECTIVES:

- Sketch the Evolution of Management.
- Extract the functions and principles of management.
- Learn the application of the principles in an organization.
- Study the various HR related activities.
- Analyze the position of self and company goals towards business.

COURSE OUTCOMES:

CO1: Upon completion of the course, students will be able to have clear understanding of managerial functions like planning, organizing, staffing, leading & controlling.

CO2: Have same basic knowledge on international aspect of management.

CO3: Ability to understand management concept of organizing.

CO4: Ability to understand management concept of directing.

CO5: Ability to understand management concept of controlling.

UNIT: I	INTRODUCTION TO MANAGEMENT AND ORGANIZATIONS	9

Definition of Management – Science or Art – Manager Vs Entrepreneur- types of managers-managerial roles and skills – Evolution of Management –Scientific, human relations, system and contingency approaches– Types of Business organization- Sole proprietorship, partnership, company-public and private sector enterprises.

UNIT: II	PLANNING	9

Nature and purpose of planning – Planning process – Types of planning – Objectives – Setting objectives – Policies – Planning premises – Strategic Management – Planning Tools and Techniques – Decision making steps and process

UNIT: III	ORGANISING	9

Nature and purpose – Formal and informal organization – Organization chart – Organization structure – Types – Line and staff authority – Job Design - Human Resource Management – HR Planning, Recruitment, selection, Training and Development

UNIT: IV	DIRECTING	9

Foundations of individual and group behaviour– Motivation – Motivation theories – Motivational techniques – Job satisfaction – Job enrichment – Leadership – types and theories of leadership

UNIT: V	CONTROLLING	9

System and process of controlling – Budgetary and non - Budgetary control techniques – Use of computers and IT in Management control – Productivity problems and management – Control and performance – Direct and preventive control – Reporting.

Total Periods :45

Pedagogical Tools:	Black board, chalk, group discussion, role play, youtube videos, NPTEL videos

TEXT BOOKS:

Sl. No	Authors	Title of the Book	Publisher	Year of publication
1	Harold Koontz and Heinz Weihrich	Essentials of management	Tata McGraw Hill	1998
2	Stephen P. Robbins and Mary Coulter	Management	Prentice Hall (India)Pvt. Ltd.	2009

REFERENCE BOOKS:

Sl. No	Authors	Title of the Book	Publisher	Year of Publication
1	Robert Kreitner and MamataMohapatra	Management	Biztantra	2008
2	Stephen A. Robbins and David A. Decenzo and Mary Coulter	Fundamentals of Management	Pearson Education	2011
3	Tripathy PC and Reddy PN	Principles of Management	Tata Mcgraw Hill	1999

CO – PO – PSO MAPPING

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	2	2	1	–	–	1	–	–	3	3	3	2	3	3	2
CO2	1	2	–	–	–	2	–	–	2	3	3	2	2	2	3
CO3	2	2	1	–	–	–	–	–	3	2	2	1	3	2	1
CO4	2	2	1	–	–	–	–	–	3	3	2	1	3	3	1
CO5	2	3	1	–	–	–	–	–	2	2	3	2	3	3	2

R 2024	AI&DS					SEMESTER: VI
24AD603	DEEP LEARNING	L	T	P	C	PC
		3	0	2	4	

COURSE OBJECTIVES:

The objectives of learning this course are to:

- Understand the fundamentals of deep learning.
- Understanding the working of Convolutional Neural Networks and RNN in decision making.
- Illustrate the strength and weaknesses of many popular deep learning approaches.
- Introduce major deep learning algorithms, the problem settings, and their applications to solve real world problems

COURSE OUTCOMES:

At the end of this course, students are able to

CO1: Analyze and interpret the concepts of neural networks relating to artificial intelligence.
CO2: Illustrate the learning processes and their statistical properties.
CO3: Design deep learning models using regularization and convolutional operations.
CO4: Analyze sequential data to build recurrent and recursive models.
CO5: Develop and analyze the applications using Autoencoders.

UNIT: I	INTRODUCTION	9

What is a Neural Network?, The Human Brain, Models of a Neuron, Neural Networks Viewed As Directed Graphs, Feedback, Network Architectures, Rosenblatt"s Perceptron: Introduction, Perceptron, The Perceptron Convergence Theorem, Relation Between the Perceptron and Bayes Classifier for a Gaussian Environment.

UNIT: II	MULTILAYER PERCEPTIONS	9

Introduction, Batch Learning and On-Line Learning, The Back-Propagation Algorithm, XOR Problem, Heuristics for Making the Back- Propagation Algorithm Perform Better, Back Propagation and Differentiation.

UNIT: III	REGULARIZATION FOR DEEP LEARNING	9

Parameter Norm Penalties - L2 Parameter Regularization, Dataset Augmentation, Semi-Supervised Learning. Optimization for Training Deep Models: Challenges in Neural Network Optimization – Ill Conditioning, Local Minima, Plateaus, Saddle Points and Other Flat Regions.

UNIT: IV	CONVOLUTION NEURAL NETWORKS	9

The Convolution Operation, Motivation, Pooling, Convolution and Pooling as an Infinitely Strong Prior, Variants of the Basic Convolution Function, Structured Outputs, Data Types, Efficient Convolution Algorithms, Convolutional Networks and the History of Deep Learning.

UNIT: V	SEQUENCE MODELING	9

Recurrent and Recursive Nets: Unfolding Computational Graphs, Recurrent Neural Networks, Bidirectional RNNs, Encoder-Decoder Sequence-to- Sequence Architectures, Deep Recurrent Networks, Recursive Neural Networks, The Long Short-Term Memory and Other Gated RNNs

Total Periods :45

Pedagogical Tools:	Black board, chalk, group discussion, role play, youtube videos, NPTEL videos

TEXT BOOKS:

Sl. No	Authors	Title of the Book	Publisher	Year of publication

1	Simon Haykin	Neural Networks and Learning Machines (3rd Edition)	Pearson	2016
2	Ian Goodfellow, Yoshua Bengio, Aaron Courville	Deep Learning	MIT Press	2016

TEXT BOOKS:

Sl. No	Authors	Title of the Book	Publisher	Year of publication
1	Christopher M. Bishop	Pattern Recognition and Machine Learning	Springer	2006
2	Charu C. Aggarwal	Neural Networks and Deep Learning	Springer	2018
3	Kevin P. Murphy	Machine Learning: A Probabilistic Perspective	MIT Press	2012

LIST OF EXPERIMENTS:	Total Periods: 30
1. Implementation of basic Image processing operations including Feature Representation and Feature Extraction	
2. Implementation of simple neural network	
3. Study of pretrained deep neural network model for Images	
4. CNN for Image classification	
5. CNN for Image segmentation	
6. RNN for video processing	
7. Implementation of Deep Generative model for Image editing	

CO – PO – PSO MAPPING

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	3	2	–	–	1	–	–	–	–	–	–	2	3	2	2
CO2	3	3	–	1	2	–	–	–	–	–	–	2	3	2	2
CO3	3	3	3	2	3	–	–	–	1	–	–	3	3	3	3
CO4	3	3	3	2	3	–	–	–	1	–	–	3	3	3	3
C05	3	3	3	2	3	–	–	–	2	1	–	3	3	3	3

R 2024	COMPUTER SCIENCE AND ENGINEERING					SEMESTER: VI
24CS602	MOBILE APPLICATION AND DEVELOPMENT	L	T	P	C	PC
		3	0	0	3	

COMMON TO ALL DEPARTMENTS

COURSE OBJECTIVES:

- Master modern cross-platform frameworks like Flutter, React Native with Expo, and .NET MAUI
- Develop native applications using Jetpack Compose (Android) and SwiftUI (iOS)
- Implement modern state management patterns and architectural approaches
- Integrate cloud services (Firebase, Supabase), RESTful APIs, and real-time databases
- Apply mobile DevOps, CI/CD pipelines, and app deployment strategies

COURSE OUTCOMES:

CO1: Build modern cross-platform applications using Flutter and React Native with Expo
CO2: Develop native Android apps using Kotlin and Jetpack Compose
CO3: Create iOS applications using Swift and SwiftUI framework
CO4: Implement cloud integration with Firebase, Supabase, and modern backend services
CO5: Apply DevOps practices including CI/CD, testing, and app store deployment

UNIT: I — MODERN MOBILE DEVELOPMENT ECOSYSTEM — 9

Evolution of mobile development – Native vs Cross-platform vs Hybrid approaches – Progressive Web Apps (PWA) – Responsive and Adaptive design principles – Mobile-first development strategy. Modern development tools and IDEs: Android Studio, Xcode, VS Code. Package managers: npm, yarn, pub, CocoaPods, Gradle. Version control with Git and GitHub. Mobile app architecture patterns: MVC, MVP, MVVM, Clean Architecture. Performance optimization fundamentals – Memory management – Battery optimization.

UNIT: II — CROSS-PLATFORM DEVELOPMENT WITH FLUTTER & REACT NATIVE — 9

Flutter Development: Dart programming fundamentals – Flutter widget tree – Stateful and Stateless widgets – Material Design and Cupertino widgets – Navigation and routing – State management with Provider, Riverpod, and BLoC pattern – Flutter animations – Custom widgets and themes.

React Native with Expo: Modern React concepts – JSX and component lifecycle – React Hooks (useState, useEffect, useContext, useReducer) – Expo CLI and managed workflow – React Navigation – State management with Redux Toolkit and Zustand – Styled Components and React Native Paper – Expo modules and APIs.

UNIT: III — NATIVE ANDROID DEVELOPMENT WITH KOTLIN & JETPACK COMPOSE — 9

Kotlin language fundamentals – Null safety and coroutines – Jetpack Compose declarative UI – Composable functions and modifiers – State and recomposition – Material Design 3 (Material You) – Navigation Component – ViewModel and LiveData – Room Database – Retrofit for networking – Dependency injection with Hilt/Koin – Android app components: Activities, Services, Broadcast Receivers, Content Providers – WorkManager for background tasks – Jetpack libraries: Paging, CameraX, Biometric authentication.

UNIT: IV — CLOUD INTEGRATION & MODERN BACKEND SERVICES — 9

Firebase Integration: Authentication (Email, Google, Phone) – Cloud Firestore real-time database – Firebase Storage – Cloud Functions – Push notifications with FCM – Firebase Analytics and Crashlytics – Remote Config – App Distribution.

Supabase (Open-source Firebase alternative): PostgreSQL database – Real-time subscriptions – Row Level Security – Storage and CDN – Supabase Auth – Edge Functions.

RESTful API Integration: HTTP clients (Dio, Axios) – GraphQL with Apollo Client – WebSockets for real-time communication – API authentication (JWT, OAuth 2.0) – Error handling and retry mechanisms.

UNIT: V — DEVOPS, TESTING & DEPLOYMENT — 9

Testing Strategies: Unit testing – Widget testing (Flutter) – Component testing (React Native) – Integration testing – End-to-end testing with Detox/Maestro – Test-driven development (TDD).

CI/CD Pipelines: GitHub Actions – Fastlane automation – App Center (Microsoft) – Bitrise – CodeMagic – Automated builds and deployments.

App Store Deployment: Google Play Console – App Store Connect – App signing and certificates – Play Store and App Store guidelines – Release management – Beta testing with TestFlight and Google Play Beta – App analytics and monitoring – Crash reporting with Sentry/BugSnag. **Emerging Technologies:** AI/ML integration (TensorFlow Lite, ML Kit) – AR capabilities (ARCore, ARKit) – Wearable app development.

Total Periods: 45

Pedagogical Tools:	Interactive coding platforms, Live coding demonstrations, GitHub repositories, Flutter DevTools, React Native Debugger, Android Studio Profiler, Xcode Instruments, Firebase Console, Video tutorials, Documentation websites (flutter.dev, reactnative.dev, developer.android.com)

TEXT BOOKS:

Sl. No	Authors	Title of the Book	Publisher	Year
1	Eric Windmill	Flutter in Action	Manning Publications	2023
2	William Candillon	React Native for Mobile Development	Packt Publishing	2024
3	Neal Mansueto	Kickstart Modern Android Development with Jetpack and Kotlin	Packt Publishing	2023

REFERENCE BOOKS:

Sl. No	Authors	Title of the Book	Publisher	Year
1	Marco L. Napoli	Beginning Flutter with Dart	Apress	2022
2	David Greenhalgh	Firebase Essentials for Android Development	Packt Publishing	2023
3	Juntao Qiu	Test-Driven Development with React and TypeScript	Packt Publishing	2023
4	Thomas Nield	Learning RxJava: Build concurrent applications using reactive programming with the latest features of RxJava 3	Packt Publishing	2020

CO PO MAPPING

COs / POs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	2	2	3	1	3	–	–	–	1	1	–	1	3	2
CO2	2	2	3	1	3	–	–	–	1	1	–	1	3	2
CO3	2	2	3	1	3	–	–	–	1	1	–	1	3	2
CO4	2	3	2	2	3	–	–	–	1	1	–	1	3	3
CO5	2	2	2	2	3	–	–	–	2	2	1	2	3	3

R 2024	CSE					SEMESTER: VI
24CS604	MOBILE APP DEVELOPMENT LAB	L	T	P	C	PC
		0	0	4	2	

COURSE OBJECTIVES:

The objectives of learning this course are:
- Master Flutter framework for building beautiful cross-platform applications
- Develop modern React Native applications using Expo and latest libraries
- Build native Android apps with Kotlin and Jetpack Compose
- Integrate cloud services like Firebase and Supabase for backend functionality
- Implement real-world features: authentication, databases, APIs, location services, and media handling

COURSE OUTCOMES:

At the end of this course, students can be able to:

CO1: Build feature-rich cross-platform mobile apps using Flutter and Dart
CO2: Develop React Native applications with Expo and modern state management
CO3: Create native Android applications using Kotlin and Jetpack Compose
CO4: Integrate Firebase/Supabase for authentication, database, and cloud storage
CO5: Deploy complete mobile applications to Google Play Store and App Store

PRACTICAL EXERCISES:

1. Build a Flutter app with Material Design widgets including AppBar, Drawer, BottomNavigationBar, FloatingActionButton, Cards, and Lists. Implement navigation between multiple screens using Navigator.

2. Create a Flutter BMI Calculator app with custom widgets, StatefulWidget for state management, Form validation, and responsive UI. Calculate and display BMI categories with color coding.

3. Develop a Flutter Weather App integrating OpenWeather API using Dio package. Implement async/await, JSON parsing, error handling, pull-to-refresh, and display current weather with 5-day forecast.

4. Build a React Native Expo app with modern React Hooks (useState, useEffect, useContext). Create a multi-screen app using React Navigation (Stack, Tab, Drawer navigators) with proper navigation flow.

5. Develop an Expense Tracker app using React Native and Expo. Implement category-wise expense tracking, charts using react-native-chart-kit, AsyncStorage for data persistence, and monthly/weekly summaries.

6. Create a News Reader app with React Native fetching data from NewsAPI. Implement FlatList with infinite scroll, image caching, WebView for article display, share functionality, and dark/light theme toggle.

7. Create an Android app with Jetpack Compose UI. Build composable functions, implement Material Design 3 components, state hoisting, and Navigation Component with multiple screens.

8. Build a Note-taking app with Room Database. Implement MVVM architecture, LiveData, ViewModel, CRUD operations, search functionality, and data persistence using Room.

9. Build a complete Authentication app using Firebase Auth (Email/Password, Google Sign-In, Phone OTP). Implement login, signup, password reset, email verification, and profile management.

10. Create a Real-time Chat application using Firebase Firestore. Implement one-on-one messaging, group chats, message timestamps, online status indicators, typing indicators, and push notifications with FCM.

11. Develop a Social Media app with Firebase. Implement user posts with images (Firebase Storage), likes/comments (Firestore), user profiles, follower system, and real-time feed updates.

12 Implement Biometric Authentication (Fingerprint/Face ID) in a Flutter/React Native app. Create a secure vault app to store passwords with encryption, biometric lock/unlock, and local secure storage.

13. Build a Fitness Tracker app integrating device sensors (Accelerometer, Pedometer, Heart Rate). Track steps, calories, distance, display charts, set goals, and send motivational notifications.

14. Create a QR/Barcode Scanner app with ML Kit. Scan QR codes, barcodes, generate custom QR codes, maintain scan history, and implement deep linking to app content.

Total Periods: 60

SOFTWARE REQUIREMENTS:

Development Tools:
- Android Studio (latest version) with Kotlin plugin
- Visual Studio Code with Flutter/React Native extensions
- Xcode (for iOS development on macOS)
- Git and GitHub Desktop

Frameworks & SDKs:
- Flutter SDK (3.x or latest)
- React Native with Expo CLI
- Node.js (LTS version) and npm/yarn
- Android SDK (API 31+)
- Firebase SDK, Supabase Client

Testing Tools:
- Android Emulator / iOS Simulator
- Expo Go app (for testing React Native apps on physical devices)
- Chrome DevTools for debugging
- Postman (for API testing)

CO PO MAPPING

COs / POs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	2	2	3	1	3	–	–	–	1	1	–	1	3	2
CO2	2	2	3	1	3	–	–	–	1	1	–	1	3	2
CO3	2	2	3	1	3	–	–	–	1	1	–	1	3	2
CO4	2	3	2	2	3	–	–	–	1	1	–	1	3	3
CO5	2	2	2	2	3	–	–	–	2	2	1	2	3	3

R 2024	COMPUTER SCIENCE AND ENGINEERING					SEMESTER: VII
24CS601	AI IN ETHICS	L	T	P	C	MC
		3	0	0	3	
COMMON TO ALL DEPARTMENTS						

COURSE OBJECTIVES:

- Understand fundamental human values, moral theories, and ethical frameworks
- Analyze ethical challenges and dilemmas emerging from AI and autonomous systems
- Explore principles of responsible AI including fairness, transparency, accountability, and privacy
- Evaluate societal impacts of AI on employment, privacy, democracy, and human rights
- Apply ethical frameworks to design and develop value-aligned AI systems

COURSE OUTCOMES:

CO1: Demonstrate understanding of core human values, ethical theories, and their application to technology
CO2: Identify and analyze ethical issues in AI systems including bias, fairness, privacy, and accountability
CO3: Apply principles of responsible AI to design trustworthy and ethical AI systems
CO4: Evaluate societal and global implications of AI on employment, governance, rights, and social justice
CO5: Propose ethical solutions and frameworks for real-world AI challenges across diverse domains

UNIT: I	FOUNDATIONS: HUMAN VALUES AND ETHICAL THEORIES	9

Core Human Values: Definition and significance of human values – Universal values: truth, righteousness, peace, love, non-violence – Personal values: integrity, honesty, empathy, compassion, responsibility – Social values: justice, equality, freedom, dignity, solidarity – Professional values in engineering and technology – Value conflicts and resolution – Role of values in decision-making.

Introduction to Ethics: Definition of ethics, morality, and moral philosophy – Distinction between ethics, morals, and values – Descriptive vs. normative ethics – Meta-ethics – Moral reasoning and judgment – Ethical relativism vs. moral absolutism – Cultural perspectives on ethics.

Classical Ethical Theories: Virtue Ethics (Aristotle) – character and moral virtues – Deontological Ethics (Kant) – duty, categorical imperative, universal laws – Consequentialism and Utilitarianism (Bentham, Mill) – greatest good for greatest number – Care Ethics – relationships and interdependence – Rights-based Ethics – human rights and dignity – Social Contract Theory – fairness and justice (Rawls).

UNIT: II	ETHICS IN TECHNOLOGY AND ARTIFICIAL INTELLIGENCE	9

Ethics in Engineering and Technology: Professional ethics for engineers – Code of ethics (IEEE, ACM, ACE) – Engineering responsibility to society – Whistleblowing and moral courage – Intellectual property rights – Patent ethics – Open source vs. proprietary software ethics – Technology and environmental sustainability.

Introduction to AI Ethics: Brief history of AI development – Ethical concerns in AI: overview – Machine ethics vs. ethics of machines – Moral agency and AI – Can machines be moral agents? – Value alignment problem – Ensuring AI systems align with human values – AI ethics frameworks and guidelines (EU AI Act, UNESCO, OECD principles).

Ethical Issues in AI Systems: Algorithmic bias and discrimination – types, sources, and impacts – Fairness in machine learning – individual fairness vs. group fairness – Transparency and explainability – black box problem – Privacy concerns – data collection, surveillance, and consent – Accountability and responsibility – who is responsible when AI fails? – Autonomy and human agency – AI decision-making vs. human autonomy – Safety and security – adversarial attacks, robustness – Dual-use dilemma – beneficial vs. harmful applications.

UNIT: III	RESPONSIBLE AI: PRINCIPLES AND PRACTICES	9

Principles of Responsible AI: Human-centered AI design – putting people first – Fairness and non-discrimination – addressing bias in datasets and algorithms – Transparency and explainability – interpretable AI, LIME, SHAP – Accountability and auditability – AI audits and impact assessments – Privacy and data protection – GDPR, differential privacy, federated learning – Security and safety – adversarial robustness, safety testing – Reliability and robustness – testing and validation – Sustainability – environmental impact of AI (carbon footprint of training models).

Fairness in AI: Defining fairness – statistical parity, equal opportunity, equalized odds – Measuring fairness – fairness metrics and trade-offs – Detecting bias – bias audits, fairness toolkits (AI Fairness 360, Fairlearn) – Mitigating bias – pre-processing, in-processing, post-processing techniques – Case studies: COMPAS (recidivism prediction), facial recognition bias, hiring algorithms.

Explainable AI (XAI): Need for explainability – trust, accountability, debugging – Levels of interpretability – global vs. local explanations – XAI techniques – feature importance, saliency maps, attention mechanisms – LIME (Local Interpretable Model-agnostic Explanations) – SHAP (SHapley Additive exPlanations) – Counterfactual explanations – Right to explanation in legal contexts.

Privacy-Preserving AI: Privacy principles – data minimization, purpose limitation, consent – Differential privacy – mathematical privacy guarantees – Federated learning – decentralized model training – Homomorphic encryption – computation on encrypted data – Secure multi-party computation – Privacy vs. utility trade-offs.

UNIT: IV	SOCIETAL IMPACTS AND GOVERNANCE OF AI	9

AI and Employment: Automation and job displacement – which jobs are at risk? – Future of work – new jobs created by AI – Reskilling and upskilling – preparing workforce for AI era – Economic inequality – digital divide, wealth concentration – Universal Basic Income (UBI) debate – Labor rights in algorithmic management – gig economy ethics.

AI and Democracy: Algorithmic governance – AI in public sector decision-making – Misinformation and deepfakes – threats to truth and trust – Social media algorithms – echo chambers, polarization, manipulation – Surveillance capitalism – data collection and behavioral modification – AI in elections – micro-targeting, voter manipulation – Democratic participation – AI for civic engagement.

AI and Human Rights: Right to privacy in the age of AI – Right to non-discrimination – Freedom of expression – content moderation dilemmas – Access to information – algorithmic censorship – Dignity and autonomy – AI manipulation and nudging – Children's rights – AI in education, parental controls – Rights of marginalized communities – algorithmic injustice.

AI Governance and Regulation: Need for AI regulation – balancing innovation and safety – International AI governance frameworks – EU AI Act, UNESCO recommendations – National AI strategies and policies – India's approach to AI governance – Industry self-regulation – AI ethics boards, ethical guidelines – Legal liability and AI – who is liable? – Intellectual property in AI – AI-generated content ownership – Algorithmic auditing and certification.

UNIT: V	DOMAIN-SPECIFIC ETHICS AND FUTURE CHALLENGES	9

Ethics in Specific AI Applications:
Healthcare AI: Medical diagnosis and treatment recommendations – Patient privacy and consent – Health equity and access – AI in drug discovery – Liability in medical AI errors.
Autonomous Vehicles: Trolley problem in self-driving cars – Programming moral decisions – Liability in accidents – Safety vs. innovation – Public acceptance and trust.
Criminal Justice AI: Predictive policing – bias and discrimination – Risk assessment tools – COMPAS and recidivism – Facial recognition by law enforcement – Surveillance and civil liberties – Due process and algorithmic decisions.

Military AI and Autonomous Weapons: Lethal autonomous weapons systems (LAWS) – International humanitarian law – Human control and meaningful human oversight – Campaign to ban killer robots – Cyber warfare ethics.

AI in Education: Personalized learning – benefits and risks – Student privacy and data collection – Automated grading and assessment fairness – Digital divide and access to AI education tools.

Total Periods: 45

Pedagogical Tools:	Case study discussions, Ethical dilemma analysis, Group debates, Role-playing exercises, Documentary screenings, Guest lectures from ethicists and industry practitioners, Real-world AI case studies (Cambridge Analytica, facial recognition controversies), Interactive ethics workshops, Research paper presentations, Policy analysis exercises

TEXT BOOKS:

Sl. No	Authors	Title of the Book	Publisher	Year
1	Stuart Russell	Human Compatible: Artificial Intelligence and the Problem of Control	Viking	2019
2	Michael Kearns, Aaron Roth	The Ethical Algorithm: The Science of Socially Aware Algorithm Design	Oxford University Press	2019
3	Mark Coeckelbergh	AI Ethics	MIT Press	2020

REFERENCE BOOKS:

Sl. No	Authors	Title of the Book	Publisher	Year
1	Cathy O'Neil	Weapons of Math Destruction: How Big Data Increases Inequality and Threatens Democracy	Crown	2016
2	Virginia Dignum	Responsible Artificial Intelligence: How to Develop and Use AI in a Responsible Way	Springer	2019
3	Brian Christian	The Alignment Problem: Machine Learning and Human Values	W. W. Norton	2020
4	Hannah Fry	Hello World: Being Human in the Age of Algorithms	W. W. Norton	2018
5	Safiya Umoja Noble	Algorithms of Oppression: How Search Engines Reinforce Racism	NYU Press	2018
6	Nick Bostrom	Superintelligence: Paths, Dangers, Strategies	Oxford University Press	2014

ADDITIONAL RESOURCES:

Ethics Guidelines and Frameworks:
• EU Ethics Guidelines for Trustworthy AI (European Commission)
• OECD AI Principles
• UNESCO Recommendation on the Ethics of AI
• IEEE Ethically Aligned Design
• Montreal Declaration for Responsible AI
• Partnership on AI Best Practices

Online Courses and Resources:
• AI Ethics (edX, Harvard University)
• Ethics, Technology and Engineering (Coursera, TU Delft)
• AI For Everyone (Coursera, Andrew Ng)
• Stanford Encyclopedia of Philosophy - Ethics sections

Case Study Resources:
• AI Incident Database (Partnership on AI)
• Ethics & Governance of Artificial Intelligence Initiative (MIT)
• AI Now Institute Reports
• Algorithm Watch case studies

CO PO MAPPING

COs / POs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	1	1	–	–	–	2	–	3	–	2	–	1	1	–
CO2	1	2	–	1	–	3	–	3	–	2	–	1	1	1
CO3	2	2	2	1	1	3	–	3	1	2	–	1	2	2
CO4	1	2	–	–	–	3	1	3	1	2	–	1	1	1
CO5	2	3	2	1	1	3	–	3	2	2	1	2	2	2

R 2024	INFORMATION TECHNOLOGY					SEMESTER: VII
24IT701	DISTRIBUTED SYSTEMS	L	T	P	C	PC
		3	0	0	3	

COURSE OBJECTIVES:

- Introduce fundamental concepts and architectures of distributed systems
- Understand communication, coordination, and synchronization
- Learn consistency, replication, and fault tolerance
- Study distributed transactions and consensus
- Explore modern distributed applications and middleware

COURSE OUTCOMES:

CO1: Understand distributed system models and architectures
CO2: Apply communication and coordination mechanisms
CO3: Analyze consistency and replication strategies
CO4: Handle faults and apply consensus methods
CO5: Build distributed applications

UNIT: I	INTRODUCTION	9

Definition – goals – challenges – types of distributed systems – architectural models – system models – fundamental issues – clocks and event ordering – logical clocks – global states.

UNIT: II	COMMUNICATION & COORDINATION	9

Inter-process communication – message passing – RPC – RMI – group communication – mutual exclusion – election algorithms – distributed deadlocks.

UNIT: III	CONSISTENCY & REPLICATION	9

Data-centric consistency models – client-centric consistency – replication techniques – quorum-based protocols – consistency in distributed databases.

UNIT: IV	FAULT TOLERANCE & CONSENSUS	9

Failure models – reliable client-server communication – Byzantine failures – agreement problem – consensus algorithms – recovery.

UNIT: V	DISTRIBUTED FILE SYSTEMS & APPLICATIONS	9

Distributed file systems – naming – security – peer-to-peer systems – cloud and web-based distributed systems – case studies.

Total Periods :45

Pedagogical Tools:	Black board, chalk, group discussion, role play, youtube videos, NPTEL videos

TEXT BOOKS:

Sl. No	Authors	Title of the Book	Publisher	Year of publication
1	Andrew S. Tanenbaum, Maarten Van Steen	Distributed Systems: Principles and Paradigms	Pearson	2007
2	George Coulouris, Jean Dollimore, Tim Kindberg,	Distributed Systems: Concepts and Design	Pearson	2012

	Gordon Blair			

REFERENCE BOOKS:				
Sl. No	**Authors**	**Title of the Book**	**Publisher**	**Year of Publication**
1	Nancy A. Lynch	Distributed Algorithms	Morgan Kaufmann	1996
2	Mukesh Singhal, Niranjan G. Shivaratri	Advanced Concepts in Operating Systems	McGraw Hill	1994
3	Pradeep K. Sinha	Distributed Operating Systems: Concepts and Design	PHI	2007

CO – PO – PSO MAPPING

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	3	2	1	–	–	–	–	–	–	–	–	1	2	1	3
CO2	3	3	2	1	2	–	–	–	–	–	–	1	3	2	3
CO3	3	3	2	2	2	–	–	–	–	–	–	1	3	2	3
CO4	3	3	3	2	2	–	–	–	–	–	–	2	3	3	3
CO5	3	3	3	2	3	–	–	–	–	–	–	2	3	3	3

R 2024	COMPUTER SCIENCE AND ENGINEERING					
	DATA VISUALIZATION TOOLS					
24ADX03	DATA VISUALIZATION TOOLS	L	T	P	C	PE
		3	0	0	3	

COURSE OBJECTIVES:

- Understand fundamental principles of data visualization and visual perception
- Master industry-standard visualization tools including Tableau, Power BI, and Python libraries
- Learn to create interactive dashboards and storytelling with data
- Apply advanced techniques including real-time visualization and AI-powered analytics
- Develop skills in choosing appropriate visualizations for different data types and audiences

COURSE OUTCOMES:

CO 1: Design effective visualizations following cognitive principles and best practices
CO 2: Build interactive dashboards using Tableau and Power BI for business intelligence
CO 3: Create visualizations with Python (Matplotlib, Seaborn, Plotly) and D3.js for web
CO 4: Implement real-time and AI-enhanced visualization solutions
CO 5: Communicate insights effectively through data storytelling and presentation

UNIT: I	FOUNDATIONS OF DATA VISUALIZATION	9

Introduction to data visualization -- history and evolution -- importance in decision-making. Visual perception principles -- pre-attentive processing -- Gestalt principles -- color theory and application. Types of data -- categorical, numerical, temporal, hierarchical, network data. Chart types and selection -- bar charts, line graphs, scatter plots, heatmaps -- choosing appropriate visualizations -- principles of effective design (clarity, accuracy, efficiency).

UNIT: II	BUSINESS INTELLIGENCE TOOLS	9

Tableau fundamentals -- connecting to data sources -- creating worksheets and dashboards -- calculated fields and parameters -- filters and actions. Advanced Tableau -- LOD expressions -- table calculations -- trend lines and forecasting -- publishing to Tableau Server. Power BI essentials -- Power Query for data transformation -- DAX formulas -- creating reports and dashboards -- Power BI Service and sharing. Dashboard design principles -- layout and composition -- KPI selection -- interactivity patterns -- mobile responsiveness.

UNIT: III	PYTHON VISUALIZATION LIBRARIES	9

Matplotlib basics -- figure and axes -- plot types -- customization and styling -- subplots and layouts. Seaborn for statistical visualization -- distribution plots -- categorical plots -- regression plots -- multi-plot grids. Plotly for interactive visualizations -- line, bar, and scatter plots -- 3D plots -- animations -- dash for web applications. Altair and declarative visualization -- grammar of graphics approach -- data transformations -- interactive selections.

UNIT: IV	WEB-BASED VISUALIZATION	9

D3.js fundamentals -- selections and data binding -- scales and axes -- SVG basics -- transitions and animations. Building interactive charts with D3.js -- bar charts, line graphs, force-directed graphs -- zoom and pan -- tooltips and legends. Chart.js and lightweight libraries -- responsive charts -- Chart.js configuration -- integration with web frameworks. Geospatial visualization -- Leaflet for interactive maps -- Mapbox integration -- choropleth maps -- GeoJSON data handling.

UNIT: V	ADVANCED TOPICS AND TRENDS	9

Real-time visualization -- streaming data handling -- WebSocket integration -- live dashboards -- performance optimization. AI-powered visualization -- automated chart selection -- natural language queries -- anomaly detection in visuals -- AI-generated narratives. Storytelling with data -- narrative structures -- presentation design -- annotation techniques -- scrollytelling. Emerging trends -- AR/VR for data visualization -- voice-activated visualizations -- embedded analytics -- ethical considerations and accessibility.

Total Periods: 45

Pedagogical Tools:	Tableau Desktop, Power BI, Python (Matplotlib, Seaborn, Plotly), D3.js, Chart.js, Jupyter Notebooks, Excel, Google Data Studio, Looker, Qlik Sense, R (ggplot2), Observable, Flourish, Datawrapper

TEXT BOOKS:

Sl. No	Authors	Title of the Book	Publisher	Year
1	Camm, Cochran, Fry, Ohlmann	Data Visualization: Exploring and Explaining with Data, 2nd Edition	Cengage	2024
2	Cole Nussbaumer Knaflic	Storytelling with Data: A Data Visualization Guide for Business Professionals	Wiley	2015

REFERENCE BOOKS:

Sl. No	Authors	Title of the Book	Publisher	Year
1	Kieran Healy	Data Visualization: A Practical Introduction	Princeton	2018
2	Nadieh Bremer	CHART: Designing Creative Data Visualizations from Charts to Art	CRC Press	2025
3	Claus O. Wilke	Fundamentals of Data Visualization	O'Reilly	2019

COs / POs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	2	2	2	1	2	–	–	–	–	2	–	1	2	2
CO2	2	2	3	1	3	–	–	–	1	2	1	1	3	3
CO3	2	2	3	2	3	–	–	–	1	2	–	1	3	3
CO4	2	3	3	2	3	1	–	–	1	2	1	1	3	3
CO5	1	2	2	1	2	1	–	–	2	3	1	1	2	3

R 2024	AI&DS					
24ADX04	NATURAL LANGUAGE PROCESSING	L	T	P	C	PE
		3	0	0	3	

COURSE OBJECTIVES:

- Master text preprocessing, tokenization, and word representation techniques
- Understand sequence models including RNNs, LSTMs, and attention mechanisms
- Learn transformer architecture and pre-trained language models
- Apply modern LLMs for text generation and natural language understanding tasks

COURSE OUTCOMES:

CO1: Implement text preprocessing pipelines and word embedding techniques
CO2: Build sequence models using RNNs and LSTMs for text processing
CO3: Apply transformer models like BERT for NLP tasks
CO4: Develop NLP applications for sentiment analysis, classification, and QA
CO5: Use modern LLMs with prompt engineering and RAG techniques

UNIT: I	NLP FUNDAMENTALS AND WORD REPRESENTATIONS	9

Introduction to NLP – challenges in language understanding – applications. Text preprocessing – tokenization, stemming, lemmatization – stop word removal – normalization. Word representations – Bag of Words (BoW) – TF-IDF – limitations of traditional methods. Word embeddings – Word2Vec (Skip-gram, CBOW) – GloVe – FastText – embedding properties and analogies. Language models – N-gram models – smoothing techniques – perplexity – part-of-speech tagging – named entity recognition basics.

UNIT: II	SEQUENCE MODELS FOR NLP	9

Recurrent Neural Networks (RNNs) – sequential processing – hidden states – vanishing gradient problem. Long Short-Term Memory (LSTM) – gates (forget, input, output) – cell state – addressing long-term dependencies. Gated Recurrent Units (GRU) – simplified architecture – comparison with LSTM. Sequence-to-sequence models – encoder-decoder architecture – applications in machine translation. Attention mechanism – Bahdanau attention – alignment and context vectors – attention weights visualization. Bidirectional RNNs – forward and backward processing – applications in text classification and sequence labeling.

UNIT: III	TRANSFORMERS AND PRE-TRAINED MODELS	9

Transformer architecture – self-attention mechanism – multi-head attention – positional encoding. Encoder and decoder stacks – layer normalization – feed-forward networks – residual connections. BERT (Bidirectional Encoder Representations from Transformers) – masked language modeling – next sentence prediction – pre-training and fine-tuning. GPT (Generative Pre-trained Transformer) – autoregressive language modeling – unidirectional attention. Transfer learning in NLP – domain adaptation – feature extraction vs fine-tuning – low-resource scenarios. Practical implementation – using Hugging Face Transformers library – model selection – tokenization – training and inference.

UNIT: IV	NLP APPLICATIONS	9

Sentiment analysis – lexicon-based approaches – machine learning methods – aspect-based sentiment analysis. Text classification – document categorization – spam detection – topic classification – multi-label classification. Named Entity Recognition (NER) – sequence labeling – CRF and LSTM-CRF – entity extraction and linking. Question answering systems – extractive QA – SQuAD dataset – BERT for QA – answer span prediction. Text summarization – extractive methods (TextRank) – abstractive methods using seq2seq – evaluation metrics (ROUGE). Chatbot development – rule-based vs data-driven approaches – intent recognition – dialogue management – response generation.

UNIT: V	LARGE LANGUAGE MODELS AND MODERN NLP	9

Large Language Models (LLMs) – GPT-3, GPT-4 architecture and capabilities – scaling laws – emergent abilities. Prompt engineering – zero-shot, one-shot, few-shot learning – chain-of-thought prompting – prompt design best practices. LangChain framework – components (models, prompts, chains, agents) – building LLM applications – memory management. Retrieval Augmented Generation (RAG) – vector databases – semantic search – combining retrieval with generation. Fine-tuning LLMs – parameter-efficient methods (LoRA, QLoRA) – instruction tuning – RLHF (Reinforcement Learning from Human Feedback). Ethical considerations – bias in language models – fairness – privacy concerns – responsible AI practices – limitations and challenges in modern NLP.

Total Periods: 45

Pedagogical Tools:	Python, NLTK, spaCy, Gensim, PyTorch, TensorFlow, Hugging Face Transformers, LangChain, Jupyter Notebooks, Google Colab, NLP datasets (GLUE, SQuAD), Mini-projects

TEXT BOOKS:

Sl. No	Authors	Title of the Book	Publisher	Year
1	Dan Jurafsky, James H. Martin	Speech and Language Processing, 3rd Edition	Pearson	2023
2	Lewis Tunstall, Leandro von Werra, Thomas Wolf	Natural Language Processing with Transformers	O'Reilly	2022

REFERENCE BOOKS:

Sl. No	Authors	Title of the Book	Publisher	Year
1	Delip Rao, Brian McMahan	Natural Language Processing with PyTorch	O'Reilly	2019
2	Steven Bird, Ewan Klein, Edward Loper	Natural Language Processing with Python	O'Reilly	2009
3	Yoav Goldberg	Neural Network Methods for Natural Language Processing	Morgan & Claypool	2017

COs / POs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	3	2	2	3	–	–	–	–	–	–	1	3	3
CO2	3	3	3	2	3	–	–	–	–	–	–	1	3	3
CO3	3	3	3	2	3	–	–	–	–	–	–	1	3	3
CO4	2	3	3	2	3	1	–	–	1	1	1	1	3	3
CO5	2	3	3	2	3	1	–	–	1	2	1	2	3	3

R 2024	AI&DS					
24ADX01	UI / UX DESIGN	L	T	P	C	PE
		3	0	0	3	
COMMON TO ALL DEPARTMENTS						

COURSE OBJECTIVES:

- To provide a sound knowledge in UI & UX
- To understand the need for UI and UX
- To understand the various Research Methods used in Design
- To explore the various Tools used in UI & UX
- Creating a wireframe and prototype

COURSE OUTCOMES:

At the end of this course, students are able to
CO1: Understand the basic grammatical structures and apply them in right context
CO2: Identify and report cause and effects in events, industrial processes through technical texts.
CO3: Apply appropriate words in a professional context.
CO4: Interpret information presented in tables, charts and other graphic forms.
CO5: Draft effective resumes in the context of job search.

UNIT: I	FOUNDATIONS OF DESIGN	9

UI vs. UX Design - Core Stages of Design Thinking - Divergent and Convergent Thinking - Brainstorming and Game storming - Observational Empathy

UNIT: II	FOUNDATIONS OF UI DESIGN	9

Visual and UI Principles - UI Elements and Patterns - Interaction Behaviors and Principles – Branding - Style Guides

UNIT: III	FOUNDATIONS OF UX DESIGN	9

Introduction to User Experience - Why You Should Care about User Experience – Understanding User Experience - Defining the UX Design Process and its Methodology - Research in User Experience Design - Tools and Method used for Research - User Needs and its Goals - Know about Business Goals

UNIT: IV	WIREFRAMING, PROTOTYPING AND TESTING	9

Sketching Principles - Sketching Red Routes - Responsive Design – Wireframing - Creating Wireflows - Building a Prototype - Building High-Fidelity Mockups - Designing Efficiently with Tools - Interaction Patterns - Conducting Usability Tests

UNIT: V	RESEARCH, DESIGNING, IDEATING, & INFORMATION ARCHITECTURE	9

Identifying and Writing Problem Statements - Identifying Appropriate Research Methods - Creating Personas - Solution Ideation - Creating User Stories - Creating Scenarios - Flow Diagrams - Flow Mapping - Information Architecture.

Total Periods :45

Pedagogical Tools:	Black board, chalk, group discussion, role play, youtube videos, NPTEL videos

TEXT BOOKS:				
Sl. No	Authors	Title of the Book	Publisher	Year of publication
1	Joel Marsh	UX for Beginners	O'Reilly	2022
2	Jon Yablonski	Laws of UX: Using Psychology to Design Better Products & Services	O'Reilly	2021

REFERENCE BOOKS:				
Sl. No	Authors	Title of the Book	Publisher	Year of publication
1	Jenifer Tidwell, Charles Brewer, Aynne Valencia	Designing Interfaces (3rd Edition)	O'Reilly	2020
2	Steve Schoger, Adam Wathan	Refactoring UI	–	2018
3	Steve Krug	Don't Make Me Think, Revisited: A Commonsense Approach to Web & Mobile (3rd Edition)	New Riders	2015

CO – PO – PSO MAPPING															
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	1	1	–	–	–	–	–	–	2	3	–	2	3	2	1
CO2	1	2	–	–	–	–	–	–	2	3	–	2	3	3	1
CO3	–	1	–	–	–	–	–	–	3	3	–	2	3	2	2
CO4	1	2	–	1	–	–	–	–	2	3	–	2	3	3	1
CO5	–	1	–	–	–	–	–	–	3	3	2	2	3	2	3

R 2024	COMPUTER SCIENCE AND ENGINEERING					
24CSX11	GAME DEVELOPMENT	L	T	P	C	PE
		3	0	0	3	

COMMON TO ALL DEPARTMENTS

COURSE OBJECTIVES:

The objectives of learning this course are to:

- The student should be made to:
- Understand the concepts of Game design and development.
- Learn the processes, mechanics and issues in Game Design.
- Be exposed to the Core architectures of Game Programming.
- Know about Game programming platforms, frame works and engines. Learn to develop games.

COURSE OUTCOMES:

Upon completion of the course, students will be able to

CO 1 Discuss the concepts of Game design and development.

CO 2 Design the processes, and use mechanics for game development.

CO 3 Explain the Core architectures of Game Programming.

CO 4 Use Game programming platforms, frame works and engines.

CO 5 Create interactive Games.

UNIT: I	GRAPHICS FOR GAME PROGRAMMING	9

3D Transformations, Quaternions, 3D Modeling and Rendering, Ray Tracing, Shader models, Lighting, Color, Texturing, Camera and Projections, Culling and Clipping, Character Animation, Physics-based Simulation, Scene Graphs.

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UNIT: II	GAME ENGINE DESIGN	9

Game engine architecture, Engine support systems, Resources and File systems, Game loop and real-time simulation, Human Interface devices, Collision and rigid body dynamics, Game profiling.

UNIT: III	GAME PROGRAMMING	9

Application layer, Game logic, Game views, managing memory, controlling the main loop, loading and caching game data, User Interface management, Game event management.

UNIT: IV	AUGMENTED REALITY	9

2D and 3D Game development using Flash, DirectX, Java, Python, Game engines - Unity. DX Studio

UNIT: V	GAME DEVELOPMENT	9

Developing 2D and 3D interactive games using DirectX or Python – Isometric and Tile Based Games, Puzzle games, Single Player games, Multi Player games.

Total Periods :45

Pedagogical Tools:	Black board, chalk, group discussion, role play, youtube videos, NPTEL videos

TEXT BOOKS:				
Sl. No	**Authors**	**Title of the Book**	**Publisher**	**Year of publication**
1	Mike McShaffry, David Graham	Game Coding Complete (4th Edition)	Cengage Learning PTR	2012
2	Jason Gregory	Game Engine Architecture	CRC Press / A K Peters	2009
3	David H. Eberly	3D Game Engine Design: A Practical Approach to Real-Time Computer Graphics (2nd Edition)	Morgan Kaufmann	2006

REFERENCE BOOKS:				
Sl. No	**Authors**	**Title of the Book**	**Publisher**	**Year of publication**
1	Ernest Adams, Andrew Rollings	Fundamentals of Game Design (2nd Edition)	Prentice Hall / New Riders	2009
2	Eric Lengyel	Mathematics for 3D Game Programming and Computer Graphics (3rd Edition)	Course Technology PTR	2011
3	Jesse Schell	The Art of Game Design: A Book of Lenses (1st Edition)	CRC Press	2008

CO – PO – PSO MAPPING

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	2	2	–	–	1	–	–	–	1	–	–	2	2	2	1
CO2	3	3	3	1	2	–	–	–	2	1	–	2	3	3	2
CO3	3	3	2	1	2	–	–	–	1	–	–	2	3	2	2
CO4	2	2	2	1	3	–	–	–	2	1	–	2	3	3	2
CO5	3	3	3	2	3	–	–	–	3	2	–	3	3	3	3

R 2024	COMPUTER SCIENCE AND ENGINEERING					
24CSX16	**AUGMENTED REALITY AND VIRTUAL REALITY**	**L**	**T**	**P**	**C**	**PE**
		3	**0**	**0**	**3**	

COURSE OBJECTIVES:

The objectives of learning this course are to:

- Enable learners use words appropriately in their communication.
- Enhance learners' grammatical accuracy in communication.
- Develop learners' ability to read and listen to texts in English.
- Strengthen the communication skills of the learners.
- Help learners write appropriately in professional contexts

COURSE OUTCOMES:

At the end of this course, students are able to

CO1: Understand the basic grammatical structures and apply them in right context

CO2: Identify and report cause and effects in events, industrial processes through technical texts.

CO3: Apply appropriate words in a professional context.

CO4: Interpret information presented in tables, charts and other graphic forms.

CO5: Draft effective resumes in the context of job search.

UNIT: I	**INTRODUCTION TO VIRTUAL REALITY**	**9**

Defining Virtual Reality, Key elements of virtual reality experience, Virtual Reality, Telepresence, Augmented Reality and Cyberspace.

UNIT: II	**INPUT AND OUTPUT DEVICES**	**9**

Three-dimensional position trackers, navigation and manipulation, interfaces and gesture interfaces. Graphics displays, sound displays & haptic feedback.

UNIT: III	**MODELING**	**9**

Geometric modeling, Kinematics modeling, Physical modeling, Behaviour modeling, Model management.

UNIT: IV	**AUGMENTED REALITY**	**9**

Taxonomy, Technology and Features of Augmented Reality, AR Vs VR, Challenges with AR, AR systems and functionality, Augmented Reality Methods, Visualization Techniques for Augmented Reality, Enhancing interactivity in AR Environments, Evaluating AR systems

UNIT: V	**VIRTUAL REALITY**	**9**

Interaction & Audio: Interaction - Motor Programs and Remapping, Locomotion, Manipulation, Social Interaction. Audio -The Physics of Sound, The Physiology of Human Hearing, Auditory Perception, Auditory Rendering. Interaction - Motor Programs and Remapping, Locomotion, Manipulation, Social Interaction. Audio -The Physics of Sound, The Physiology of Human Hearing, Auditory Perception, Auditory Rendering

Total Periods :45

Pedagogical Tools:	Black board, chalk, group discussion, role play, youtube videos, NPTEL videos

TEXT BOOKS:				
Sl. No	Authors	Title of the Book	Publisher	Year of publication
1	Gregory C. Burdea, Philippe Coiffet	Virtual Reality Technology (2nd Edition)	John Wiley & Sons, Inc.	2017
2	Steven M. LaValle	Virtual Reality	Cambridge University Press	2016

REFERENCE BOOKS:				
Sl. No	Authors	Title of the Book	Publisher	Year of publication
1	Rajesh K. Maurya	Computer Graphics with Virtual Reality System (3rd Edition)	Wiley	2018
2	William R. Sherman, Alan B. Craig	Understanding Virtual Reality: Interface, Application, and Design (2nd Edition)	Morgan Kaufmann (Elsevier)	2019
3	Grigore C. Burdea, Philippe Coiffet	Virtual Reality Technology (2nd Edition)	Wiley	2017
4	K. S. Hale, K. M. Stanney	Handbook of Virtual Environments (2nd Edition)	CRC Press	2015

CO – PO – PSO MAPPING															
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	1	1	–	–	–	–	–	–	2	3	–	2	3	2	1
CO2	1	2	–	–	–	–	–	–	2	3	–	2	3	3	1
CO3	–	1	–	–	–	–	–	–	3	3	–	2	3	2	2
CO4	1	2	–	1	–	–	–	–	2	3	–	2	3	3	1
CO5	–	1	–	–	–	–	–	–	3	3	2	2	3	2	3

R 2024	AI&DS				
24ADX02	COMPUTATIONAL THINKING	L ˉ P	C	PE
		3 (0	3	

COURSE OBJECTIVES:

• Develop problem-solving skills using computational thinking principles
• Master algorithmic design techniques and complexity analysis
• Understand data structures and their applications in problem solving
• Apply computational thinking to real-world problems in AI and ML

COURSE OUTCOMES:

CO1: Apply decomposition, abstraction, and pattern recognition to solve problems
CO2: Design and analyze algorithms using various paradigms
CO3: Implement efficient data structures for problem solving
CO4: Evaluate time and space complexity of algorithms
CO5: Solve computational problems using Python programming

UNIT: I	FOUNDATIONS OF COMPUTATIONAL THINKING	9

Introduction to computational thinking – problem-solving fundamentals – computational vs analytical thinking. Core concepts – decomposition (breaking down complex problems) – abstraction (identifying patterns and generalizing) – pattern recognition – algorithm design. Problem formulation – defining inputs and outputs – constraints and requirements – solution strategies. Computational modeling – representing real-world problems – state space representation – search strategies.

UNIT: II	ALGORITHMIC PROBLEM SOLVING	9

Algorithm design strategies – brute force – divide and conquer (merge sort, quick sort, binary search). Greedy algorithms – activity selection – fractional knapsack – Huffman coding – optimization problems. Dynamic programming – principles of optimality – memoization – tabulation – solving recursive problems (Fibonacci, longest common subsequence). Backtracking – N-Queens problem – subset sum – constraint satisfaction problems. Recursion – recursive thinking – base cases and recursive cases – tree recursion – tail recursion.

UNIT: III	DATA STRUCTURES AND COMPLEXITY	9

Linear data structures – arrays, linked lists, stacks, queues – operations and applications. Trees – binary trees, binary search trees – tree traversals (inorder, preorder, postorder) – AVL trees basics. Graphs – representation (adjacency matrix, adjacency list) – graph traversals (BFS, DFS) – shortest path algorithms (Dijkstra). Hash tables – hash functions – collision resolution (chaining, open addressing) – applications. Algorithm complexity – time complexity – space complexity – Big-O, Big-Theta, Big-Omega notation – best, worst, and average case analysis – amortized analysis.

UNIT: IV	LOGIC AND MATHEMATICAL THINKING	9

Boolean logic – truth tables – logical operators (AND, OR, NOT, XOR) – De Morgan's laws – propositional logic. Set theory – sets, subsets, power sets – set operations (union, intersection, difference) – Venn diagrams. Relations and functions – types of relations – equivalence relations – functions and mappings – composition of functions. Graph theory basics – paths and cycles – connectivity – Euler and Hamiltonian paths – graph coloring problems. Finite state machines – deterministic and non-deterministic automata – state diagrams – applications in pattern matching.

UNIT: V	PRACTICAL APPLICATIONS AND PROGRAMMING	9

Python programming for computational thinking – implementing algorithms – data structure implementations. Problem-solving with code – coding best practices – debugging strategies – testing and validation. Computational thinking in AI/ML – feature extraction as abstraction – algorithm selection – optimization problems in machine learning. Real-world applications – computational biology – cryptography basics – scheduling and optimization – game theory. Competitive programming – problem-solving techniques – online judges (LeetCode, HackerRank) – time and memory constraints – practice problems and challenges.

Total Periods: 45

Pedagogical Tools:	Python, Jupyter Notebooks, Algorithm visualization tools, Online coding platforms (LeetCode, HackerRank), Flowchart tools, Pseudocode practice

TEXT BOOKS:

Sl. No	Authors	Title of the Book	Publisher	Year
1	John V. Guttag	Introduction to Computation and Programming Using Python, 3rd Edition	MIT Press	2021
2	Thomas H. Cormen, Charles E. Leiserson, Ronald L. Rivest, Clifford Stein	Introduction to Algorithms, 4th Edition	MIT Press	2022

REFERENCE BOOKS:

Sl. No	Authors	Title of the Book	Publisher	Year
1	Paul Curzon, Peter W. McOwan	The Power of Computational Thinking	World Scientific	2017
2	Aditya Bhargava	Grokking Algorithms	Manning	2016
3	Steven S. Skiena	The Algorithm Design Manual, 3rd Edition	Springer	2020

COs / POs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	3	2	1	2	–	–	–	–	–	–	1	3	2
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CO4	3	3	2	2	2	–	–	–	–	–	–	1	3	2
CO5	2	3	2	1	3	–	–	–	1	1	–	1	3	3

R 2024	AI&DS					
24ADX05	WEB TECHNOLOGIES	L	T	P	C	PE
		3	0	0	3	

COURSE OBJECTIVES:

1. To teach students the basics of server side scripting using PHP
2. To explain web application development procedures
3. To impart servlet technology for writing business logic
4. To facilitate students to connect to databases using JDBC
5. To familiarize various concepts of application development using JSP

COURSE OUTCOMES:

At the end of this course, students are able to

CO1: Create web pages using PHP

CO2: Identify the difference between the HTML PHP and XML documents.

CO3: Identify the engineering structural design of XML and parse tree

CO4: Analyze the difference between and PHP and XML.

CO5: Understand the concept of JAVA SCRIPTS.

UNIT: I	INTRODUCTION TO PHP	9

Declaring variables, data types, arrays, strings, operations, expressions, control structures, functions, Reading data from web form controls like Text Boxes, radio buttons, lists etc., Handling File Uploads, Connecting to database (My SQL as reference), executing simple queries, handling results, Handling sessions and cookies.

UNIT: II	CLIENT SIDE SCRIPTING	9

Introduction to JavaScript: JavaScript language – declaring variables, scope of variables functions, event handlers (on click, on submit etc.), Document Object Model, Form validations. Simple AJAX applications.

UNIT: III	XML	9

Introduction to XML, Defining XML tags, their attributes and values, Document type definition, XML Schemas, Document Object model, XHTML Parsing XML Data - DOM and SAX parsers in java

UNIT: IV	INTRODUCTION TO SERVLETS	9

Sketching Principles - Sketching Red Routes - Responsive Design – Wireframing - Creating Wireflows - Building a Prototype - Building High-Fidelity Mockups - Designing Efficiently with Tools - Interaction Patterns - Conducting Usability Tests

UNIT: V	INTRODUCTION TO JSP	9

Identifying and Writing Problem Statements - Identifying Appropriate Research Methods - Creating Personas - Solution Ideation - Creating User Stories - Creating Scenarios - Flow Diagrams - Flow Mapping - Information Architecture.

Total Periods :45

Pedagogical Tools:	Black board, chalk, group discussion, role play, youtube videos, NPTEL videos

List of Experiments:

1. Create a web page with the following using HTML.
    - To embed an image map in a web page.
    - To fix the hot spots.

• Show all the related information when the hot spots are clicked.

2. Create a web page with all types of Cascading style sheets.
3. Client Side Scripts for Validating Web Form Controls using DHTML.
4. Installation of Apache Tomcat web server.
5. Write programs in Java using Servlets:
    - To invoke servlets from HTML forms.
    - Session Tracking.
6. Write programs in Java to create three-tier applications using JSP and Databases in a database which has been stored in a database server.
7. Programs using XML – Schema – XSLT/XSL.

TEXT BOOKS:				
Sl. No	**Authors**	**Title of the Book**	**Publisher**	**Year of publication**
1	Uttam K. Roy	Web Technologies	Oxford University Press	2010
2	Steven Holzner	The Complete Reference PHP	Tata McGraw-Hill	2007

REFERENCE BOOKS:				
Sl. No	**Authors**	**Title of the Book**	**Publisher**	**Year of publication**
1	Chris Bates	Web Programming: Building Internet Applications (2nd Edition)	Wiley Dreamtech	2006
2	Hans Bergsten	Java Server Pages	SPD O'Reilly	2003
3	David Flanagan	JavaScript	O'Reilly, SPD	2006
4	Jon Duckett	Beginning Web Programming	WROX	2004
5	Robert W. Sebesta	Programming the World Wide Web (4th Edition)	Pearson	2008
6	H.M. Deitel and P.J. Deitel, A.B. Goldberg	Internet & World Wide Web: How to Program	Pearson	2012

COs / POs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	2	2	3	1	3	–	–	–	–	–	–	1	3	2
CO2	2	3	2	1	2	–	–	–	–	–	–	1	2	2
CO3	2	3	2	2	2	–	–	–	–	–	–	1	2	2
CO4	2	3	2	2	2	–	–	–	–	–	–	1	2	2
CO5	2	2	2	1	3	–	–	–	–	–	–	1	3	2

R 2024	AI&DS					
24ADX09	AI AGENTS	L	T	P	C	PE
		3	0	0	3	

COURSE OBJECTIVES:

- Understand the fundamentals of autonomous AI agents and their architectures
- Master agentic workflows, planning, and reasoning techniques for intelligent systems
- Learn to build single-agent and multi-agent systems with LLMs
- Implement tool-use, memory management, and agent orchestration patterns
- Explore real-world applications of AI agents across various domains

COURSE OUTCOMES:

CO 1: Design and develop autonomous AI agents with planning and reasoning capabilities
CO 2: Build agentic workflows using frameworks like LangChain, LangGraph, and AutoGen
CO 3: Implement tool-calling, memory systems, and agent orchestration for complex tasks
CO 4: Create multi-agent systems with collaboration and coordination mechanisms
CO 5: Deploy production-ready AI agents for real-world business and research applications

UNIT: I	FOUNDATIONS OF AI AGENTS	9

Introduction to AI agents -- definition and characteristics -- agent types (reflex, model-based, goal-based, utility-based). Agent environments -- properties (observable, deterministic, episodic, static, discrete) -- performance measures. Foundation models for agents -- LLMs as reasoning engines -- prompt engineering for agents -- in-context learning and few-shot prompting.

UNIT: II	AGENT ARCHITECTURE AND REASONING	9

Agent reasoning patterns -- ReAct (Reasoning + Acting) -- Chain-of-Thought (CoT) -- Tree-of-Thought -- self-reflection. Planning and task decomposition -- hierarchical task networks -- goal-based planning -- plan-and-execute pattern. Tool use and function calling -- tool selection -- API integration -- code execution agents. Building agents with LangChain -- chains -- agents -- callbacks -- implementing basic ReAct agents.

UNIT: III	AGENTIC WORKFLOWS AND ORCHESTRATION	9

Agent workflows -- sequential -- conditional -- parallel execution patterns. LangGraph for agentic workflows -- state graphs -- nodes and edges -- cyclic workflows -- human-in-the-loop patterns. Memory systems -- short-term and long-term memory -- vector stores for memory -- conversation history management -- semantic memory with embeddings. Observability and debugging -- logging agent decisions -- trajectory visualization -- error handling and recovery strategies.

UNIT: IV	MULTI-AGENT SYSTEMS	9

Multi-agent architectures -- hierarchical -- swarm -- peer-to-peer coordination. Agent collaboration patterns -- supervisor-worker -- hand-off -- debate -- consensus-building. Building multi-agent systems -- AutoGen framework -- CrewAI -- agent communication protocols. Multi-agent coordination -- task allocation -- conflict resolution -- resource sharing -- implementing team workflows.

UNIT: V	ADVANCED TOPICS AND APPLICATIONS	9

Model Context Protocol (MCP) -- standardized agent-tool communication -- server-client architecture. Agent evaluation -- success metrics -- benchmark datasets -- trajectory analysis -- measuring agent reliability. Real-world applications -- coding agents (software engineering assistants) -- research agents (scientific

discovery) -- business process automation. Ethical considerations -- agent safety -- alignment -- transparency -- responsible deployment -- case studies of deployed agent systems.

Total Periods: 45

Pedagogical Tools:	Python, LangChain, LangGraph, AutoGen, CrewAI, OpenAI API, Anthropic API, Vector Databases (Pinecone, Weaviate), Jupyter Notebooks, VS Code, Docker, Model Context Protocol tools, Agent evaluation frameworks

TEXT BOOKS:

Sl. No	Authors	Title of the Book	Publisher	Year
1	Micheal Lanham	AI Agents in Action	Manning	2025
2	Victor Dibia	Designing Multi-Agent Systems	Self-published	2025

REFERENCE BOOKS:

Sl. No	Authors	Title of the Book	Publisher	Year
1	Michael Albada	Building Applications with AI Agents	O'Reilly	2025
2	Roberto Infante	AI Agents and Applications: With LangChain, LangGraph, and MCP	Manning	2026
3	Stuart Russell, Peter Norvig	Artificial Intelligence: A Modern Approach, 4th Edition	Pearson	2020

COs / POs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	3	3	2	3	–	–	–	1	1	–	1	3	3
CO2	2	3	3	2	3	–	–	–	1	1	–	1	3	3
CO3	2	3	3	2	3	–	–	–	1	1	1	1	3	3
CO4	2	3	3	2	3	–	–	–	2	1	1	1	3	3
CO5	2	3	3	2	3	1	–	–	2	2	2	2	3	3

R 2024	AI&DS					
24ADX08	AI IN E-COMMERCE	L	T	P	C	PE
		3	0	0	3	

COURSE OBJECTIVES:

• Understand the fundamentals of AI and its applications in e-commerce ecosystems
• Master machine learning techniques for personalization and recommendation systems
• Learn to implement conversational AI, chatbots, and virtual shopping assistants
• Apply AI for fraud detection, pricing optimization, and supply chain management
• Explore emerging AI technologies like visual search, voice commerce, and autonomous agents

COURSE OUTCOMES:

CO 1: Design and implement AI-powered recommendation systems for e-commerce platforms
CO 2: Build conversational AI solutions including chatbots and virtual assistants for customer service
CO 3: Develop predictive analytics models for demand forecasting and inventory management
CO 4: Implement AI-based fraud detection and dynamic pricing strategies
CO 5: Create visual search, voice commerce, and AR-enabled shopping experiences

UNIT: I	AI FOUNDATIONS FOR E-COMMERCE	9

Introduction to AI in e-commerce -- evolution and market trends -- AI ecosystem components. Types of AI: machine learning, deep learning, NLP, computer vision -- e-commerce landscape (B2B, B2C, C2C models). Data collection and management -- ETL processes -- Python for AI (NumPy, Pandas, Matplotlib) -- working with e-commerce APIs.

UNIT: II	PERSONALIZATION AND RECOMMENDATION SYSTEMS	9

Machine learning fundamentals -- supervised vs unsupervised learning -- model evaluation. Collaborative filtering (user-based and item-based) -- content-based filtering -- hybrid systems. Deep learning for recommendations -- neural collaborative filtering -- embedding layers. Customer segmentation -- RFM analysis -- A/B testing -- real-time personalization with online learning.

UNIT: III	CONVERSATIONAL AI AND CUSTOMER EXPERIENCE	9

NLP basics -- tokenization, NER -- chatbot architecture (rule-based vs AI-powered). Building chatbots with Rasa and Dialogflow -- LLM integration (GPT-4, Claude API). Prompt engineering -- RAG with vector databases -- sentiment analysis. Voice commerce -- speech recognition -- text-to-speech -- generative AI for content creation.

UNIT: IV	PREDICTIVE ANALYTICS AND OPERATIONS	9

Demand forecasting -- time series analysis (ARIMA, Prophet) -- inventory optimization. Dynamic pricing -- price elasticity modeling -- reinforcement learning for pricing. Customer lifetime value prediction -- churn prediction with XGBoost. Fraud detection -- anomaly detection -- isolation forest -- autoencoders -- handling imbalanced data.

UNIT: V	EMERGING AI TECHNOLOGIES AND FUTURE TRENDS	9

Computer vision -- CNNs for image classification -- visual search with ResNet and CLIP. AR shopping -- virtual try-on -- 3D product visualization with WebXR. AI agents and autonomous commerce -- agentic AI architecture -- auto-replenishment systems. Explainable AI -- SHAP and LIME -- ethical AI -- bias mitigation -- privacy-preserving ML.

Total Periods: 45

Pedagogical Tools:	Python, Jupyter Notebooks, Scikit-learn, TensorFlow/Keras, PyTorch, Rasa, LangChain, OpenAI API, Pandas, NumPy, Matplotlib, Seaborn, VS Code, Google Colab, E-commerce APIs (Shopify, Amazon), Vector databases, A/B testing platforms

TEXT BOOKS:

Sl. No	Authors	Title of the Book	Publisher	Year
1	Ramgopal Prajapat	AI-Powered Ecommerce: How Machine Learning Is Transforming Online Shopping	Springer	2024
2	Henry Perlmutter	AI Ecommerce: Mastering Ecommerce with AI and ChatGPT	Self-published	2024

REFERENCE BOOKS:

Sl. No	Authors	Title of the Book	Publisher	Year
1	Loveleen Gaur, et al.	Role of Explainable Artificial Intelligence in E-Commerce	Springer	2024
2	Yiliu Paul Tu, Maomao Chi (Eds.)	E-Business: Generative AI and Management Transformation (WHICEB 2025)	Springer	2025
3	Aurélien Géron	Hands-On Machine Learning with Scikit-Learn, Keras, and TensorFlow, 3rd Edition	O'Reilly	2022

COs / POs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	3	3	2	3	–	–	–	1	1	–	1	3	3
CO2	2	3	3	2	3	–	–	–	1	2	–	1	3	3
CO3	3	3	3	2	3	1	–	–	1	1	1	1	3	3
CO4	3	3	3	2	3	1	–	–	1	1	1	1	3	3
CO5	2	2	3	2	3	1	–	–	1	2	1	1	3	3

R 2024	AI&DS					
24ADX07	DATA EXPLORATION TECHNIQUES	L	T	P	C	PE
		3	0	0	3	

COURSE OBJECTIVES:

- Understand the fundamentals of exploratory data analysis and statistical techniques
- Master data wrangling, cleaning, and preprocessing methods for various data types
- Learn to identify patterns, trends, and anomalies through statistical analysis
- Apply dimensionality reduction and feature engineering techniques
- Develop skills in automated data profiling and modern exploration tools

COURSE OUTCOMES:

CO 1: Perform comprehensive exploratory data analysis using statistical methods
CO 2: Clean and transform complex datasets with advanced data wrangling techniques
CO 3: Apply multivariate analysis and correlation techniques to uncover relationships
CO 4: Implement feature engineering and dimensionality reduction for machine learning pipelines
CO 5: Utilize modern tools for automated data profiling and quality assessment

UNIT: I	FOUNDATIONS OF DATA EXPLORATION	9

Introduction to exploratory data analysis (EDA) -- importance and workflow -- data types (structured, unstructured, semi-structured). Descriptive statistics -- measures of central tendency, dispersion, and shape -- univariate analysis techniques. Python for data exploration -- NumPy arrays -- Pandas DataFrames -- basic operations and data manipulation. Data profiling -- automated profiling with pandas-profiling and ydata-profiling -- summary statistics -- data quality metrics.

UNIT: II	DATA CLEANING AND PREPROCESSING	9

Handling missing data -- detection techniques -- imputation methods (mean, median, KNN, MICE) -- deletion strategies. Outlier detection and treatment -- statistical methods (IQR, Z-score) -- visualization techniques -- domain-specific handling. Data transformation -- normalization and standardization -- log transformation -- Box-Cox transformation -- encoding categorical variables (one-hot, label, target encoding). Data consistency -- duplicate detection -- format standardization -- string manipulation -- regular expressions for data cleaning.

UNIT: III	STATISTICAL ANALYSIS AND PATTERN DISCOVERY	9

Bivariate analysis -- correlation analysis (Pearson, Spearman, Kendall) -- cross-tabulation and contingency tables -- chi-square tests. Distribution analysis -- probability distributions -- normality tests (Shapiro-Wilk, Kolmogorov-Smirnov) -- Q-Q plots. Hypothesis testing -- t-tests, ANOVA -- non-parametric tests -- p-values and statistical significance. Time series exploration -- trend analysis -- seasonality detection -- autocorrelation -- moving averages and rolling statistics.

UNIT: IV	DIMENSIONALITY REDUCTION AND FEATURE ENGINEERING	9

Dimensionality reduction -- PCA (Principal Component Analysis) -- t-SNE -- UMAP -- explained variance and scree plots. Feature engineering -- feature creation techniques -- polynomial features -- interaction features -- binning and discretization. Feature selection -- filter methods (variance threshold, correlation) -- wrapper methods (forward/backward selection) -- embedded methods (Lasso, Ridge). Advanced techniques -- feature importance from tree models -- automated feature engineering with Featuretools -- handling high-cardinality categorical variables.

UNIT: V	ADVANCED EXPLORATION AND MODERN TOOLS	9

Text data exploration -- text preprocessing -- TF-IDF -- word clouds -- topic modeling basics. Exploring large datasets -- Dask for parallel computing -- sampling strategies -- working with SQL databases -- chunking and batch processing. Automated ML tools -- AutoML for feature discovery -- DataRobot -- H2O.ai -- automated insights generation. Data quality frameworks -- Great Expectations -- data validation -- schema enforcement -- monitoring data drift -- modern EDA tools (Sweetviz, D-Tale).

Total Periods: 45

Pedagogical Tools:	Python, Pandas, NumPy, Scikit-learn, Matplotlib, Seaborn, Jupyter Notebooks, ydata-profiling, Sweetviz, D-Tale, Featuretools, Great Expectations, Dask, SQL, Excel, Scipy, Statsmodels

TEXT BOOKS:

Sl. No	Authors	Title of the Book	Publisher	Year
1	Wes McKinney	Python for Data Analysis, 3rd Edition	O'Reilly	2022
2	Jake VanderPlas	Python Data Science Handbook, 2nd Edition	O'Reilly	2023

REFERENCE BOOKS:

Sl. No	Authors	Title of the Book	Publisher	Year
1	Alice Zheng, Amanda Casari	Feature Engineering for Machine Learning	O'Reilly	2018
2	Max Kuhn, Kjell Johnson	Feature Engineering and Selection: A Practical Approach	CRC Press	2019
3	Brett Lantz	Machine Learning with R, 4th Edition	Packt	2023

COs / POs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	3	2	2	2	–	–	–	–	–	–	1	3	2
CO2	3	3	2	2	3	–	–	–	–	–	–	1	3	3
CO3	3	3	2	3	2	–	–	–	–	–	–	1	3	3
CO4	3	3	3	2	3	–	–	–	–	–	–	1	3	3
CO5	2	3	2	2	3	–	–	–	–	–	–	1	3	3

R 2024	AI&DS					
24ADX08	AI IN E-COMMERCE	L	T	P	C	PE
		3	0	0	3	

COURSE OBJECTIVES:

- Understand the fundamentals of AI and its applications in e-commerce ecosystems
- Master machine learning techniques for personalization and recommendation systems
- Learn to implement conversational AI, chatbots, and virtual shopping assistants
- Apply AI for fraud detection, pricing optimization, and supply chain management
- Explore emerging AI technologies like visual search, voice commerce, and autonomous agents

COURSE OUTCOMES:

CO 1: Design and implement AI-powered recommendation systems for e-commerce platforms
CO 2: Build conversational AI solutions including chatbots and virtual assistants for customer service
CO 3: Develop predictive analytics models for demand forecasting and inventory management
CO 4: Implement AI-based fraud detection and dynamic pricing strategies
CO 5: Create visual search, voice commerce, and AR-enabled shopping experiences

UNIT: I	AI FOUNDATIONS FOR E-COMMERCE	9
Introduction to AI in e-commerce -- evolution and market trends -- AI ecosystem components. Types of AI: machine learning, deep learning, NLP, computer vision -- e-commerce landscape (B2B, B2C, C2C models). Data collection and management -- ETL processes -- Python for AI (NumPy, Pandas, Matplotlib) -- working with e-commerce APIs.		
UNIT: II	**PERSONALIZATION AND RECOMMENDATION SYSTEMS**	**9**
Machine learning fundamentals -- supervised vs unsupervised learning -- model evaluation. Collaborative filtering (user-based and item-based) -- content-based filtering -- hybrid systems. Deep learning for recommendations -- neural collaborative filtering -- embedding layers. Customer segmentation -- RFM analysis -- A/B testing -- real-time personalization with online learning.		
UNIT: III	**CONVERSATIONAL AI AND CUSTOMER EXPERIENCE**	**9**
NLP basics -- tokenization, NER -- chatbot architecture (rule-based vs AI-powered). Building chatbots with Rasa and Dialogflow -- LLM integration (GPT-4, Claude API). Prompt engineering -- RAG with vector databases -- sentiment analysis. Voice commerce -- speech recognition -- text-to-speech -- generative AI for content creation.		
UNIT: IV	**PREDICTIVE ANALYTICS AND OPERATIONS**	**9**
Demand forecasting -- time series analysis (ARIMA, Prophet) -- inventory optimization. Dynamic pricing -- price elasticity modeling -- reinforcement learning for pricing. Customer lifetime value prediction -- churn prediction with XGBoost. Fraud detection -- anomaly detection -- isolation forest -- autoencoders -- handling imbalanced data.		
UNIT: V	**EMERGING AI TECHNOLOGIES AND FUTURE TRENDS**	**9**
Computer vision -- CNNs for image classification -- visual search with ResNet and CLIP. AR shopping -- virtual try-on -- 3D product visualization with WebXR. AI agents and autonomous commerce -- agentic AI architecture -- auto-replenishment systems. Explainable AI -- SHAP and LIME -- ethical AI -- bias mitigation -- privacy-preserving ML.		

Total Periods: 45

Pedagogical Tools:	Python, Jupyter Notebooks, Scikit-learn, TensorFlow/Keras, PyTorch, Rasa, LangChain, OpenAI API, Pandas, NumPy, Matplotlib, Seaborn, VS Code, Google Colab, E-commerce APIs (Shopify, Amazon), Vector databases, A/B testing platforms

TEXT BOOKS:

Sl. No	Authors	Title of the Book	Publisher	Year
1	Ramgopal Prajapat	AI-Powered Ecommerce: How Machine Learning Is Transforming Online Shopping	Springer	2024
2	Henry Perlmutter	AI Ecommerce: Mastering Ecommerce with AI and ChatGPT	Self-published	2024

REFERENCE BOOKS:

Sl. No	Authors	Title of the Book	Publisher	Year
1	Loveleen Gaur, et al.	Role of Explainable Artificial Intelligence in E-Commerce	Springer	2024
2	Yiliu Paul Tu, Maomao Chi (Eds.)	E-Business: Generative AI and Management Transformation (WHICEB 2025)	Springer	2025
3	Aurélien Géron	Hands-On Machine Learning with Scikit-Learn, Keras, and TensorFlow, 3rd Edition	O'Reilly	2022

COs / POs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	3	3	2	3	–	–	–	1	1	–	1	3	3
CO2	2	3	3	2	3	–	–	–	1	2	–	1	3	3
CO3	3	3	3	2	3	1	–	–	1	1	1	1	3	3
CO4	3	3	3	2	3	1	–	–	1	1	1	1	3	3
CO5	2	2	3	2	3	1	–	–	1	2	1	1	3	3

R 2024	AI&DS					
24ADX10	GENERATIVE AI	L	T	P	C	PE
		3	0	0	3	

COURSE OBJECTIVES:

- Understand the fundamentals of generative AI models and their architectures
- Master large language models, transformers, and prompt engineering techniques
- Learn to build applications using generative models for text, image, audio, and video
- Implement fine-tuning, RAG, and advanced techniques for customizing generative models
- Explore ethical considerations, safety, and responsible deployment of generative AI

COURSE OUTCOMES:

CO 1: Design and develop applications using large language models and transformers

CO 2: Build text generation systems with advanced prompt engineering and RAG techniques

CO 3: Create image, audio, and video generation applications using diffusion models

CO 4: Implement model fine-tuning, evaluation, and optimization strategies

CO 5: Deploy generative AI systems with ethical considerations and safety measures

UNIT: I	FOUNDATIONS OF GENERATIVE AI	9

Introduction to generative AI -- discriminative vs generative models -- applications across domains. Neural network fundamentals -- deep learning basics -- backpropagation -- optimization algorithms (SGD, Adam). Generative model families -- autoregressive models -- VAEs -- GANs -- diffusion models -- transformer architectures. Ethics and safety -- bias in generative models -- deepfakes -- misinformation -- responsible AI principles.

UNIT: II	LARGE LANGUAGE MODELS AND TRANSFORMERS	9

Transformer architecture -- self-attention mechanism -- multi-head attention -- positional encoding. Large language models -- GPT family -- BERT -- T5 -- Claude -- LLaMA -- model scaling laws. Tokenization -- BPE -- WordPiece -- SentencePiece -- vocabulary construction. Prompt engineering -- zero-shot, few-shot, chain-of-thought prompting -- prompt design patterns -- system prompts -- temperature and sampling strategies.

UNIT: III	TEXT GENERATION AND ADVANCED TECHNIQUES	9

Retrieval Augmented Generation (RAG) -- vector databases -- semantic search -- embedding models -- chunking strategies. Fine-tuning LLMs -- full fine-tuning vs LoRA/QLoRA -- instruction tuning -- RLHF (Reinforcement Learning from Human Feedback). Building LLM applications -- API integration (OpenAI, Anthropic, Hugging Face) -- LangChain for application development -- streaming responses. Advanced techniques -- function calling -- structured output generation -- model distillation -- quantization for efficient deployment.

UNIT: IV	MULTIMODAL GENERATIVE AI	9

Image generation -- diffusion models (Stable Diffusion, DALL-E, Midjourney) -- U-Net architecture -- noise scheduling -- conditioning techniques. Text-to-image generation -- CLIP embeddings -- prompt engineering for images -- ControlNet for controlled generation -- image editing with inpainting. Audio and speech -- text-to-speech (TTS) models -- speech recognition -- voice cloning -- music generation with MusicGen. Video generation -- Sora, Runway, Pika -- temporal consistency -- video editing -- frame interpolation -- multimodal models like GPT-4V and Gemini.

UNIT: V	EVALUATION, DEPLOYMENT, AND FUTURE TRENDS	9
Model evaluation -- perplexity -- BLEU, ROUGE scores -- human evaluation -- benchmark datasets (MMLU, HumanEval). Safety and alignment -- jailbreaking -- adversarial attacks -- red teaming -- content filtering -- Constitutional AI. Deployment strategies -- model serving -- API design -- cost optimization -- latency reduction -- caching strategies. Emerging trends -- mixture of experts (MoE) -- multimodal foundation models -- AI-generated synthetic data -- neuromorphic computing -- future of generative AI and AGI.		

Total Periods: 45

Pedagogical Tools:	Python, PyTorch, TensorFlow, Hugging Face Transformers, OpenAI API, Anthropic API, Stable Diffusion, LangChain, Vector Databases (Pinecone, ChromaDB), Jupyter Notebooks, Google Colab, Weights & Biases, Gradio, Streamlit

TEXT BOOKS:

Sl. No	Authors	Title of the Book	Publisher	Year
1	Mark Liu	Learn Generative AI with PyTorch	Manning	2024
2	Takeshi Okadome	Essentials of Generative AI	Springer	2025

REFERENCE BOOKS:

Sl. No	Authors	Title of the Book	Publisher	Year
1	Ben Auffarth, Leonid Kuligin	Generative AI with LangChain, 2nd Edition	Packt	2024
2	Sebastian Raschka	Build a Large Language Model (From Scratch)	Manning	2024
3	Ian Goodfellow, et al.	Deep Learning	MIT Press	2016

COs / POs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	3	3	2	3	–	–	–	1	1	–	1	3	3
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CO3	2	3	3	2	3	–	–	–	1	1	–	1	3	3
CO4	3	3	3	2	3	–	–	–	1	1	1	1	3	3
CO5	2	2	2	2	3	1	–	3	1	2	1	2	3	3

R 2024	CSE					
24CSX22	WEB APPLICATION SECURITY	L	T	P	C	PE
		3	0	2	3	

COURSE OBJECTIVES:

1. To reveal the underlying in web application.
2. To identify and aid in fixing any security vulnerabilities during the web development process.
3. To understand the security principles in developing a reliable web application.
4. Secure software requirements gathering to design, development, configuration, deployment, and ongoing maintenance
5. Security of enterprise information systems.

COURSE OUTCOMES:

CO1: Identify the vulnerabilities in the web applications.
CO2: Identify the various types of threats and mitigation measures of web applications.
CO3: Apply the security principles in developing a reliable web application.
CO4: Use industry standard tools for web application security.
CO5: Apply penetration testing to improve the security of web applications.

UNIT: I	OVERVIEW OF WEB APPLICATIONS	9

Introduction history of web applications interface ad structure benefits and drawbacks of web applications Web application Vs Cloud application.

UNIT: II	WEB APPLICATION SECURITY FUNDAMENTALS	9

Security Fundamentals: Input Validation - Attack Surface Reduction Rules of Thumb- Classi- fying and Prioritizing Threads

UNIT: III	BROWSER SECURITY PRINCIPLES	9

Origin Policy - Exceptions to the Same-Origin Policy - Cross-Site Scripting and Cross-Site Request Forgery - Reflected XSS - HTML Injection

UNIT: IV	WEB APPLICATION VULNERABILITIES	9

Understanding vulnerabilities in traditional client server application and web applications, client state manipulation, cookie based attacks, SQL injection, cross domain attack (XSS/XSRF/XSSI) http header injection. SSL vulnerabilities and testing

UNIT: V	WEB APPLICATION MITIGATIONS	9

Http request , http response, rendering and events , html image tags, image tag security, issue, java script on error , Javascript timing , port scanning , remote scripting , running remotecode, frame and iframe, browser sandbox, policy goals, same origin policy, library import, domain relaxation

Total Periods: 45

PRACTICAL EXERCISES

1. Install wireshark and explore the various protocols
    a. Analyze the difference between HTTP vs HTTPS
    b. Analyze the various security mechanisms embedded with different protocols.
2. Identify the vulnerabilities using OWASP ZAP tool
3. Create simple REST API using python for following operation.
    a. PUSH b. POST c. DELETE
4. Install Burp Suite to do following vulnerabilities: SQL injection
    a. cross-site scripting (XSS)
5. Attack the website using Social Engineering method

Pedagogical Tools:	Python, Jupyter Notebooks, Algorithm visualization tools, Online coding platforms (LeetCode, HackerRank), Flowchart tools, Pseudocode practice

TEXT BOOKS:

Sl. No	Authors	Title of the Book	Publisher	Year
1	Bryan Sullivan & Vincent Liu	Web Application Security: A Beginner's Guide	McGraw Hill Professional	2011

REFERENCE BOOKS:

Sl. No	Authors	Title of the Book	Publisher	Year
1	Dafydd Stuttard & Marcus Pinto	The Web Application Hacker's Handbook: Finding and Exploiting Security Flaws	John Wiley & Sons	2011

COs / POs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
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CO5	2	1	2	1	3	–	–	–	1	1	–	1	3	3

R 2024	AI&DS					
24CSX17	SECURITY ASSESSMENT AND RISK ANALYSIS	L	T	P	C	PE
		3	0	0	3	

COURSE OBJECTIVES:

1. The course takes a software development perspective to the challenges of engineering software systems that are secure.
2. This course addresses design and implementation issues critical to producing secure software systems.
3. The course deals with the question of how to make the requirements for confidentiality, integrity, and availability integral to the software development process.
4. Secure software requirements gathering to design, development, configuration, deployment, and ongoing maintenance
5. Security of enterprise information systems.

COURSE OUTCOMES:

CO1: Understand various aspects and principles of software security.
CO2: Devise security models for implementing at the design level.
CO3: Identify and analyze the risks associated with s/w engineering and use relevant models to mitigate the risks.
CO4: Understand the various security algorithms to implement for secured computing and computer networks
CO5: Explain different security frameworks for different types of systems including electronic systems.

UNIT: I	**Fundamentals of Computer and Software Security**	**9**

Defining computer security, the principles of secure software, trusted computing base, etc, threat modeling, advanced techniques for mapping security requirements into design specifications. Secure software implementation, deployment and ongoing management.

UNIT: II	**Secure Software Design and Design Patterns**	**9**

Software design and an introduction to hierarchical design representations. Difference between high-level and detailed design. Handling security with high-level design. General Design Notions. Security concerns designs at multiple levels of abstraction, Design patterns.

UNIT: III	**Software Assurance and Risk Management**	**9**

Software Assurance Model: Identify project security risks & selecting risk management strategies, Risk Management Framework, Security Best practices/ Known Security Flaws, Architectural risk analysis

UNIT: IV	**Enterprise Information Security and Cryptography**	**9**

Software Security in Enterprise Business: Identification and authentication, Enterprise Information Security, Symmetric and asymmetric cryptography, including public key cryptography, data encryption standard (DES), advanced encryption standard (AES).

UNIT: V	**Security Frameworks and Secure Information Systems**	**9**

Security development frameworks. Security issues associated with the development and deployment of information systems, including Internet-based e-commerce, e-business, and e-service systems.

Total Periods: 45

Pedagogical Tools:	Python, Jupyter Notebooks, Algorithm visualization tools, Online coding platforms (LeetCode, HackerRank), Flowchart tools, Pseudocode practice

TEXT BOOKS:

Sl. No	Authors	Title of the Book	Publisher	Year
1	W. Stallings	Cryptography and Network Security: Principles and Practice, 5th Edition	Prentice Hall, Upper Saddle River, NJ	2011
2	C. Kaufman, R. Perlman & M. Speciner	Network Security: Private Communication in a Public World, 2nd Edition	Prentice Hall, Upper Saddle River, NJ	2002

3	C. P. Pfleeger & S. L. Pfleeger	Security in Computing, 4th Edition	Prentice Hall, Upper Saddle River, NJ	2007
4	M. Merkow & J. Breithaupt	Information Security: Principles and Practices	Prentice Hall, Upper Saddle River, NJ	2005

REFERENCE BOOKS:

Sl. No	Authors	Title of the Book	Publisher	Year
1	Gary McGraw	Software Security: Building Security In	Addison-Wesley	2006

COs / POs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	3	2	1	2	–	–	–	–	–	–	1	3	2
CO2	3	3	3	2	2	–	–	–	–	–	–	1	3	3
CO3	3	3	3	1	3	–	–	–	–	–	–	1	3	3
CO4	3	3	2	2	2	–	–	–	–	–	–	1	3	2
CO5	2	3	2	1	3	–	–	–	1	1	–	1	3	3

R 2024	AI&DS					
24CSX18	ETHICAL HACKING	L	T	P	C	PE
		3	0	2	3	

COURSE OBJECTIVES:

- To understand the basics of computer based vulnerabilities.
- To explore different foot printing, reconnaissance and scanning methods.
- To expose the enumeration and vulnerability analysis methods.
- To understand hacking options available in Web and wireless applications.
- To explore the options for network protection.

COURSE OUTCOMES:

CO1: To express knowledge on basics of computer based vulnerabilities
CO2: To gain understanding on different foot printing, reconnaissance and scanning methods.
CO3: To demonstrate the enumeration and vulnerability analysis methods
CO4: To gain knowledge on hacking options available in Web and wireless applications.
CO5: To acquire knowledge on the options for network protection.

UNIT: I	INTRODUCTION	9

Ethical Hacking Overview - Role of Security and Penetration Testers .- Penetration-Testing Methodologies- Laws of the Land - Overview of TCP/IP- The Application Layer - The Transport Layer - The Internet Layer - IP Addressing .- Network and Computer Attacks - Malware

UNIT: II	FOOT PRINTING, RECONNAISSANCE AND SCANNING NETWORKS	9

Footprinting Concepts - Footprinting through Search Engines, Web Services, Social Networking Sites, Website, Email - Competitive Intelligence - Footprinting through Social Engineering - Footprinting Tools - Network Scanning Concepts - Port-Scanning Tools - Scanning Techniques

UNIT: III	ENUMERATION AND VULNERABILITY ANALYSIS	9

Enumeration Concepts - NetBIOS Enumeration – SNMP, LDAP, NTP, SMTP and DNS Enumeration - Vulnerability Assessment Concepts - Desktop and Server OS Vulnerabilities - Windows OS Vulnerabilities

UNIT: IV	SYSTEM HACKING	9

Hacking Web Servers - Web Application Components- Vulnerabilities - Tools for Web Attackers and Security Testers Hacking Wireless Networks - Components of a Wireless Network – Wardriving Wireless Hacking

UNIT: V	NETWORK PROTECTION SYSTEMS	9

Access Control Lists. - Cisco Adaptive Security Appliance Firewall - Configuration and Risk Analysis Tools for Firewalls and Routers - Intrusion Detection and Prevention Systems - Network-Based and Host-Based IDSs and IPSs

Total Periods: 45

PRACTICAL EXERCISES:

1. Install Kali or Backtrack Linux / Metasploitable/ Windows XP
2. Practice the basics of reconnaissance.
3. Using FOCA / SearchDiggity tools, extract metadata and expanding the target list.
4. Aggregates information from public databases using online free tools like Paterva's Maltego.
5. Information gathering using tools like Robtex.
6. Scan the target using tools like Nessus.
7. View and capture network traffic using Wireshark.
8. Automate dig for vulnerabilities and match exploits using Armitage

Pedagogical Tools:	Python, Jupyter Notebooks, Algorithm visualization tools, Online coding platforms (LeetCode, HackerRank), Flowchart tools, Pseudocode practice

TEXT BOOKS:

Sl. No	Authors	Title of the Book	Publisher	Year
1	Michael T. Simpson, Kent Backman & James E. Corley	Hands-On Ethical Hacking and Network Defense	Course Technology, Delmar Cengage Learning	2010

REFERENCE BOOKS:

Sl. No	Authors	Title of the Book	Publisher	Year
1	Patrick Engebretson	The Basics of Hacking and Penetration Testing	Syngress, Elsevier	2013
2	Dafydd Stuttard & Marcus Pinto	The Web Application Hacker’s Handbook: Finding and Exploiting Security Flaws	Wiley	2011

COs / POs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	2	2	1	2	–	–	–	–	–	–	1	3	2
CO2	3	2	3	2	2	–	–	–	–	–	–	1	3	3
CO3	3	2	3	1	3	–	–	–	–	–	–	1	3	3
CO4	3	2	2	2	2	–	–	–	–	–	–	1	3	2
CO5	2	2	2	1	3	–	–	–	1	1	–	1	3	3

R 2024	CSE					
24CSX19	DIGITAL MOBILE FORENSICS	L	T	P	C	PE
		3	0	0	3	

COURSE OBJECTIVES:

1. This course will cover the fundamentals of digital forensics.
2. Provides an in-depth study of the rapidly changing and fascinating field of computer forensics.
3. Combines both the technical expertise and the knowledge required to investigate, detect and prevent digital crimes.
4. Knowledge on digital forensics legislations, digital crime, forensics processes and procedures, data acquisition and validation, e-discovery tools E-evidence collection
5. It provides preservation, investigating operating systems and file systems, network forensics, art of steganography and mobile device forensics.

COURSE OUTCOMES:

CO1: Understand relevant legislation and codes of ethics.
CO2: Investigate computer forensics and digital detective and various processes, policies and procedures data acquisition and validation, e-discovery tools.
CO3: Analyze E-discovery, guidelines and standards, E-evidence, tools and environment.
CO4: Apply the underlying principles of Email, web and network forensics to handle real life problems
CO5: Use IT Acts and apply mobile forensics techniques

UNIT: I	DIGITAL FORENSICS SCIENCE	9

Forensics science, computer forensics, and digital forensics. Computer Crime: Criminalistics as it relates to the investigative process, analysis of cyber-criminalistics area, challenges faced by digital forensics.

UNIT: II	CYBER CRIME SCENE ANALYSIS	9

Identifying digital evidence, collecting evidence in private-sector incident scenes, processing law enforcement crime scenes, preparing for a search, securing a computer incident or crime scene, seizing digital evidence at the scene.

UNIT: III	EVIDENCE MANAGEMENT & PRESENTATION	9

Create and manage shared folders using operating system, importance of the forensic mindset, define the workload of law enforcement, Types of Evidence, Define who should be notified of a crime, parts of gathering evidence.

UNIT: IV	COMPUTER FORENSICS	9

Preparing a computer case investigation, Procedures for corporate hi-tech investigations, conducting an investigation, Complete and critiquing the case. Network Forensics: Overview of network forensics, open-source security tools for network forensic analysis.

UNIT: V	MOBILE FORENSICS	9

Mobile forensics techniques, mobile forensics tools, recent trends in mobile forensic technique and methods to search and seizure electronic evidence. Legal Aspects of Digital Forensics: IT Act 2000, amendment of IT Act 2008.

Total Periods: 45

LAB EXPERIMENTS:

1. Installation of Sleuth Kit on Linux. List all data blocks. Analyze allocated as well as unallocated blocks of a disk image.
2. Data extraction from call logs using Sleuth Kit.
3. Data extraction from SMS and contacts using Sleuth Kit.
4. Install Mobile Verification Toolkit or MVT and decrypt encrypted iOS backups.
5. Process and parse records from the iOS system.
6. Extract installed applications from Android devices.
7. Extract diagnostic information from Android devices through the adb protocol.
8. Generate a unified chronological timeline of extracted records

Pedagogical Tools:	Python, Jupyter Notebooks, Algorithm visualization tools, Online coding platforms (LeetCode, HackerRank), Flowchart tools, Pseudocode practice

TEXT BOOKS:

Sl. No	Authors	Title of the Book	Publisher	Year
1	B. Nelson, A. Phillips & C. Steuart	Guide to Computer Forensics and Investigations, 4th Edition	Course Technology	2010

REFERENCE BOOKS:

Sl. No	Authors	Title of the Book	Publisher	Year
1	John Sammons	The Basics of Digital Forensics, 2nd Edition	Elsevier	2014
2	John Vacca	Computer Forensics: Computer Crime Scene Investigation, 2nd Edition	Laxmi Publications	2005

COs / POs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	2	2	1	2	–	–	–	–	–	–	1	2	2
CO2	3	3	3	2	2	–	–	–	–	–	–	1	2	3
CO3	3	3	3	1	3	–	–	–	–	–	–	1	2	3
CO4	3	2	2	2	2	–	–	–	–	–	–	1	3	2
CO5	2	3	2	1	3	–	–	–	1	1	–	1	1	3

R 2024	CSE					
24CSX20	CRYPTOCURRENCY AND BLOCK CHAIN TECHNOLOGIES	L	T	P	C	PE
		3	0	2	3	

COURSE OBJECTIVES:

- To understand the basics of Blockchain
- To learn Different protocols and consensus algorithms in Blockchain
- To learn the Blockchain implementation frameworks
- To understand the Blockchain Applications
- To experiment the Hyperledger Fabric, Ethereum networks

COURSE OUTCOMES:

CO1: Understand emerging abstract models for Blockchain Technology
CO2: Identify major research challenges and technical gaps existing between theory and practice in the crypto currency domain.
CO3: It provides conceptual understanding of the function of Blockchain as a method of securing distributed ledgers.
CO4: Apply hyperledger Fabric and Ethereum platform to implement the Block chain Application.
CO5: How consensus on their contents is achieved, and the new applications that they enable.

UNIT: I	INTRODUCTION TO BLOCKCHAIN	9

Blockchain- Public Ledgers, Blockchain as Public Ledgers - Block in a Blockchain, TransactionsThe Chain and the Longest Chain - Permissioned Model of Blockchain, Cryptographic -Hash Function, Properties of a hash function-Hash pointer and Merkle tree

UNIT: II	BITCOIN AND CRYPTOCURRENCY	9

A basic crypto currency, Creation of coins, Payments and double spending, FORTH – the precursor for Bitcoin scripting, Bitcoin Scripts , Bitcoin P2P Network, Transaction in Bitcoin Network, Block Mining, Block propagation and block relay

UNIT: III	BITCOIN CONSENSUS	9

Bitcoin Consensus, Proof of Work (PoW)- Hashcash PoW , Bitcoin PoW, Attacks on PoW ,monopoly problem- Proof of Stake- Proof of Burn - Proof of Elapsed Time - Bitcoin Miner, Mining Difficulty, Mining Pool-Permissioned model and use cases.

UNIT: IV	HYPERLEDGER FABRIC & ETHEREUM	9

Architecture of Hyperledger fabric v1.1- chain code- Ethereum: Ethereum network, EVM, Transaction fee, Mist Browser, Ether, Gas, Solidity.

UNIT: V	BLOCKCHAIN APPLICATIONS	9

Smart contracts, Truffle Design and issue- DApps- NFT. Blockchain Applications in Supply Chain Management, Logistics, Smart Cities, Finance and Banking, Insurance,etc- Case Study.

Total Periods: 45

Practical Experiments:
1. Install and understand Docker container, Node.js, Java and Hyperledger Fabric, Ethereum and perform necessary software installation on local machine/create instance on cloud to run.
2. Create and deploy a blockchain network using Hyperledger Fabric SDK for Java Set up and initialize the channel, install and instantiate chain code, and perform invoke and query on your blockchain network.
3. Interact with a blockchain network. Execute transactions and requests against a blockchain network by creating an app to test the network and its rules.
4. Deploy an asset-transfer app using blockchain. Learn app development within a Hyperledger Fabric network.
5. Use blockchain to track fitness club rewards. Build a web app that uses Hyperledger Fabric to track and trace member rewards.
6. Car auction network: A Hello World example with Hyperledger Fabric Node SDK and IBM Blockchain Starter Plan. Use Hyperledger Fabric to invoke chain code while storing results and data in the starter plan

Total Periods: 30

Pedagogical Tools:	Python, Jupyter Notebooks, Algorithm visualization tools, Online coding platforms (LeetCode, HackerRank), Flowchart tools, Pseudocode practice

TEXT BOOKS:

Sl. No	Authors	Title of the Book	Publisher	Year
1	Imran Bashir	Mastering Blockchain: Deeper Insights into Decentralization, Cryptography, Bitcoin, and Popular Blockchain Frameworks	Packt Publishing	2017
2	Andreas M. Antonopoulos	Mastering Bitcoin: Unlocking Digital Cryptocurrencies	O'Reilly Media	2014

REFERENCE BOOKS:

Sl. No	Authors	Title of the Book	Publisher	Year
1	Daniel Drescher	Blockchain Basics	Apress	2017
2	Arvind Narayanan, Joseph Bonneau, Edward Felten, Andrew Miller & Steven Goldfeder	Bitcoin and Cryptocurrency Technologies: A Comprehensive Introduction	Princeton University Press	2016
3	Melanie Swan	Blockchain: Blueprint for a New Economy	O'Reilly Media	2015
4	Ritesh Modi	Solidity Programming Essentials: A Beginner's Guide to Build Smart Contracts for Ethereum and Blockchain	Packt Publishing	2018

COs / POs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	2	2	2	1	2	–	–	–	–	–	–	1	3	2
CO2	3	3	3	2	2	–	–	–	–	–	–	1	3	3
CO3	3	3	3	1	3	–	–	–	–	–	–	1	3	3
CO4	3	3	2	2	2	–	–	–	–	–	–	1	3	2
CO5	2	3	2	1	3	–	–	–	1	1	–	1	3	3

R 2024	CSE					
24CSX21	SECURITY AND PRIVACY IN CLOUD	L	T	P	C	PE
		3	0	0	3	

COURSE OBJECTIVES:

- To understand the fundamental concepts of cloud computing and security challenges in cloud environments.
- To study security architectures, threats, and vulnerabilities associated with cloud computing.
- To learn privacy protection mechanisms and data security techniques in cloud platforms.
- To explore identity management, authentication, and access control mechanisms in cloud systems.
- To analyze security standards, compliance issues, and risk management in cloud-based applications.

COURSE OUTCOMES:

CO1 Explain the fundamentals of cloud computing and the security issues associated with cloud environments.

CO2 Analyze cloud threats, vulnerabilities, and security risks in different cloud service models.

CO3 Apply cryptographic techniques and privacy protection methods to secure cloud data.

CO4 Implement identity management, authentication, and access control mechanisms in cloud systems.

CO5 Evaluate cloud security frameworks, standards, and compliance requirements for secure cloud deployment.

UNIT: I	Fundamentals of Cloud Computing	9

Cloud Computing Defined, The SPI Framework for Cloud Computing, Relevant Technologies in Cloud Computing, The Traditional Software Model,The Cloud Services Delivery Model, Cloud Deployment Models, Key Drivers to Adopting the Cloud.

UNIT: II	Cloud Infrastructure and Data Security	9

Infrastructure Security: The Network Level, Infrastructure Security: The Host Level, Infrastructure Security: The Application Level

Data Security and Storage: Aspects of Data Security, Data Security Mitigation, Provider Data and Its Security

UNIT: III	Identity and Access Management in Cloud	9

Identity and Access Management: Trust Boundaries and IAM, Why IAM? IAM Challenges, IAM Definitions, IAM Architecture and Practice, Getting Ready for the Cloud, Relevant IAM Standards and Protocols for Cloud Services, IAM Practices in the Cloud, Cloud Authorization Management, Cloud Service Provider IAM Practice

UNIT: IV	Cloud Security Management and Access Control	9

Security Management in the Cloud: Security Management Standards, Security Management in the Cloud Availability Management, SaaS Availability Management PaaS Availability Management, IaaS Availability Management, Access Control.

UNIT: V	Privacy and Compliance in Cloud Computing	9

Privacy: What Is Privacy, What Is the Data Life Cycle, What Are the Key Privacy Concerns in the Cloud, Who Is Responsible for Protecting Privacy, Changes to Privacy Risk Management and Compliance in Relation to Cloud Computing

Total Periods: 45

Pedagogical Tools:	Python, Jupyter Notebooks, Algorithm visualization tools, Online coding platforms (LeetCode, HackerRank), Flowchart tools, Pseudocode practice

TEXT BOOKS:

Sl. No	Authors	Title of the Book	Publisher	Year
1	Tim Mather, Subra Kumaraswamy & Shahed Latif	Cloud Security and Privacy: An Enterprise Perspective on Risks and Compliance	O’Reilly Media	2009

REFERENCE BOOKS:

Sl. No	Authors	Title of the Book	Publisher	Year
1	Ronald L. Krutz & Russell Dean Vines	Cloud Security	Wiley	2010
2	John Rittinghouse & James Ransome	Cloud Computing: Implementation, Management, and Security	CRC Press	2009
3	Vic (J. R.) Winkler	Securing the Cloud	Syngress	2011

COs / POs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	3	2	1	2	–	–	–	–	–	–	1	3	2
CO2	3	3	3	2	2	–	–	–	–	–	–	1	3	3
CO3	3	3	3	1	3	–	–	–	–	–	–	1	3	3
CO4	3	3	2	2	2	–	–	–	–	–	–	1	3	2
CO5	2	3	2	1	3	–	–	–	1	1	–	1	3	3

R 2024	IT					
24ITX06	**3D MODELLING & ANIMATION**	L	T	P	C	PE
		3	0	0	3	

COURSE OBJECTIVES:

- To understand the basic concepts of Robotic Process Automation.
- To expose to the key RPA design and development strategies and methodologies.
- To learn the fundamental RPA logic and structure.
- To explore the Exception Handling, Debugging and Logging operations in RPA.
- To learn to deploy and maintain the software bot.

COURSE OUTCOMES:

CO1: Enunciate the key distinctions between RPA and existing automation techniques and platforms.
CO2: Use UiPath to design control flows and work flows for the target process
CO3: Implement recording, web scraping andprocess mining by automation
CO4: Use UIPath Studio to detect, and handle exceptions in automation processes
CO5: Implement and use Orchestrator for creation, monitoring, scheduling, and controlling of automated bots and processes.

UNIT: I	COMPUTER-BASED ANIMATION & GETTING STARTED WITH MAX	9

Definition of Computer-based Animation, Basic Types of Animation: Real Time, Non-real-time, Definition of Modelling, Creation of 3D objects. Exploring the Max Interface, Controlling & Configuring the Viewports, Customizing the Max Interface & Setting Preferences.

UNIT: II	2D SPLINES & SHAPES & COMPOUND OBJECT	9

Understanding 2D Splines& shape, Extrude & Bevel 2D object to 3D, Understanding Loft & terrain, Modeling simple 4 objects with splines, Understanding morph, scatter, conform, connect compound objects, blobmesh, Boolean, Proboolean & procutter compound object

UNIT: III	3D MODELLING	

Modeling with Polygons, using the graphite, working with XRefs, Building simple scenes, Building complex scenes with XRefs, using assets tracking, deforming surfaces & using the mesh modifiers, modeling with patches & NURBS

UNIT: IV	KEYFRAME ANIMATION	

Creating Keyframes, Auto Keyframes, Move & Scale Keyframe on the timeline, Animating with constraints & simple controllers, animation Modifiers & complex controllers, function curves in the track view, motion mixer etc.

UNIT: V	LIGHTING & CAMERA	9

Configuring & Aiming Cameras, camera motion blur, camera depth of field, camera tracking, using basic lights & lighting Techniques, working with advanced lighting, Light Tracing, Radiosity, video post, mental ray lighting etc.

Total Periods: 45

Pedagogical Tools:	Python, Jupyter Notebooks, Algorithm visualization tools, Online coding platforms (LeetCode, HackerRank), Flowchart tools, Pseudocode practice

TEXT BOOKS:

Sl. No	Authors	Title of the Book	Publisher	Year
1	Kelly L. Murdock	Autodesk 3ds Max Complete Reference Guide	McGraw-Hill Education	2014
2	Dariush Derakhshani	Introducing Autodesk 3ds Max	Sybex (Wiley)	2016

REFERENCE BOOKS:

Sl. No	Authors	Title of the Book	Publisher	Year
1	Jeremy Birn	Digital Lighting and Rendering	New Riders Publishing	2013
2	Isaac V. Kerlow	The Art of 3D Computer Animation and Effects	Wiley	2009
3	Tom Capizzi	Autodesk 3ds Max Modeling for Games	Focal Press	2012

COs / POs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	3	2	1	2	–	–	–	–	–	–	1	3	2
CO2	3	3	3	2	2	–	–	–	–	–	–	1	3	3
CO3	3	3	3	1	3	–	–	–	–	–	–	1	3	3
CO4	3	3	2	2	2	–	–	–	–	–	–	1	3	2
CO5	2	3	2	1	3	–	–	–	1	1	–	1	3	3

R 2024	IT					
24ITX01	CLOUD SERVICE MANAGEMENT	L	T	P	C	PE
		3	0	0	3	

COURSE OBJECTIVES:

- To understand the fundamentals of cloud computing and cloud service models.
- To learn cloud service management concepts, architectures, and deployment strategies.
- To study monitoring, provisioning, and management of cloud resources and services.
- To analyze security, risk management, and compliance issues in cloud services.
- To explore service level agreements (SLA), governance, and performance management in cloud environments.

COURSE OUTCOMES:

CO1 Explain the fundamentals of cloud computing and cloud service models.

CO2 Analyze cloud service architectures and deployment models.

CO3 Apply techniques for managing cloud resources, monitoring, and provisioning services.

CO4 Evaluate security, risk, and compliance aspects in cloud service management.

CO5 Assess service level agreements, governance, and performance management in cloud environments.

UNIT: I	Fundamentals of Cloud Computing	9

Cloud computing concepts and characteristics – Evolution of cloud computing – Cloud service models (IaaS, PaaS, SaaS) – Cloud deployment models (Public, Private, Hybrid, Community) – Virtualization concepts – Benefits and challenges of cloud computing.

UNIT: II	Cloud Service Architecture	9

Cloud architecture and components – Resource pooling and elasticity – Cloud infrastructure management – Virtual machines and containers – Cloud platforms and tools – Multi-cloud and hybrid cloud architectures.

UNIT: III	Cloud Service Provisioning and Management	9

Cloud resource provisioning – Monitoring and performance management – Auto-scaling and load balancing – Cloud orchestration and automation – Cloud service lifecycle management – Cost management and billing models.

UNIT: IV	Security and Risk Management in Cloud	9

Cloud security architecture – Identity and access management – Data protection and encryption – Risk management and compliance – Security threats and vulnerabilities in cloud – Disaster recovery and business continuity.

UNIT: V	Cloud Governance and Service Level Management	9

Service Level Agreements (SLA) – Cloud governance frameworks – Performance monitoring and reporting – Cloud auditing and compliance – Best practices for cloud service management – Future trends in cloud services.

Total Periods: 45

Pedagogical Tools:	Python, Jupyter Notebooks, Algorithm visualization tools, Online coding platforms (LeetCode, HackerRank), Flowchart tools, Pseudocode practice

TEXT BOOKS:

Sl. No	Authors	Title of the Book	Publisher	Year
1	Thomas Erl, Ricardo Puttini & Zaigham Mahmood	Cloud Computing: Concepts, Technology & Architecture	Pearson	2013
2	George Reese	Cloud Application Architectures	O’Reilly Media	2009

REFERENCE BOOKS:

Sl. No	Authors	Title of the Book	Publisher	Year
1	Rajkumar Buyya, Christian Vecchiola & S. Thamarai Selvi	Mastering Cloud Computing	McGraw Hill	2013
2	Judith Hurwitz, Robin Bloor, Marcia Kaufman & Fern Halper	Cloud Computing for Dummies	Wiley	2010

COs / POs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	3	2	1	2	–	–	–	–	–	–	1	3	2
CO2	3	3	3	2	2	–	–	–	–	–	–	1	3	3
CO3	3	3	3	1	3	–	–	–	–	–	–	1	3	3
CO4	3	3	2	2	2	–	–	–	–	–	–	1	3	2
CO5	2	3	2	1	3	–	–	–	1	1	–	1	3	3

R 2024	AI&DS					
24ITX02	ROBOTICS PROCESS AUTOMATION	L	T	P	C	PE
		3	0	0	3	

COURSE OBJECTIVES:

- To understand the basic concepts of Robotic Process Automation.
- To expose to the key RPA design and development strategies and methodologies.
- To learn the fundamental RPA logic and structure.
- To explore the Exception Handling, Debugging and Logging operations in RPA.
- To learn to deploy and maintain the software bot.

COURSE OUTCOMES:

CO1: Enunciate the key distinctions between RPA and existing automation techniques and platforms.
CO2: Use UiPath to design control flows and work flows for the target process
CO3: Implement recording, web scraping andprocess mining by automation
CO4: Use UIPath Studio to detect, and handle exceptions in automation processes
CO5: Implement and use Orchestrator for creation, monitoring, scheduling, and controlling of automated bots and processes.

UNIT: I	**INTRODUCTION TO ROBOTIC PROCESS AUTOMATION**	**9**

Emergence of Robotic Process Automation (RPA), Evolution of RPA, Differentiating RPA from Automation - Benefits of RPA - Application areas of RPA, Components of RPA, RPA Platforms.

UNIT: II	**AUTOMATION PROCESS ACTIVITIES**	**9**

Sequence, Flowchart & Control Flow: Sequencing the Workflow, Activities, Flowchart, Control Flow for Decision making. Data Manipulation: Variables, Collection, Arguments, Data Table, Clipboard management.

UNIT: III	**APP INTEGRATION, RECORDING AND SCRAPING**	**9**

App Integration, Recording, Scraping, Selector, Workflow Activities. Recording mouse and keyboard actions to perform operation, scraping data from website and writing to CSV. Process Mining.

UNIT: IV	**EXCEPTION HANDLING AND CODE MANAGEMENT**	**9**

Exception handling, Common exceptions, Logging- Debugging techniques, collecting crash dumps, Error reporting. Code management and maintenance: Project organization, Nesting workflows, Reusability, Templates, Commenting techniques, State Machine.

UNIT: V	**DEPLOYMENT AND MAINTENANCE**	**9**

Publishing using publish utility, Orchestration Server, Control bots, Orchestration Server to deploy bots, License management, Publishing and managing updates.

Total Periods: 45

PRACTICAL EXERCISES:

Setup and Configure a RPA tool and understand the user interface of the tool:

1. Create a Sequence to obtain user inputs display them using a message box
2. Create a Flowchart to navigate to a desired page based on a condition
3. Create a State Machine workflow to compare user input with a random number.
4. Build a process in the RPA platform using UI Automation Activities.
5. Automate login to (web)Email account
6. Recording mouse and keyboard actions.
7. Scraping data from website and writing to CSV

Pedagogical Tools:	Python, Jupyter Notebooks, Algorithm visualization tools, Online coding platforms (LeetCode, HackerRank), Flowchart tools, Pseudocode practice

TEXT BOOKS:

Sl. No	Authors	Title of the Book	Publisher	Year
1	Alok Mani Tripathi	Learning Robotic Process Automation: Create Software Robots and Automate Business Processes with the Leading RPA Tool – UiPath	Packt Publishing	2018
2	Tom Taulli	The Robotic Process Automation Handbook: A Guide to Implementing RPA Systems	Apress Publications	2020

REFERENCE BOOKS:

Sl. No	Authors	Title of the Book	Publisher	Year
1	Frank Casale, Rebecca Dilla, Heidi Jaynes & Lauren Livingston	Introduction to Robotic Process Automation: A Primer	Institute of Robotic Process Automation, Amazon Asia-Pacific Holdings Private Limited	2018
2	Richard Murdoch	Robotic Process Automation: Guide to Building Software Robots, Automate Repetitive Tasks & Become an RPA Consultant	Amazon Asia-Pacific Holdings Private Limited	2018
3	Gerardus Blokdyk	Robotic Process Automation (RPA): A Complete Guide	5STARCooks	2020

COs / POs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	3	2	1	2	–	–	–	–	–	–	1	3	2
CO2	3	3	3	2	2	–	–	–	–	–	–	1	3	3
CO3	3	3	3	1	3	–	–	–	–	–	–	1	3	3
CO4	3	3	2	2	2	–	–	–	–	–	–	1	3	2
CO5	2	3	2	1	3	–	–	–	1	1	–	1	3	3

R 2024	IT					
24ITX03	AI BASED CONVERSATIONAL SYSTEM	L	T	P	C	PE
		3	0	0	3	

COURSE OBJECTIVES:

- To understand the fundamental concepts of conversational systems and basic programming structures.
- To learn and apply Natural Language Processing (NLP) techniques used in conversational technologies.
- To design and develop chatbots and conversational intelligent systems for real-world applications.
- To analyze the role of Machine Learning and Artificial Intelligence in conversational platforms.
- To evaluate the performance of conversational systems using appropriate metrics and analytics.

COURSE OUTCOMES:

CO1: Understand the fundamentals of conversational systems and foundational blocks of programming.
CO2: Apply the natural language processing techniques in building conversational systems.
CO3: Design and build chatbots and conversational intelligent systems.
CO4: Analyse the significance of machine learning methods and artificial intelligence in conversational technologies.
CO5: Perform the analytics on conversational systems using performance metrics.

UNIT: I	FUNDAMENTALS OF CONVERSATIONAL SYSTEMS	9

Introduction: Overview, Case studies, Explanation about different modes of engagement for a human being, History and impact of AI. Underlying technologies: Natural Language Processing, Artificial Intelligence and Machine Learning, NLG, Speech-To-Text, Text-To-Speech, Computer Vision etc.

UNIT: II	NATURAL LANGUAGE PROCESSING	9

Introduction: Brief history, Basic Concepts, Phases of NLP, Application of chatbots etc. General chatbot architecture, Basic concepts in chatbots: Intents, Entities, Utterances, Variables and Slots, Fulfilment, Lexical Knowledge Networks (WordNet, Verbnet, PropBank, etc.)

UNIT: III	BUILDING A CHATBOT/CONVERSATIONAL AI SYSTEMS	9

Fundamentals of Conversational Systems (NLU, DM and NLG). Chatbot framework & Architecture, Conversational Flow & Design, Intent Classification (ML and DL based techniques), Dialogue Management Strategies, Natural Language Generation.UX design, APIs and SDKs, Usage of Conversational Design Tools. Introduction to popular chatbot frameworks – Google Dialog flow, Microsoft Bot Framework, Amazon Lex.

UNIT: IV	ROLE OF ML/AI IN CONVERSATIONAL TECHNOLOGIES	9

Understanding on how Conversational Systems uses ML technologies in ASR, NLP, Advanced Dialog management, Language Translation, Emotion/Sentiment Analysis, Information extraction, etc. to effectively converse. Case Study.

UNIT: V	OVERVIEW ON CONVERSATIONAL ANALYTICS	9

Conversation Analytics: The need of it ,Introduction to Conversational Metrics, Summary, Robots and Sensory Applications overview, XR Technologies in Conversational Systems , XR-Commerce, Future technologies and market innovations overview.

Total Periods: 45

PRACTICAL EXPERIMENTS

1. Study of basics of python programming related to conversational AI
2. Implementation of lexical analysis
3. Implementation of syntactic analysis
4. Implementation of Sentimental Analysis
5. Implementation of natural language processing using python libraries.
6. Testing of chatbot frameworks
7. Implementation of voice bots
8. Implementation of a generic chat bot
9. Implementation of a bot for a class room discussion application.

Pedagogical Tools:	Python, Jupyter Notebooks, Algorithm visualization tools, Online coding platforms (LeetCode, HackerRank), Flowchart tools, Pseudocode practice

TEXT BOOKS:

Sl. No	Authors	Title of the Book	Publisher	Year
1	Michael McTear	Conversational AI: Dialogue Systems, Conversational Agents and Chatbots	Morgan & Claypool Publishers	2020
2	Luis Fernando D. Haro, Zoraida Callejas & Satoshi Nakamura	Conversational Dialogue Systems for the Next Decade	Springer	2021

REFERENCE BOOKS:

Sl. No	Authors	Title of the Book	Publisher	Year
1	Srini Janarthanam	Chatbots and Conversational UI Development	Packt Publishers	2017
2	Diana Pérez-Marín & Ismael Pascual-Nieto	Conversational Agents and Natural Language Interaction	IGI Global Publishers	2011

COs / POs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	2	2	1	2	–	–	–	–	–	–	1	3	2
CO2	3	3	3	2	2	–	–	–	–	–	–	1	3	3
CO3	3	3	3	1	3	–	–	–	–	–	–	1	3	3
CO4	3	2	2	2	2	–	–	–	–	–	–	1	3	2
CO5	3	2	2	1	3	–	–	–	1	1	–	1	3	3

R 2024	IT					
24ITX04	**BIG DATA INTEGRATION AND PROCESSING**	L	T	P	C	PE
		3	0	0	3	

COURSE OBJECTIVES:

- To understand the fundamentals of big data and distributed computing frameworks.
- To learn big data storage technologies and data integration techniques.
- To study big data processing frameworks such as Hadoop and Spark.
- To analyze methods for data processing, transformation, and analytics in big data environments.
- To explore real-time data processing and big data applications in various domains

COURSE OUTCOMES:

CO1 Explain the concepts and characteristics of big data and its ecosystem.

CO2 Apply big data storage and integration techniques for large datasets.

CO3 Implement big data processing using Hadoop and related frameworks.

CO4 Analyze big data processing techniques using tools such as Spark and MapReduce.

CO5 Evaluate real-time big data applications and processing frameworks.

UNIT: I	Introduction to Big Data	9
Big data concepts and characteristics – Big data sources – Challenges of big data – Big data ecosystem – Distributed computing fundamentals – Overview of Hadoop architecture – Hadoop Distributed File System (HDFS).		
UNIT: II	**Big Data Integration**	**9**
Data integration concepts – Data ingestion techniques – ETL and ELT processes – Data cleaning and transformation – Data warehousing and data lakes – Tools for big data integration.		
UNIT: III	**Big Data Processing Frameworks**	**9**
MapReduce programming model – Hadoop MapReduce architecture – YARN resource management – Introduction to Apache Spark – Spark architecture and components.		
UNIT: IV	**Data Processing and Analytics**	**9**
Batch processing and stream processing – Data transformation techniques – Spark RDD and DataFrame – Query processing with Hive and Pig – Data visualization and analytics.		
UNIT: V	**Real-Time Big Data Processing**	**9**
Real-time data processing concepts – Apache Kafka and streaming systems – Spark Streaming – NoSQL databases – Applications of big data in healthcare, finance, and social media.		
		Total Periods: 45

Pedagogical Tools:	Python, Jupyter Notebooks, Algorithm visualization tools, Online coding platforms (LeetCode, HackerRank), Flowchart tools, Pseudocode practice

TEXT BOOKS:

Sl. No	Authors	Title of the Book	Publisher	Year
1	Tom White	Hadoop: The Definitive Guide	O'Reilly Media	2015
2	Bill Chambers & Matei Zaharia	Spark: The Definitive Guide	O'Reilly Media	2018

REFERENCE BOOKS:

Sl. No	Authors	Title of the Book	Publisher	Year
1	Judith Hurwitz, Alan Nugent, Fern Halper & Marcia Kaufman	Big Data for Dummies	Wiley	2013
2	Nathan Marz & James Warren	Big Data: Principles and Best Practices of Scalable Realtime Data Systems	Manning Publications	2015

3	Jiawei Han, Micheline Kamber & Jian Pei	Data Mining: Concepts and Techniques	Morgan Kaufmann	2022

COs / POs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	2	2	1	2	–	–	–	–	–	–	1	3	2
CO2	3	3	3	2	2	–	–	–	–	–	–	1	3	3
CO3	3	3	3	1	3	–	–	–	–	–	–	1	3	3
CO4	3	2	2	2	2	–	–	–	–	–	–	1	3	2
CO5	2	3	2	1	3	–	–	–	1	1	–	1	3	3

<table>
<tr><td>R 2024</td><td colspan="5">IT</td><td></td></tr>
<tr><td rowspan="2">24ITX05</td><td rowspan="2">PREDICTIVE ANALYTICS</td><td>L</td><td>T</td><td>P</td><td>C</td><td rowspan="2">PE</td></tr>
<tr><td>3</td><td>0</td><td>0</td><td>3</td></tr>
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COURSE OBJECTIVES:

- To understand the process of defining business objectives and collecting relevant data for predictive analytics.
- To learn the steps involved in designing, building, evaluating, and implementing predictive models for business applications.
- To study and compare different predictive modeling techniques used in data analysis.
- To identify and select appropriate predictive modeling approaches for solving business problems.
- To apply predictive modeling techniques using tools such as SPSS Modeler for practical data analysis.

COURSE OUTCOMES:

CO1: Understand the process of formulating business objectives, data selection/collection.
CO2: Preparation and process to successfully design, build, evaluate and implement predictive models for a various business application.
CO3: Compare the underlying predictive modeling techniques.
CO4: Select appropriate predictive modeling approaches to identify cases to progress with.
CO5: Apply predictive modeling approaches using a suitable package such as SPSS Modeler

UNIT: I	**INTRODUCTION TO DATA MINING**	**9**

what is Data Mining? Concepts of Data mining, Technologies Used, Data Mining Process, KDD Process Model, CRISP – DM, Mining on various kinds of data, Applications of Data Mining, Challenges of Data Mining.

UNIT: II	**DATA UNDERSTANDING AND PREPARATION**	**9**

Introduction, Reading data from various sources, Data visualization, Distributions and summary statistics, Relationships among variables, Extent of Missing Data. Segmentation, Outlier detection, Automated Data Preparation, Combining data files, Aggregate Data, Duplicate Removal, Sampling DATA, Data Caching, Partitioning data, Missing Values.

UNIT: III	**MODEL DEVELOPMENT & TECHNIQUES**	**9**

Data Partitioning, Model selection, Model Development Techniques, Neural networks, Decision trees, Logistic regression, Discriminant analysis, Support vector machine, Bayesian Networks, Linear Regression, Cox Regression, Association rules.

UNIT: IV	**MODEL EVALUATION AND DEPLOYMENT**	**9**

Introduction, Model Validation, Rule Induction Using CHAID, Automating Models for Categorical and Continuous targets, Comparing and Combining Models, Evaluation Charts for Model Comparison, MetaLevel Modeling, Deploying Model, Assessing Model Performance, Updating a Model.

UNIT: V	**ADVANCED TRENDS & ETHICS**	**9**

Complex Data: Mining text, web, spatial, and multimedia data streams. Big Data: Real-time analytics, cloud mining, and scalability. Governance: Data privacy (GDPR compliance), ethics, and bias mitigation.

Total Periods: 45

Pedagogical Tools:	Python, Jupyter Notebooks, Algorithm visualization tools, Online coding platforms (LeetCode, HackerRank), Flowchart tools, Pseudocode practice

TEXT BOOKS:

Sl. No	Authors	Title of the Book	Publisher	Year
1	John D. Kelleher, Brian Mac Namee & Aoife D'Arcy	Fundamentals of Machine Learning for Predictive Data Analytics: Algorithms, Worked Examples, and Case Studies (2nd Edition)	MIT Press	2020

REFERENCE BOOKS:

Sl. No	Authors	Title of the Book	Publisher	Year
1	Jiawei Han, Micheline Kamber & Jian Pei	Data Mining: Concepts and Techniques (4th Edition)	Morgan Kaufmann	2022
2	Pang-Ning Tan, Michael Steinbach, Anuj Karpatne & Vipin Kumar	Introduction to Data Mining (2nd Edition)	Pearson	2021

COs / POs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	2	1	2	1	2	–	–	–	–	–	–	1	3	2
CO2	3	2	3	2	2	–	–	–	–	–	–	1	3	3
CO3	3	1	3	1	3	–	–	–	–	–	–	1	3	3
CO4	2	2	2	2	2	–	–	–	–	–	–	1	3	2
CO5	2	3	2	1	3	–	–	–	1	1	–	1	3	3

R 2024	CSE					
24CSX36	EXPLAINABLE AI	L	T	P	C	PE
		3	0	0	3	

COURSE OBJECTIVES:

- Understand the fundamentals and importance of explainable AI in modern machine learning systems
- Master interpretable model architectures and post-hoc explanation techniques
- Learn visualization methods and interactive tools for model transparency
- Implement fairness, bias detection, and ethical AI practices in real-world applications
- Explore regulatory compliance and best practices for deploying explainable AI systems

COURSE OUTCOMES:

CO 1: Design and develop interpretable machine learning models with transparency requirements

CO 2: Apply post-hoc explanation methods like LIME, SHAP, and attention mechanisms to complex models

CO 3: Create effective visualizations and interactive dashboards for model explanation

CO 4: Evaluate and mitigate bias in AI systems while ensuring fairness across diverse populations

CO 5: Deploy explainable AI solutions that meet regulatory standards and ethical guidelines

UNIT: I	FOUNDATIONS OF EXPLAINABLE AI	9

Introduction to XAI -- need for explainability in AI systems -- interpretability vs explainability -- transparency and trust. Types of explanations -- global vs local explanations -- model-agnostic vs model-specific approaches. Interpretable models -- linear regression -- decision trees -- rule-based systems -- generalized additive models (GAMs). Model complexity trade-offs -- accuracy vs interpretability -- selecting appropriate models for different contexts.

UNIT: II	POST-HOC EXPLANATION METHODS	9

Feature importance methods -- permutation importance -- SHAP (SHapley Additive exPlanations) -- integrated gradients. Local explanation techniques -- LIME (Local Interpretable Model-agnostic Explanations) -- counterfactual explanations -- anchor explanations. Attention mechanisms -- attention visualization in neural networks -- transformer interpretability -- attention weights analysis. Saliency maps -- gradient-based methods -- CAM and Grad-CAM -- guided backpropagation for vision models.

UNIT: III	VISUALIZATION AND INTERACTIVE EXPLANATIONS	9

Visualization techniques -- feature importance plots -- partial dependence plots -- individual conditional expectation (ICE) plots. Interactive explanation tools -- What-If Tool -- InterpretML -- Alibi -- building custom dashboards with Streamlit and Plotly. Model debugging -- identifying failure modes -- error analysis -- detecting spurious correlations. Explanation quality metrics -- faithfulness -- consistency -- user studies and human evaluation of explanations.

UNIT: IV	FAIRNESS AND BIAS IN AI	9

Fairness definitions -- demographic parity -- equalized odds -- individual fairness -- causal fairness. Bias detection methods -- fairness metrics -- disparate impact analysis -- intersectional fairness. Bias mitigation techniques -- pre-processing (data resampling, reweighting) -- in-processing (adversarial debiasing, fairness constraints) -- post-processing (threshold optimization). Fairness toolkits -- AI Fairness 360 -- Fairlearn -- implementing bias audits in production systems.

UNIT: V	ETHICS, REGULATIONS, AND APPLICATIONS	9
AI ethics frameworks -- responsible AI principles -- accountability and governance -- ethical design practices. Regulatory landscape -- GDPR right to explanation -- EU AI Act -- FDA guidelines for medical AI -- industry-specific compliance. Domain-specific XAI -- healthcare (clinical decision support) -- finance (credit scoring, fraud detection) -- autonomous systems (self-driving cars) -- legal applications. Deployment best practices -- documentation standards -- model cards -- explainability audits -- case studies of successful XAI implementations.		

Total Periods: 45

Pedagogical Tools:	Python, Scikit-learn, SHAP, LIME, InterpretML, Alibi, AI Fairness 360, Fairlearn, TensorFlow, PyTorch, Captum, What-If Tool, Streamlit, Plotly, Jupyter Notebooks

TEXT BOOKS:

Sl. No	Authors	Title of the Book	Publisher	Year
1	Christoph Molnar	Interpretable Machine Learning: A Guide for Making Black Box Models Explainable	Self-published	2024
2	Ajay Thampi	Interpretable AI: Building Explainable Machine Learning Systems	Manning	2024

REFERENCE BOOKS:

Sl. No	Authors	Title of the Book	Publisher	Year
1	Alejandro Barredo Arrieta et al.	Explainable Artificial Intelligence (XAI): Concepts, Taxonomies, Opportunities and Challenges	Information Fusion	2024
2	Cynthia Rudin et al.	Interpretable Machine Learning: Fundamental Principles and 10 Grand Challenges	Statistics Surveys	2024
3	Finale Doshi-Velez, Been Kim	Towards A Rigorous Science of Interpretable Machine Learning	arXiv	2024

CO / PO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	3	3	2	3	–	–	2	–	–	–	1	3	2
CO2	3	3	2	3	3	–	–	2	–	–	–	1	3	3
CO3	2	2	2	2	3	–	–	–	2	3	–	1	2	3
CO4	3	3	2	3	2	2	–	3	–	–	–	1	3	2
CO5	3	3	3	3	3	2	–	3	2	2	–	2	3	3

R 2024	CSE					
24CSX37	Human Computer Interaction	L	T	P	C	PE
		3	0	0	3	

COURSE OBJECTIVES:

- Understand the principles and theories of human-computer interaction and user experience design
- Master user research methods, usability evaluation techniques, and user-centered design processes
- Learn interaction design patterns, interface design principles, and prototyping methodologies
- Implement accessibility standards and design inclusive interfaces for diverse user populations
- Explore emerging interaction paradigms including AR/VR, voice interfaces, and AI-powered UX

COURSE OUTCOMES:

CO 1: Conduct comprehensive user research and apply findings to create user-centered design solutions

CO 2: Design and prototype intuitive, accessible interfaces following established HCI principles and guidelines

CO 3: Evaluate interface usability through heuristic evaluation, user testing, and analytics-driven methods

CO 4: Implement accessibility standards (WCAG) and design inclusive experiences for users with disabilities

CO 5: Create innovative interaction designs for emerging technologies including voice, gesture, and immersive interfaces

UNIT: I	FOUNDATIONS OF HCI	9

Introduction to HCI -- history and evolution -- importance in software development -- interdisciplinary nature of HCI. Human factors -- perception and cognition -- attention and memory -- human error and limitations -- mental models. Interaction paradigms -- command-line vs GUI -- direct manipulation -- WIMP interfaces (windows, icons, menus, pointers). Design principles -- affordances -- feedback -- consistency -- visibility -- mapping -- Norman's design principles -- Gestalt principles of visual perception.

UNIT: II	USER RESEARCH AND REQUIREMENTS ANALYSIS	9

User research methods -- interviews and surveys -- contextual inquiry -- ethnographic studies -- diary studies -- participatory design. User personas and scenarios -- creating data-driven personas -- user journey mapping -- task analysis -- scenario-based design. Requirements gathering -- functional vs non-functional requirements -- user stories -- use cases -- prioritization techniques (MoSCoW method). Competitive analysis -- heuristic evaluation of competitors -- feature comparison -- identifying market gaps and opportunities.

UNIT: III	INTERACTION DESIGN AND PROTOTYPING	9

Interaction design process -- ideation techniques -- sketching and wireframing -- information architecture -- navigation design. Prototyping methods -- low-fidelity prototypes (paper, sketches) -- high-fidelity prototypes (Figma, Adobe XD) -- interactive prototypes -- wizard of oz technique. Design patterns -- UI patterns for common tasks -- navigation patterns -- form design -- responsive design patterns -- mobile-first design. Visual design -- typography -- color theory -- layout and grid systems -- creating design systems and component libraries.

UNIT: IV	USABILITY EVALUATION AND TESTING	9

Usability evaluation methods -- heuristic evaluation (Nielsen's 10 heuristics) -- cognitive walkthrough -- expert review. User testing -- planning user tests -- recruiting participants -- moderated vs unmoderated testing -- think-aloud protocol -- A/B testing. Quantitative metrics -- task completion rate -- time on task -- error rate -- System Usability Scale (SUS) -- Net Promoter Score (NPS). Analytics and data-driven design -- heatmaps and clickmaps -- session recordings -- funnel analysis -- using analytics to inform design decisions.

UNIT: V	ACCESSIBILITY AND EMERGING TECHNOLOGIES	9
Accessibility principles -- WCAG guidelines -- ARIA roles and landmarks -- keyboard navigation -- screen reader compatibility -- assistive technologies. Inclusive design -- designing for disabilities (visual, auditory, motor, cognitive) -- color contrast and text sizing -- captioning and transcripts -- testing with accessibility tools (axe, WAVE). Emerging interaction paradigms -- voice user interfaces (VUI) -- conversational AI and chatbots -- gesture-based interfaces -- virtual reality (VR) and augmented reality (AR) interfaces. Future of HCI -- brain-computer interfaces -- haptic feedback -- affective computing -- AI-powered personalization -- ethical considerations in AI-driven UX.		

Total Periods: 45

Pedagogical Tools:	Figma, Adobe XD, Sketch, InVision, Axure RP, Balsamiq, UserTesting, Hotjar, Google Analytics, Optimal Workshop, Marvel, Principle, Miro, JAWS, NVDA, axe DevTools, WAVE

TEXT BOOKS:

Sl. No	Authors	Title of the Book	Publisher	Year
1	Alan Dix, Janet Finlay, Gregory Abowd, Russell Beale	Human-Computer Interaction, 4th Edition	Pearson	2024
2	Jenny Preece, Helen Sharp, Yvonne Rogers	Interaction Design: Beyond Human-Computer Interaction, 6th Edition	Wiley	2024

REFERENCE BOOKS:

Sl. No	Authors	Title of the Book	Publisher	Year
1	Donald Norman	The Design of Everyday Things, Revised and Expanded Edition	Basic Books	2024
2	Steve Krug	Don't Make Me Think, Revisited: A Common Sense Approach to Web Usability, 4th Edition	New Riders	2024
3	Jakob Nielsen, Rolf Molich	Usability Engineering and Heuristic Evaluation	Academic Press	2024

CO / PO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	2	3	2	3	2	–	–	–	2	2	–	1	2	3
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CO4	2	2	2	2	2	2	–	3	2	2	–	1	2	3
CO5	2	2	3	2	3	–	–	–	2	2	–	2	2	3

R 2024	COMPUTER SCIENCE AND ENGINEERING					
		L	T	P	C	
24CSX38	**IMAGE PROCESSING AND ANALYTICS**	3	0	0	3	**PE**

COURSE OBJECTIVES:

• Understand digital image fundamentals, color models, and enhancement techniques
• Master edge detection, segmentation, and feature extraction methods
• Learn CNNs and deep learning architectures for computer vision
• Apply object detection, face recognition, and image generation techniques

COURSE OUTCOMES:

CO1: Implement image enhancement, filtering, and frequency domain processing techniques
CO2: Apply edge detection, segmentation, and morphological operations
CO3: Design and train CNNs for image classification using modern architectures
CO4: Develop object detection and face recognition systems
CO5: Build computer vision applications using OpenCV, PyTorch, and TensorFlow

UNIT: I	**DIGITAL IMAGE FUNDAMENTALS AND ENHANCEMENT**	**9**

Digital image representation – sampling and quantization – color models (RGB, HSV, YCbCr). Image enhancement – histogram equalization and matching – point operations – intensity transformations. Spatial filtering – smoothing (mean, Gaussian) and sharpening filters (Laplacian, unsharp masking)..

UNIT: II	**FEATURE EXTRACTION AND SEGMENTATION**	**9**

Edge detection – gradient operators (Sobel, Prewitt, Roberts) – Canny edge detector with hysteresis thresholding. Feature detection – Harris corner detector – SIFT fundamentals – feature descriptors and matching. Image segmentation – thresholding techniques (Otsu's method, adaptive) – region-based segmentation (region growing, watershed). Clustering-based segmentation – K-means and mean shift for image segmentation. Morphological operations – erosion, dilation, opening, closing – morphological gradient – applications in image preprocessing.

UNIT: III	**CONVOLUTIONAL NEURAL NETWORKS**	**9**

CNN fundamentals – convolution and pooling layers – activation functions (ReLU, LeakyReLU). CNN architectures – LeNet, AlexNet, VGGNet, ResNet – residual connections for deeper networks. Training CNNs – backpropagation – loss functions – optimization algorithms (SGD, Adam) – batch normalization and dropout. Transfer learning – using pre-trained models – feature extraction vs fine-tuning – domain adaptation. Image classification – building custom classifiers – data augmentation – model evaluation metrics – implementation using PyTorch and TensorFlow.

UNIT: IV	**OBJECT DETECTION AND RECOGNITION**	**9**

Object detection – R-CNN family (R-CNN, Fast R-CNN, Faster R-CNN) – Region Proposal Networks. YOLO architecture – single-stage detection – grid-based approach – anchor boxes – non-maximum suppression – real-time implementation. Semantic segmentation – Fully Convolutional Networks (FCN) – U-Net architecture for medical imaging – encoder-decoder design. Face detection and recognition – Haar cascades – modern deep learning methods – face embeddings – FaceNet and triplet loss. Image generation basics – Generative Adversarial Networks (GANs) – generator-discriminator architecture – applications in image synthesis.

UNIT: V	**APPLICATIONS AND TOOLS**	**9**

Medical image analysis – X-ray, CT, MRI processing – tumor detection – organ segmentation – CAD systems. Optical Character Recognition (OCR) – text detection – Tesseract OCR – CRNN for text recognition – scene text understanding. Video processing – optical flow – motion detection and tracking – action recognition basics. OpenCV programming – image I/O – applying filters – feature detection – object detection implementation – real-time applications. PyTorch and TensorFlow frameworks – building CNN models – training pipelines – pre-trained models – deployment – practical projects in autonomous vehicles, surveillance, and industrial applications.

Total Periods: 45

Pedagogical Tools:	Python, OpenCV, PyTorch, TensorFlow/Keras, NumPy, Matplotlib, Jupyter Notebooks, Google Colab, Kaggle datasets, Mini-projects

TEXT BOOKS:

Sl. No	Authors	Title of the Book	Publisher	Year
1	Rafael C. Gonzalez, Richard E. Woods	Digital Image Processing, 4th Edition	Pearson	2018
2	Richard Szeliski	Computer Vision: Algorithms and Applications, 2nd Edition	Springer	2022

REFERENCE BOOKS:

Sl. No	Authors	Title of the Book	Publisher	Year
1	Adrian Rosebrock	Deep Learning for Computer Vision with Python	PyImageSearch	2023
2	François Chollet	Deep Learning with Python, 2nd Edition	Manning	2021
3	Joseph Howse, Joe Minichino	Learning OpenCV 4 Computer Vision with Python 3	Packt	2020

CO / PO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
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CO4	3	3	3	3	3	–	–	–	–	–	–	1	3	3
CO5	3	3	3	2	3	–	–	–	2	2	–	2	3	3

R 2024	AI&DS					
24CSX33	AI IN DRONES DEVELOPMENT	L	T	P	C	PE
		3	0	0	3	

COURSE OBJECTIVES:

- Understand drone hardware architectures, flight control systems, and sensor integration
- Master autonomous navigation techniques including path planning and obstacle avoidance
- Learn computer vision and deep learning methods for drone-based perception and object detection
- Implement multi-drone systems with swarm intelligence and collaborative decision-making
- Explore real-world applications of AI-powered drones across various domains

COURSE OUTCOMES:

At the end of this course, students are able to

CO 1: Design and develop autonomous drone systems with intelligent flight control and navigation
CO 2: Implement computer vision algorithms for drone-based object detection, tracking, and mapping
CO 3: Build path planning and obstacle avoidance systems using AI and sensor fusion techniques
CO 4: Create multi-drone coordination systems with swarm intelligence and distributed control
CO 5: Deploy AI-powered drone solutions for applications in agriculture, surveillance, delivery, and rescue operations

UNIT: I	FOUNDATIONS OF DRONE TECHNOLOGY	9

Drone fundamentals -- types of drones (fixed-wing, rotary-wing, hybrid) -- components and subsystems -- propulsion systems. Flight control systems -- flight controllers (Pixhawk, ArduPilot) -- sensors (IMU, GPS, barometer, magnetometer) -- PID controllers and flight stabilization. Drone hardware platforms -- DJI development kits -- Raspberry Pi and NVIDIA Jetson integration -- communication protocols (MAVLink, ROS). Simulation environments -- Gazebo -- AirSim -- PX4 SITL -- testing and validation in simulated environments before real flights.

UNIT: II	AUTONOMOUS NAVIGATION AND PATH PLANNING	9

Localization and mapping -- GPS-based navigation -- visual odometry -- simultaneous localization and mapping (SLAM) -- sensor fusion with Kalman filters. Path planning algorithms -- A* and Dijkstra -- rapidly-exploring random trees (RRT) -- potential fields -- trajectory optimization. Obstacle avoidance -- depth cameras and LiDAR -- reactive vs deliberative approaches -- dynamic obstacle detection and avoidance. Mission planning -- waypoint navigation -- coverage path planning -- multi-objective optimization -- integrating mission planners with flight control systems.

UNIT: III	COMPUTER VISION AND DEEP LEARNING FOR DRONES	9

Image processing for drones -- aerial image characteristics -- perspective and distortion correction -- image stabilization techniques. Object detection and tracking -- YOLO and faster R-CNN for aerial images -- single object tracking -- multi-object tracking -- real-time processing on edge devices. Semantic segmentation -- land use classification -- building detection -- road extraction -- U-Net and DeepLab architectures for aerial imagery. 3D reconstruction -- structure from motion (SfM) -- photogrammetry -- point cloud generation -- applications in mapping and surveying.

UNIT: IV	MULTI-DRONE SYSTEMS AND SWARM INTELLIGENCE	9

Swarm robotics principles -- decentralized control -- self-organization -- emergence -- bio-inspired algorithms (ant colony, particle swarm). Multi-drone coordination -- formation control -- consensus algorithms -- distributed task allocation -- leader-follower architectures. Communication and networking -- ad-hoc networks -- delay-tolerant networking -- bandwidth optimization -- mesh networking for drone swarms. Collision avoidance in swarms -- geometric approaches -- velocity obstacles -- reciprocal collision avoidance -- coordination in crowded airspace.

UNIT: V	APPLICATIONS AND EMERGING TECHNOLOGIES	9

Precision agriculture -- crop monitoring and health assessment -- variable rate application -- multispectral and hyperspectral imaging -- yield prediction. Surveillance and security -- perimeter monitoring -- crowd monitoring -- search and rescue operations -- thermal imaging for night operations. Delivery and logistics -- autonomous package delivery -- route optimization -- urban air mobility -- regulatory considerations (FAA, DGCA). Emerging applications -- drone-based 5G

networks -- infrastructure inspection -- environmental monitoring -- disaster response -- ethical and safety considerations in autonomous drone deployment.
Total Periods :45

Pedagogical Tools:	Black board, chalk, group discussion, role play, youtube videos, NPTEL videos

TEXT BOOKS:

Sl. No	Authors	Title of the Book	Publisher	Year of publication
1	Randal W. Beard, Timothy W. McLain	Small Unmanned Aircraft: Theory and Practice	Princeton University Press	2024
2	Kimon P. Valavanis, George J. Vachtsevanos	Handbook of Unmanned Aerial Vehicles	Springer	2024

REFERENCE BOOKS:

Sl. No	Authors	Title of the Book	Publisher	Year of publication
1	Yunhao Chen	Artificial Intelligence in Unmanned Aerial Vehicles: Applications and Challenges	Springer	2025
2	Andreas Birk et al.	Autonomous Robots and Multi-Robot Systems	Springer	2024
3	Morgan Quigley et al.	Programming Robots with ROS: A Practical Introduction	O'Reilly Media	2024

COs / POs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	3	3	2	3	1	–	–	1	1	–	1	3	3
CO2	3	3	3	2	3	–	–	–	1	1	–	1	3	3
CO3	3	3	3	2	3	–	–	–	1	1	1	1	3	3
CO4	2	3	3	2	3	–	–	–	2	1	1	1	3	3
CO5	2	3	3	2	3	2	–	–	2	2	1	2	3	3

R 2024	CSE					
24CSX34	COMPUTER VISION	L	T	P	C	PE
		3	0	0	3	

COURSE OBJECTIVES:

• Understand the fundamentals of image formation, processing, and feature extraction
• Master classical computer vision techniques and modern deep learning approaches
• Learn to build applications for image classification, object detection, and segmentation
• Apply advanced architectures including CNNs, transformers, and diffusion models
• Develop real-world computer vision systems for various industry applications

COURSE OUTCOMES:

CO 1: Implement image processing and feature extraction techniques for vision tasks
CO 2: Build image classification systems using CNNs and transfer learning
CO 3: Develop object detection and segmentation models with YOLO, R-CNN, and U-Net
CO 4: Apply vision transformers and modern architectures for advanced applications
CO 5: Deploy computer vision solutions for real-world use cases including autonomous systems

UNIT: I	IMAGE FUNDAMENTALS AND PROCESSING	9

Introduction to computer vision -- applications across industries -- image formation and camera models. Digital image fundamentals -- pixels, resolution, color spaces (RGB, HSV, LAB) -- image representation. Image processing basics -- filtering (Gaussian, median, bilateral) -- edge detection (Sobel, Canny) -- morphological operations. Feature extraction -- corner detection (Harris, FAST) -- SIFT and SURF features -- HOG descriptors -- blob detection.

UNIT: II	DEEP LEARNING FOR IMAGE CLASSIFICATION	9

Convolutional Neural Networks -- convolution layers -- pooling -- activation functions -- backpropagation for CNNs. CNN architectures -- LeNet, AlexNet, VGGNet -- ResNet and skip connections -- Inception and GoogleNet -- EfficientNet. Transfer learning -- pre-trained models -- fine-tuning strategies -- domain adaptation -- data augmentation techniques. Training best practices -- batch normalization -- dropout -- learning rate scheduling -- handling overfitting -- evaluation metrics (accuracy, precision, recall, F1).

UNIT: III	OBJECT DETECTION AND LOCALIZATION	9

Object detection fundamentals -- sliding window -- region proposals -- bounding box regression. Two-stage detectors -- R-CNN, Fast R-CNN, Faster R-CNN -- Region Proposal Networks -- ROI pooling. One-stage detectors -- YOLO (v3, v4, v5, v8) -- SSD -- RetinaNet and focal loss -- real-time detection trade-offs. Advanced techniques -- non-maximum suppression -- anchor boxes -- feature pyramid networks -- multi-scale detection -- evaluation metrics (IoU, mAP).

UNIT: IV	IMAGE SEGMENTATION AND MODERN ARCHITECTURES	9

Semantic segmentation -- FCN (Fully Convolutional Networks) -- U-Net architecture -- encoder-decoder networks -- DeepLab and atrous convolution. Instance segmentation -- Mask R-CNN -- YOLACT -- segmentation evaluation metrics (Dice coefficient, IoU). Vision transformers -- self-attention for vision -- ViT (Vision Transformer) -- DETR for detection -- Swin Transformer. Advanced architectures -- ConvNeXt -- EfficientNetV2 -- MobileNet for edge devices -- neural architecture search.

UNIT: V	ADVANCED TOPICS AND APPLICATIONS	9

Generative models -- GANs for image generation -- StyleGAN -- diffusion models for image synthesis. Video analysis -- optical flow -- action recognition -- video object tracking -- temporal models. 3D vision -- depth estimation -- stereo vision -- point clouds -- 3D reconstruction basics. Real-world applications -- autonomous vehicles -- medical imaging -- facial recognition -- OCR and document analysis -- industrial inspection -- deployment with ONNX and TensorRT.

Total Periods: 45

Pedagogical Tools:	Python, OpenCV, PyTorch, TensorFlow/Keras, scikit-image, PIL, NumPy, Jupyter Notebooks, Google Colab, Pre-trained models (ImageNet), YOLO, Detectron2, MMDetection, Roboflow, Hugging Face, ONNX

TEXT BOOKS:

Sl. No	Authors	Title of the Book	Publisher	Year
1	Torralba, Isola, Freeman	Foundations of Computer Vision	MIT Press	2024
2	V Kishore Ayyadevara, Yeshwanth Reddy	Modern Computer Vision with PyTorch, 2nd Edition	Packt	2024

REFERENCE BOOKS:

Sl. No	Authors	Title of the Book	Publisher	Year
1	Richard Szeliski	Computer Vision: Algorithms and Applications, 2nd Edition	Springer	2022
2	Simon J. D. Prince	Computer Vision: Models, Learning, and Inference	Cambridge	2012
3	Mohamed Elgendy	Deep Learning for Vision Systems	Manning	2020

CO / PO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	3	2	2	3	–	–	–	–	–	–	1	3	2
CO2	3	3	3	2	3	–	–	–	–	–	–	1	3	3
CO3	3	3	3	3	3	–	–	–	–	–	–	1	3	3
CO4	3	3	2	3	3	–	–	–	–	–	–	2	3	3
CO5	3	3	3	3	3	2	–	–	2	2	–	2	3	3

R 2024	CSE					
24CSX35	COMPUTER VISION	L	T	P	C	PE
		3	0	0	3	

COURSE OBJECTIVES:

- Understand the fundamentals of edge computing architectures and their role in distributed systems
- Master edge device programming, resource optimization, and real-time processing techniques
- Learn edge-cloud integration patterns, data management, and synchronization strategies
- Implement security frameworks and privacy-preserving techniques for edge environments
- Explore IoT integration, 5G networks, and real-world edge computing applications

COURSE OUTCOMES:

CO 1: Design and deploy edge computing architectures for low-latency, distributed applications
CO 2: Develop and optimize applications for resource-constrained edge devices
CO 3: Implement edge-cloud orchestration and data management strategies for hybrid systems
CO 4: Apply security best practices and privacy-preserving techniques in edge deployments
CO 5: Build production-ready edge computing solutions for IoT, AI, and real-time analytics

UNIT: I	FOUNDATIONS OF EDGE COMPUTING	9

Introduction to edge computing -- definitions and characteristics -- edge vs cloud vs fog computing -- motivation and use cases. Edge computing architectures -- three-tier architecture (cloud-edge-device) -- multi-access edge computing (MEC) -- cloudlets and micro data centers. Edge infrastructure -- edge servers -- gateways -- smart sensors -- hardware platforms (Raspberry Pi, NVIDIA Jetson, Intel NUC). Performance metrics -- latency -- bandwidth -- energy efficiency -- comparing edge and cloud performance for different workloads.

UNIT: II	EDGE DEVICE PROGRAMMING AND OPTIMIZATION	9

Edge programming models -- containerization with Docker -- lightweight containers (Balena, K3s) -- serverless at the edge (OpenFaaS, Knative). Resource management -- CPU and memory optimization -- power management -- thermal management -- scheduling and task allocation. Edge AI and ML -- TensorFlow Lite -- PyTorch Mobile -- ONNX Runtime -- model quantization and pruning for edge deployment. Real-time processing -- stream processing frameworks -- Apache Kafka at the edge -- time-series data handling -- latency optimization techniques.

UNIT: III	EDGE-CLOUD INTEGRATION AND DATA MANAGEMENT	9

Edge orchestration platforms -- Kubernetes at the edge (K3s, MicroK8s) -- Azure IoT Edge -- AWS Greengrass -- Google Distributed Cloud Edge. Data management strategies -- edge caching -- data aggregation and preprocessing -- hierarchical storage management. Edge-cloud offloading -- computation offloading decisions -- adaptive offloading -- workload partitioning between edge and cloud. Data synchronization -- conflict resolution -- consistency models -- edge databases (SQLite, LevelDB) -- distributed data systems.

UNIT: IV	SECURITY AND PRIVACY IN EDGE COMPUTING	9

Edge security challenges -- attack vectors -- physical security -- secure boot and attestation -- trusted execution environments (TEE). Authentication and access control -- device identity management -- zero-trust architectures -- blockchain for edge security. Privacy-preserving techniques -- federated learning -- differential privacy -- secure multi-party computation -- homomorphic encryption at the edge. Network security -- VPNs and secure tunneling -- firewall configurations -- intrusion detection systems (IDS) for edge networks.

UNIT: V	APPLICATIONS AND EMERGING TRENDS	9
IoT and edge computing -- smart cities -- industrial IoT (IIoT) -- connected vehicles -- smart healthcare systems. 5G and edge computing -- mobile edge computing (MEC) -- network slicing -- ultra-reliable low-latency communications (URLLC). AI at the edge -- edge inference -- real-time computer vision -- natural language processing on edge devices -- AI model deployment pipelines. Case studies and deployment -- smart manufacturing -- autonomous vehicles -- augmented reality (AR) / virtual reality (VR) -- video analytics -- retail and logistics applications.		

Total Periods: 45

Pedagogical Tools:	Python, Docker, Kubernetes (K3s), TensorFlow Lite, PyTorch Mobile, ONNX Runtime, Raspberry Pi, NVIDIA Jetson, AWS Greengrass, Azure IoT Edge, Apache Kafka, MQTT, InfluxDB, Grafana, Node-RED

TEXT BOOKS:

Sl. No	Authors	Title of the Book	Publisher	Year
1	Blesson Varghese, Rajkumar Buyya	Edge Computing: Models, Technologies and Applications	Springer	2024
2	Schahram Dustdar, Rajkumar Buyya	Fog and Edge Computing: Principles and Paradigms	Wiley	2024

REFERENCE BOOKS:

Sl. No	Authors	Title of the Book	Publisher	Year
1	Jianhua He, Kun Yang	Edge Computing: A Primer	Springer	2024
2	Rajkumar Buyya et al.	Edge Computing: From Cloud to Things	Morgan Kaufmann	2025
3	Weisong Shi, Schahram Dustdar	Edge Computing: Vision and Challenges	IEEE Internet of Things Journal	2024

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CO4	3	2	2	2	3	2	–	2	–	–	–	1	3	2
CO5	3	3	3	3	3	2	–	–	2	2	–	2	3	3